

Protein Design

Part 3: *de novo* design

Anastassia Vorobieva

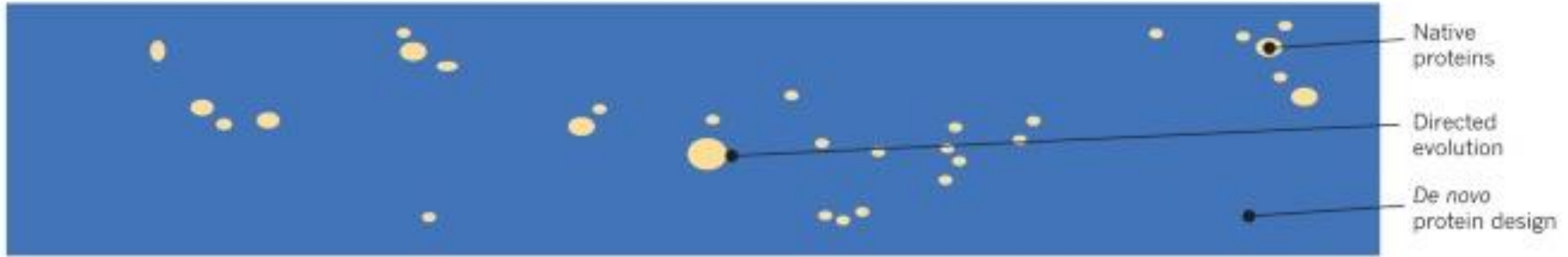


TIME	Wednesday 6/5	Thursday 7/5	Friday 8/5
09:00 – 10:30	Protein modelling and structure prediction: Intro to key concepts, from physics-based modelling to AI	De novo protein design: introduction, minimal sequence design, structure-based design principles	De novo design with AI models: RFDiffusion, ProteinMPNN, and ColabFold
10:30 – 11:00	Break and questions	Break and questions	Break and questions
11:00 – 12:00	Introduction to protein design: predicting the effect of mutations on protein stability	Structure-based de novo design: How to generate new structures? The chicken-and-egg problem	Practical session: De novo design of a SARS-CoV-2 RBD binder using RFDiffusion and ProteinMPNN
12:00 – 13:30	Lunch	Lunch	Lunch
13:30 – 15:00	Practical session: AlphaFold hands-on	Practical session: Parametric design of alpha-helical bundles	Practical session: Data analysis and group slide preparation
15:15 – 18:30	Practical session: In silico mutational scanning and $\Delta\Delta G$ calculations	Practical session: Sequence design for parametric bundles with PyRosetta	Practical session: Group presentations and results discussion

Introduction - protein design

Generating new-to-nature protein structure and sequences.

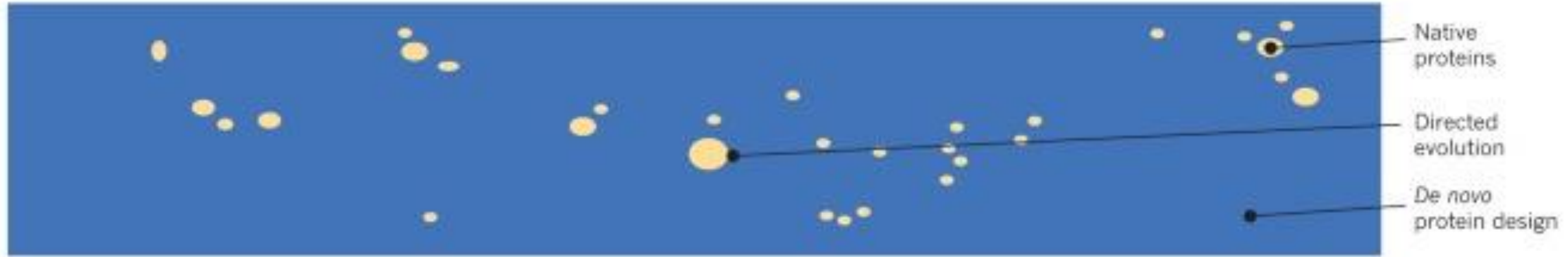
De novo protein design – exploring the dark protein sequence space



Single helix of 30 residues: $20^{30} = 10^{39}$ possible sequences

Which of these sequences encode foldable proteins?

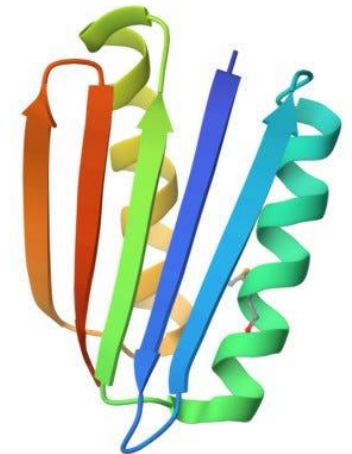
De novo protein design – exploring the dark protein sequence space

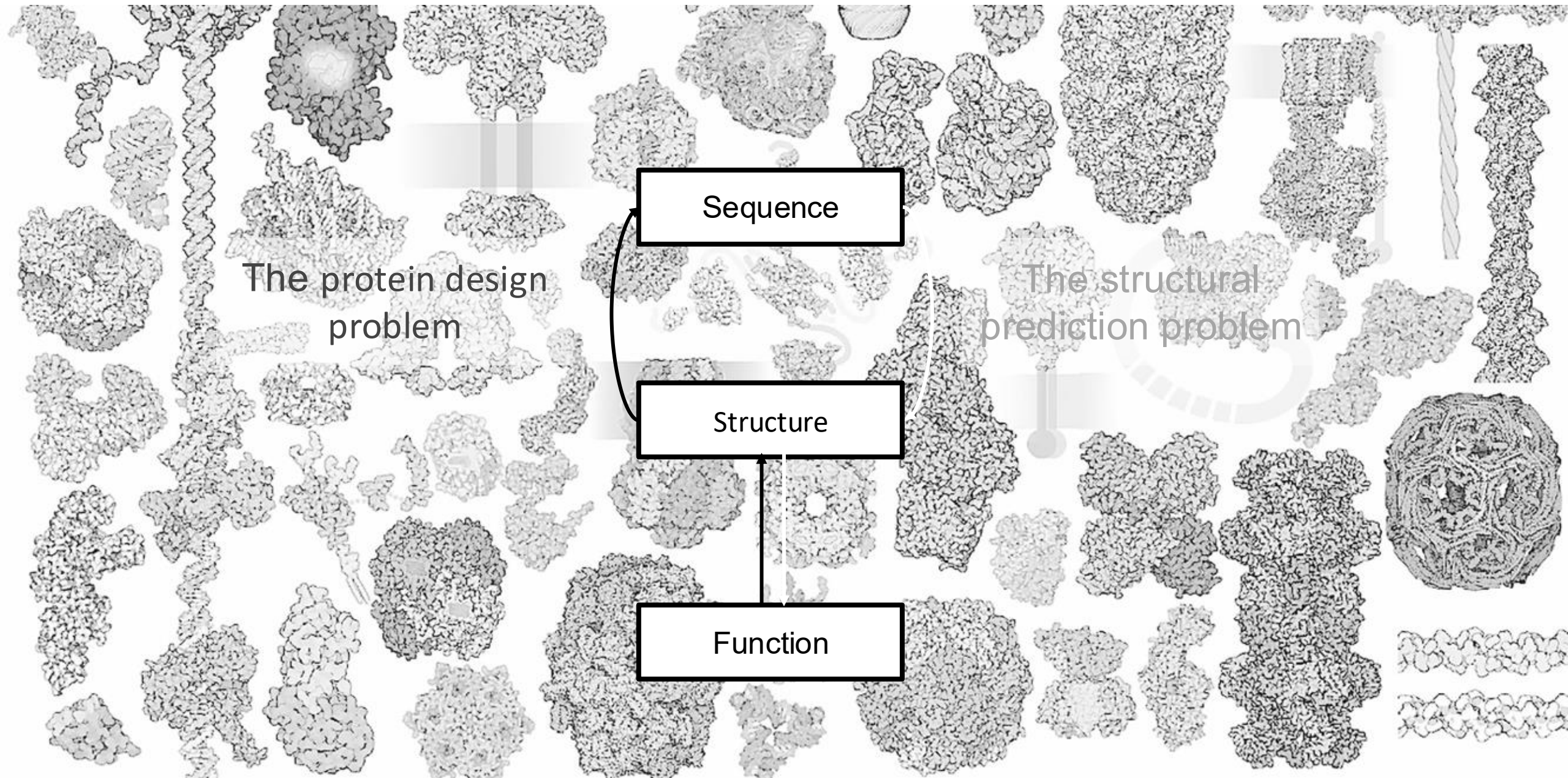


Single helix of 30 residues: $20^{30} = 10^{39}$ possible sequences

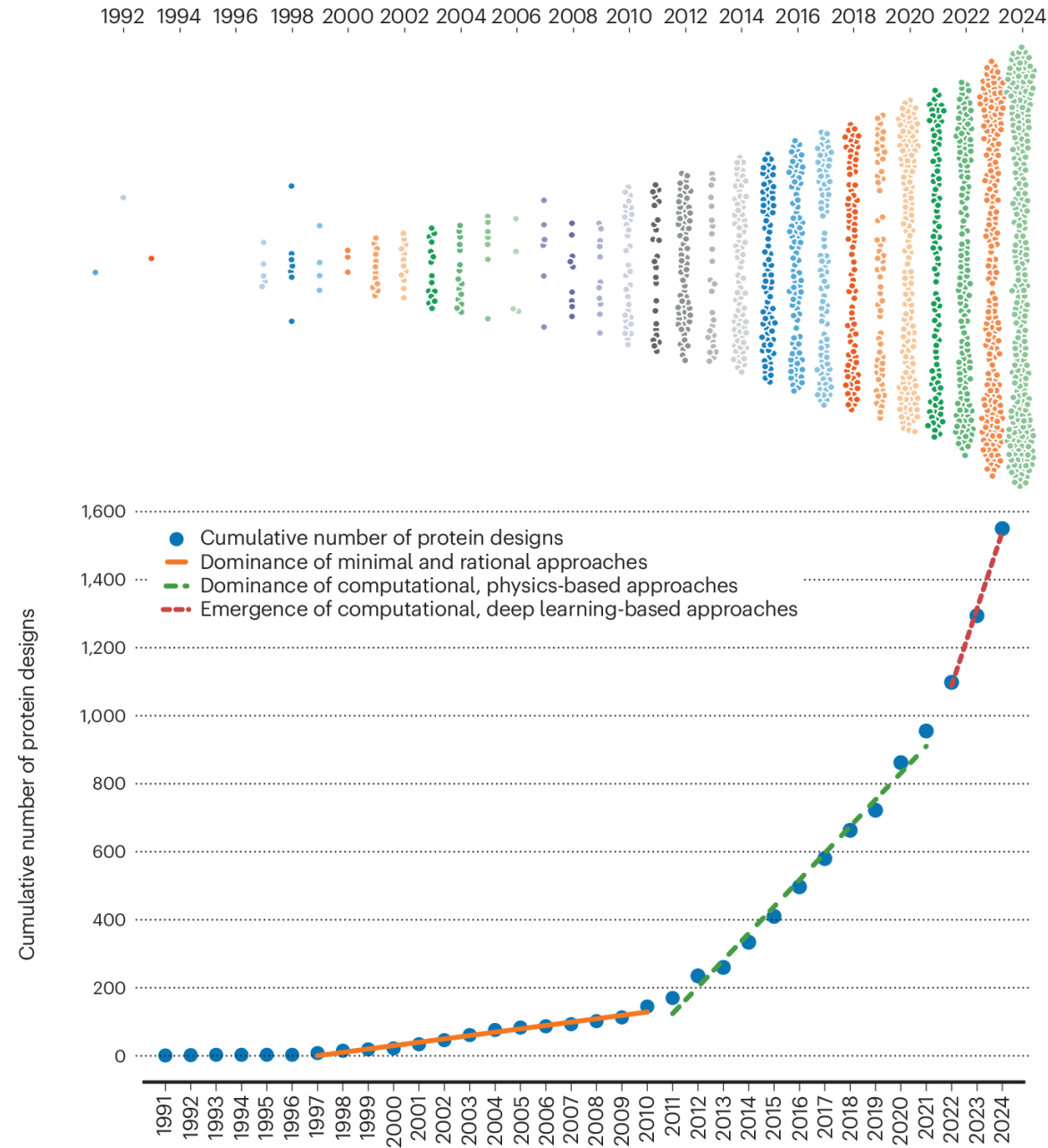
Which of these sequences encode foldable proteins?

Top7 – first time that a new-to-nature fold was designed from scratch



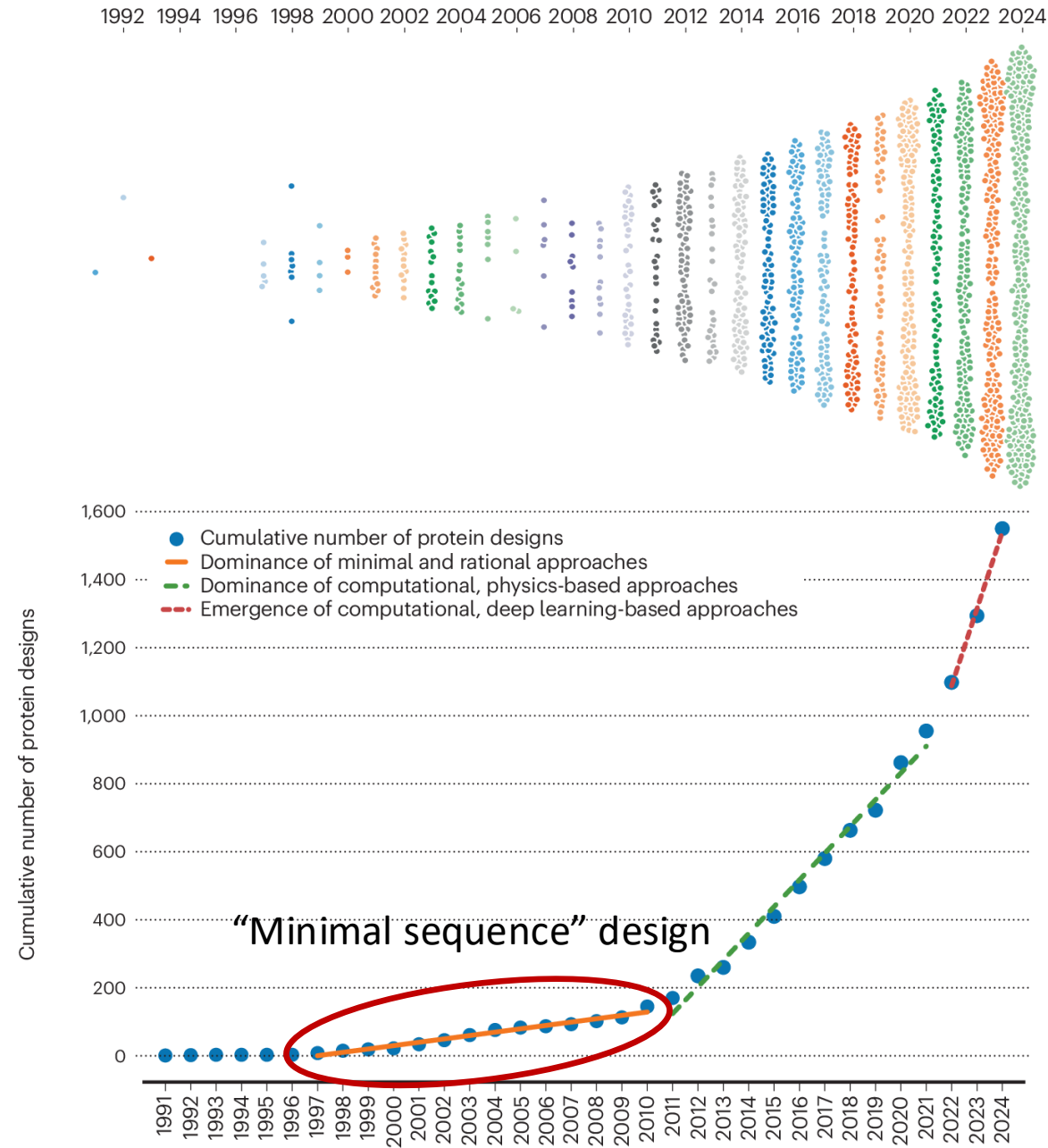


Evolution of *de novo* protein design



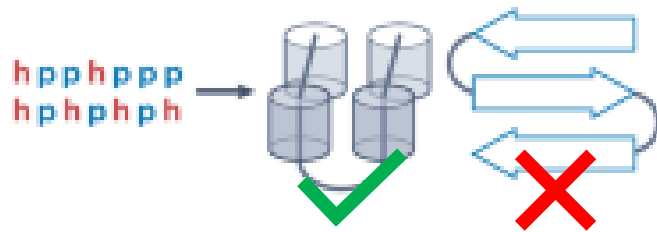
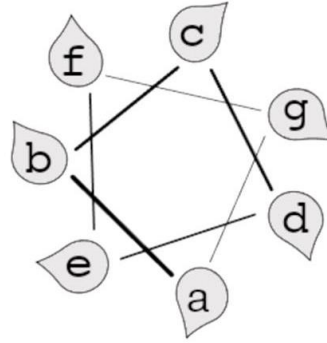
Minimal sequence design

Evolution of *de novo* protein design



Minimal sequence design

TM helix 1 ...MPYIYLAIAIAAEVVGTSALK...
TM helix 2 ...LIPSVGTLVGYGASFYLLSLT...
TM helix 3 ...YALWSGIGIVAIISLVGVWILF...
TM helix 4 ...LDLMKIVGLALIVAGVVILNL...
Pattern ...hhhhpGhGhhhhphhGhhhhp...
Idealized ...LLLLSGLGLLLLSELLGLLLLS...
Position ...abcdefgabcdefgabcdefg...



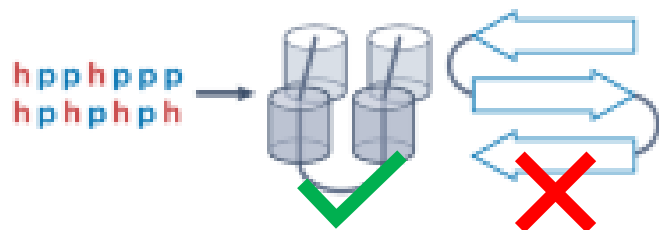
Typical design:

1. Glycine
2. Proline
3. One hydrophobic amino acid
(Leucine or Valine)
4. One polar amino acid (Serine)

Minimal sequence design

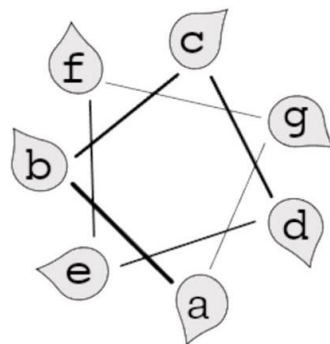
TM helix 1 ...MPYIYLAIAIAAEVVGTSALK...
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 TM helix 4 ...LDLMKIVGLALIVAGVVILNL...

Pattern ...h h h h p G h G h h h h p h h G h h h h p...
 Idealized ...L L L L S G L G L L L L S L L G L L L L S...
 Position ...a b c d e f g a b c d e f g a b c d e f g...

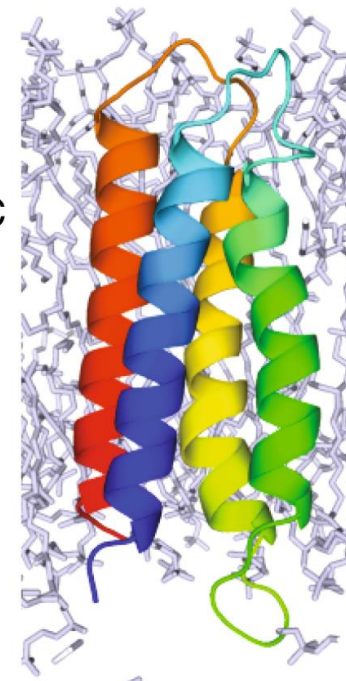
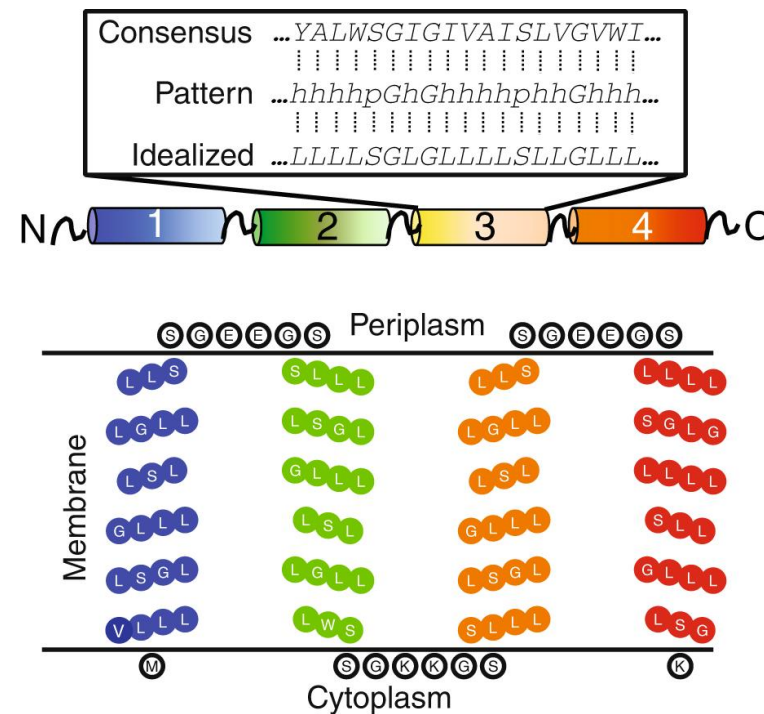


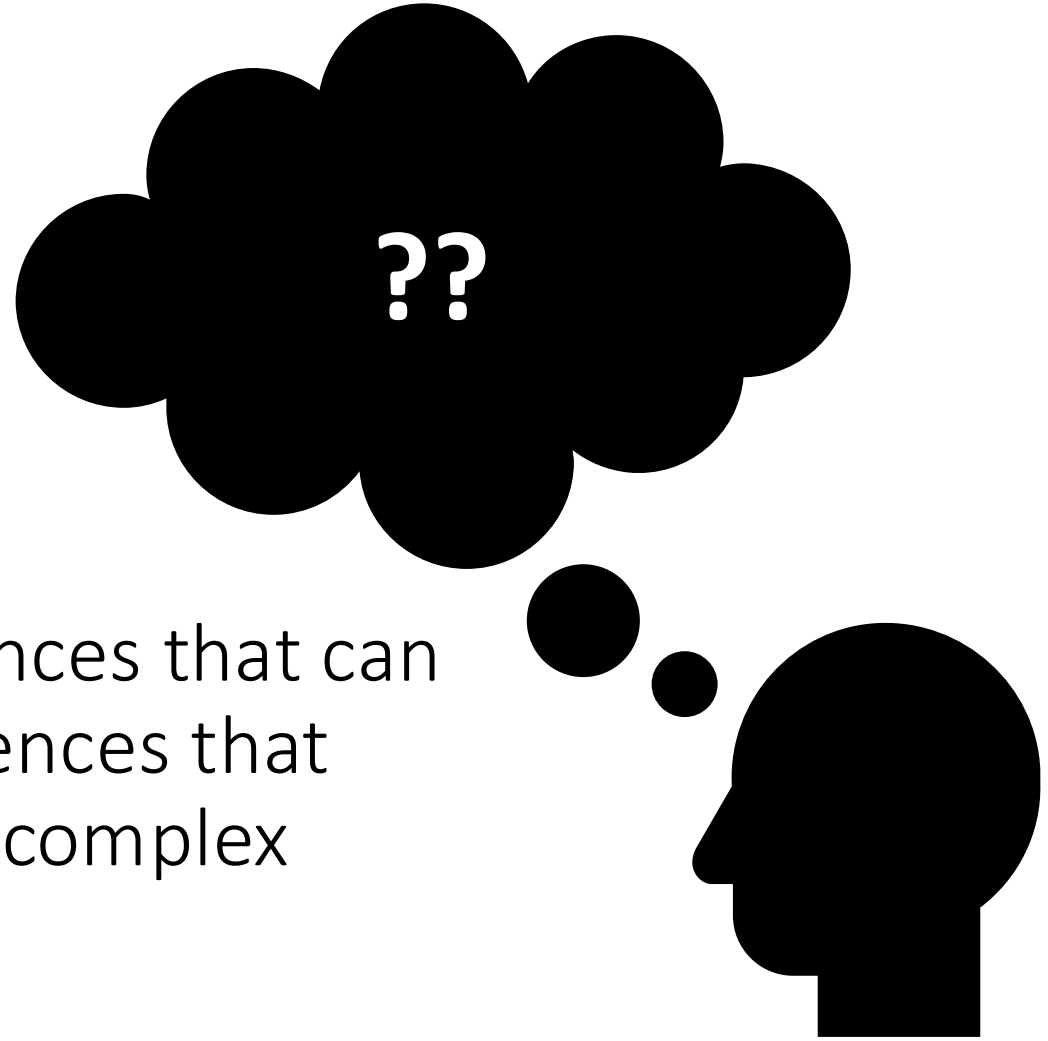
Typical design:

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3. One hydrophobic amino acid (Leucine or Valine)
4. One polar amino acid (Serine)



Example: Minimal membrane protein

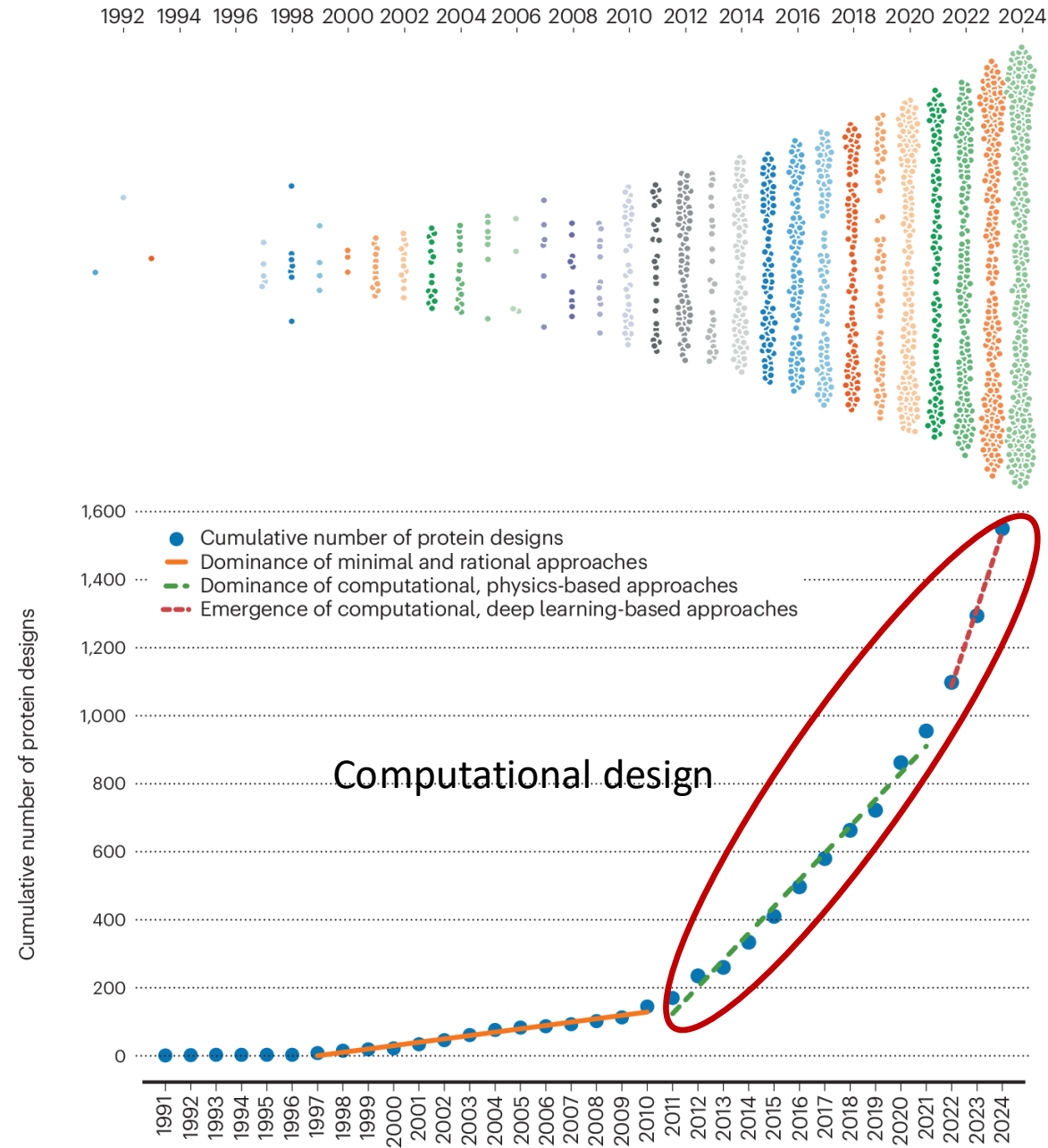




Minimal design generates sequences that can just fold? Can we generate sequences that have functions or fold into more complex structures?

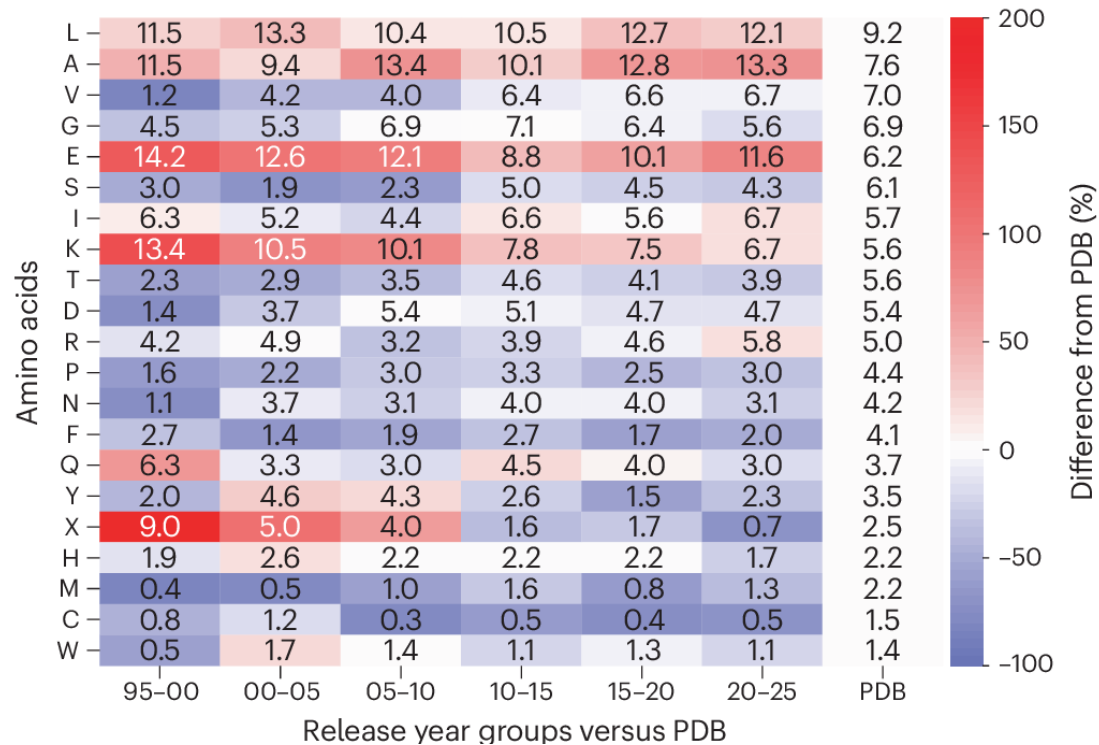
Computational design (structure-based)

Evolution of *de novo* protein design

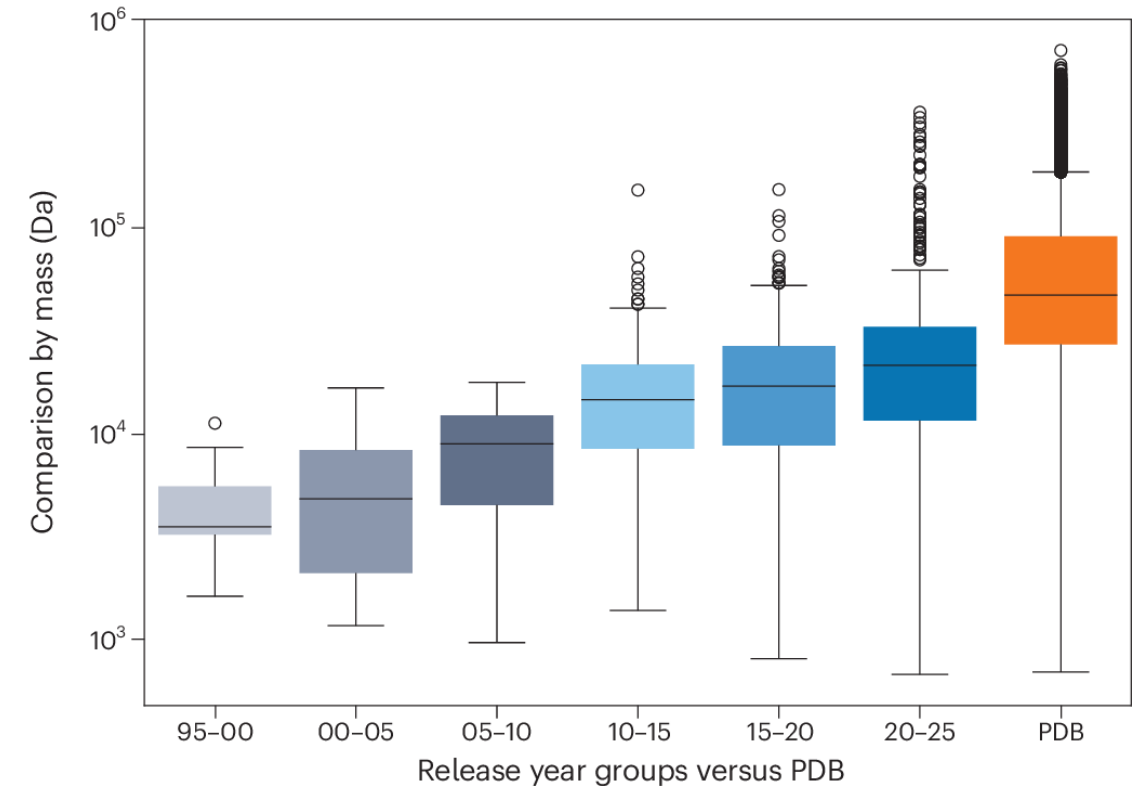


Computational design enabled the generation of more diverse and longer sequences

a Design amino acid percentages by year group versus PDB

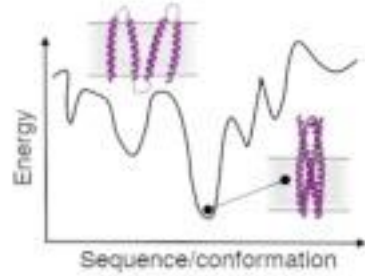


c Comparison by mass (Da)

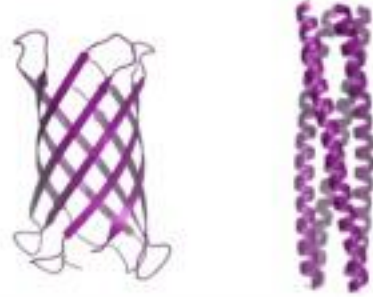


De novo design pipeline

Backbone generation



Scoring

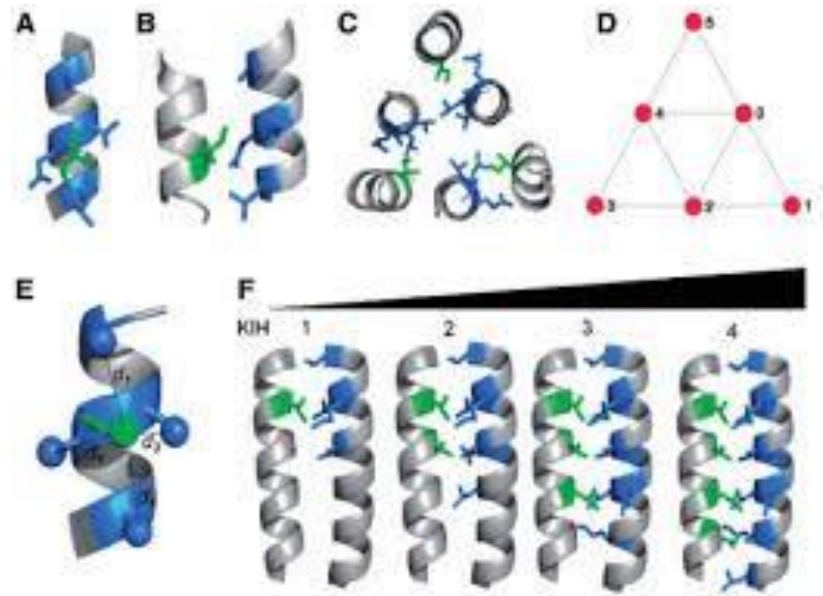
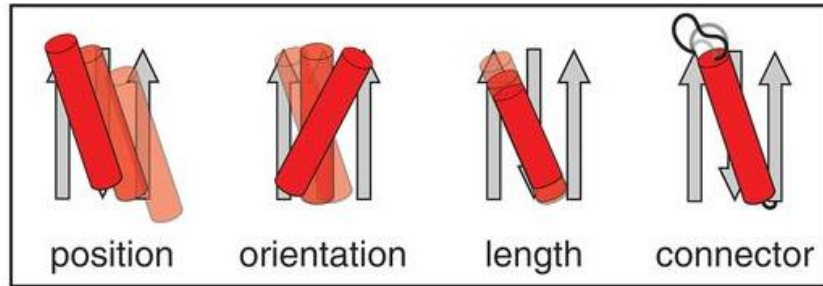


Backbone definition

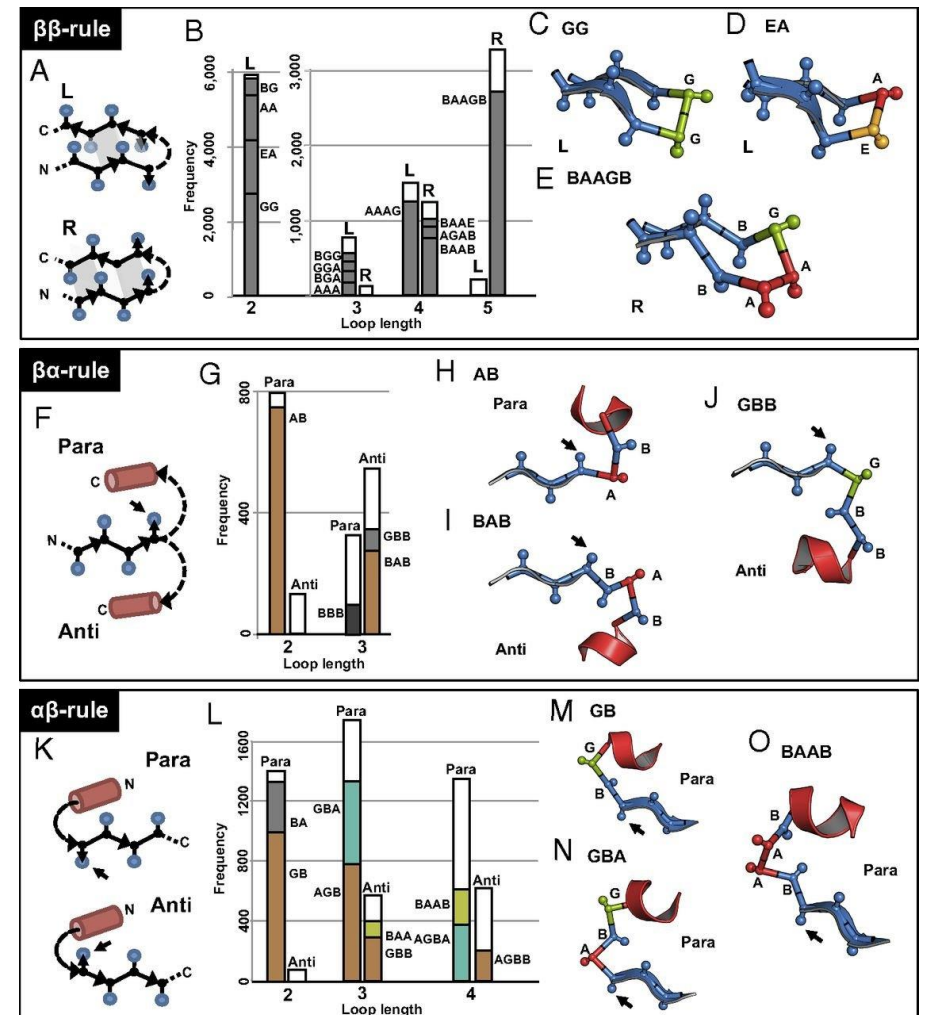
**Generate “designable”
sequence-free protein
backbones**

Designable backbones: new structures incorporating native-like features

1. Side-chain packing: knob-into-hole

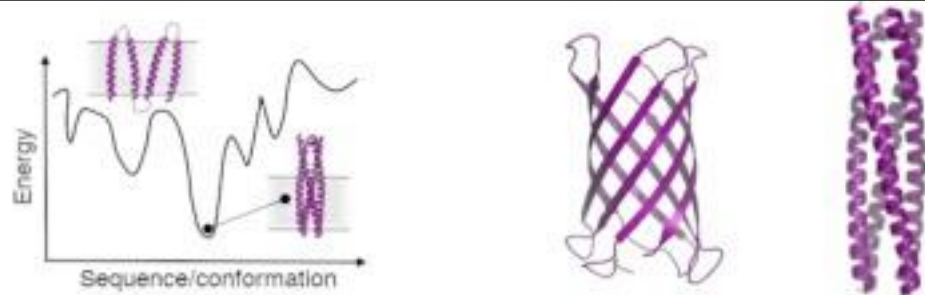


2. Loops with preferred geometries



De novo design pipeline

Backbone generation

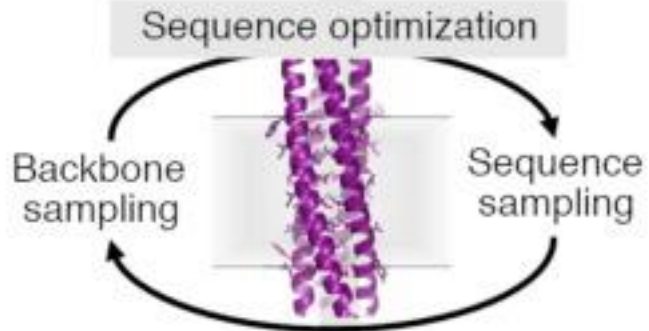


Scoring

Backbone definition

**Generate “designable”
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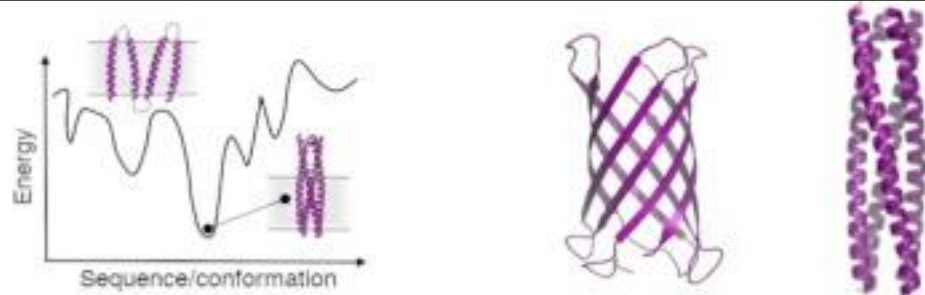
Sequence design



**Design new sequences
compatible with the
backbones**

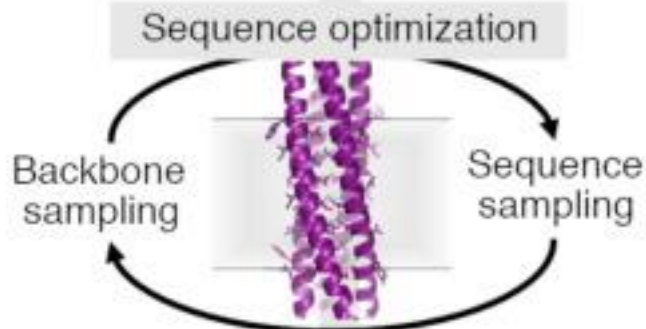
De novo design pipeline

Backbone generation



**Generate “designable”
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Sequence design

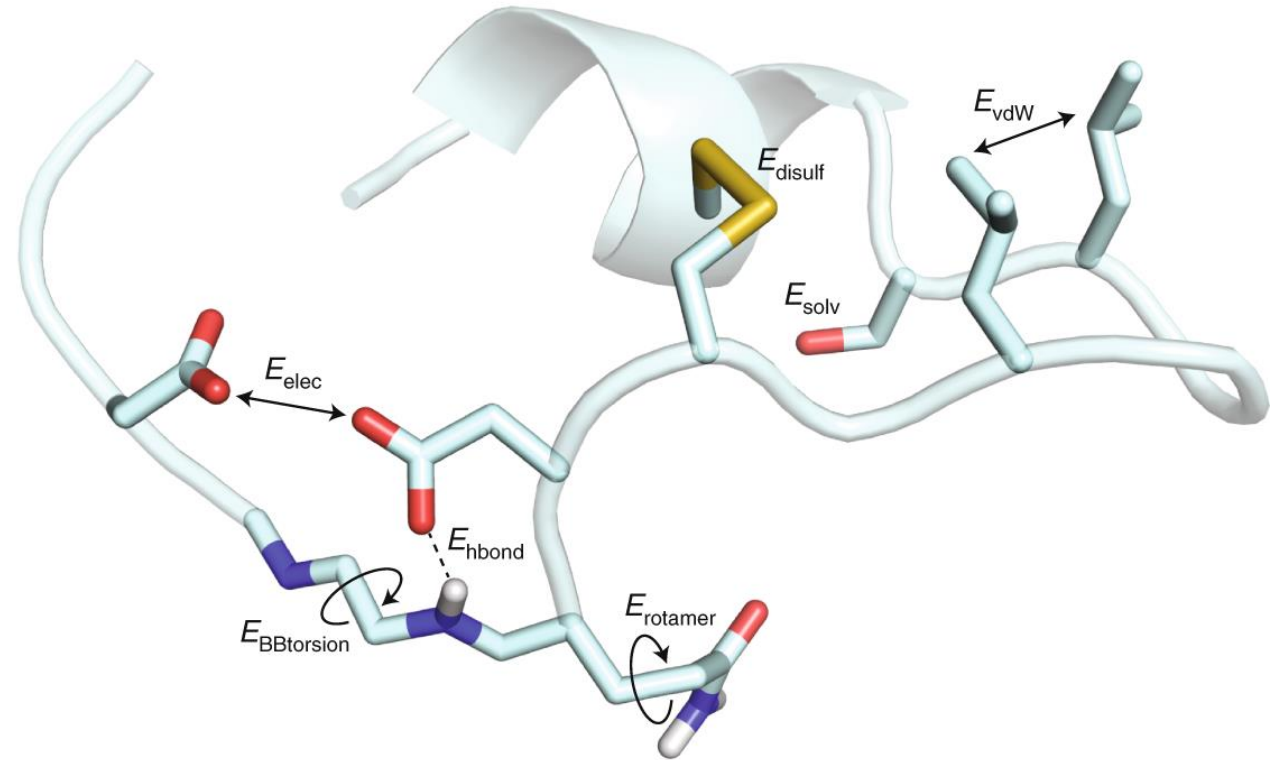
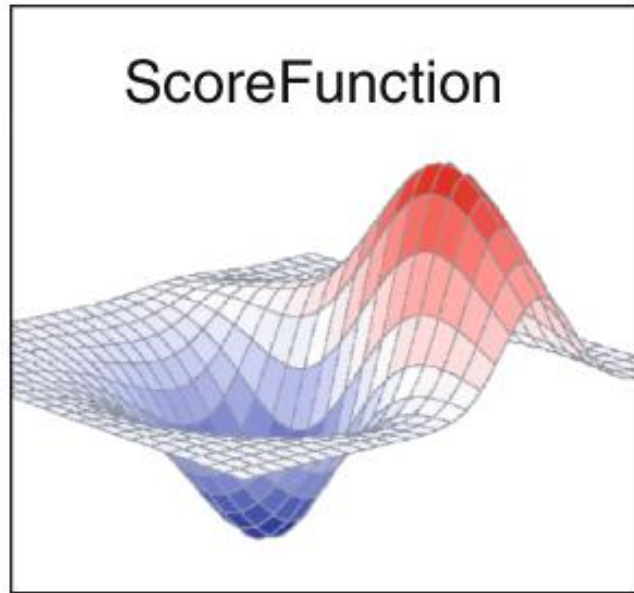


**Design new sequences
compatible with the
backbones**

**Reminder: Two major components
of physics based models?**

Scoring – energy functions

$$\Delta E_{\text{total}} = \sum_i \overset{\text{weight}}{w_i} E_i(\Theta_i, \text{aa}_i)$$



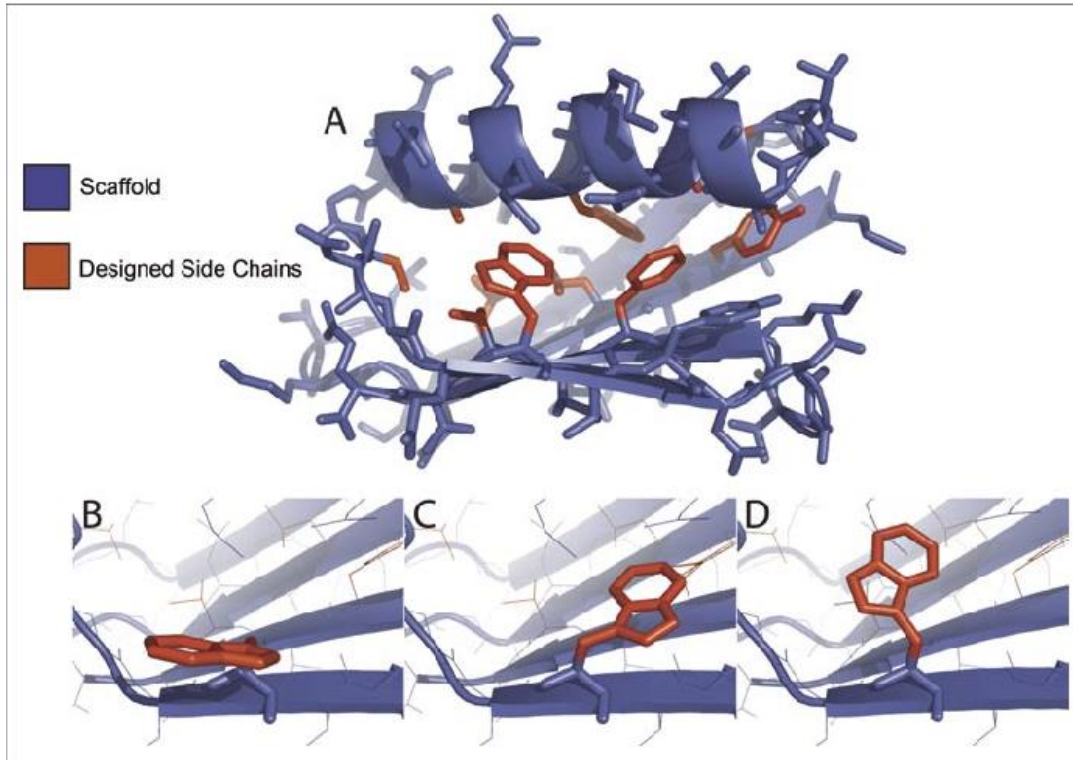
E_{vdW} Lennard–Jones for attractive or repulsive interaction
 E_{hbond} Hydrogen bonding allows buried polar atoms
 E_{elec} Electrostatic interaction between charges
 E_{disulf} Disulfide bonds between cysteines

E_{solv} Implicit solvation model penalizes buried polar atoms
 $E_{\text{BBtorsion}}$ Backbone torsion preferences from main-chain potential
 E_{rotamer} Side-chain torsion angles from rotamer library
 E_{ref} Unfolded state reference energy for design

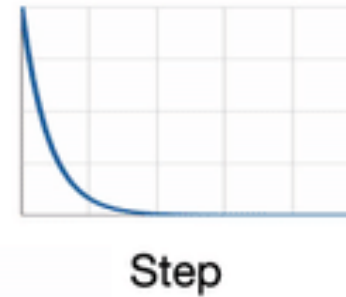
Sampling methods

Sampling residue type and rotamers

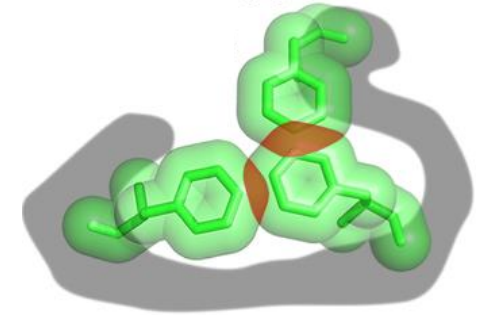
- **Simulated annealer – Rosetta design**



Temperature



Low repulsive energy –
many moves are allowed



High repulsive energy –
less moves are allowed



Temperature

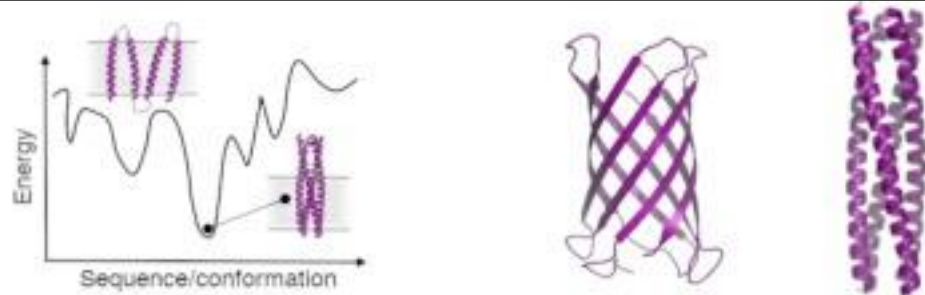
Analogy: avalanche rescue search

Your search strategy will depend on your objectives and of the landscape



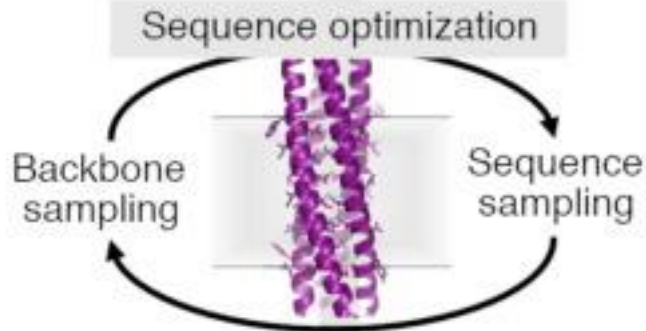
De novo design pipeline

Backbone generation



**Generate “designable”
sequence-free protein
backbones**

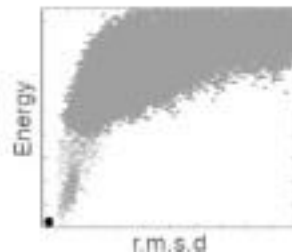
Sequence design



**Design new sequences
compatible with the
backbones**

Validation

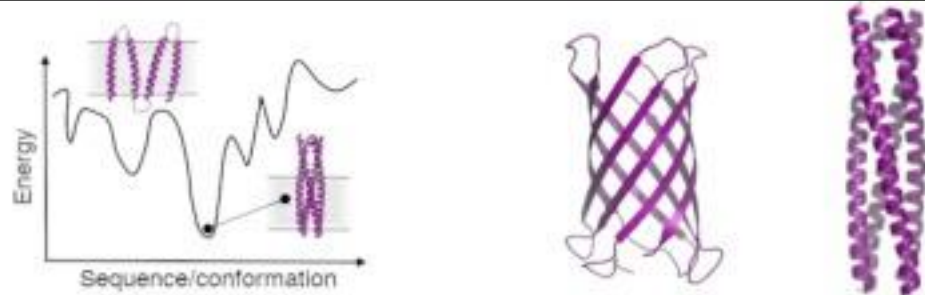
Energy landscape of the
designed sequence
converges on target structure



**Verify sequence/structure
compatibility with *ab
initio* structure prediction**

De novo design pipeline

Backbone generation



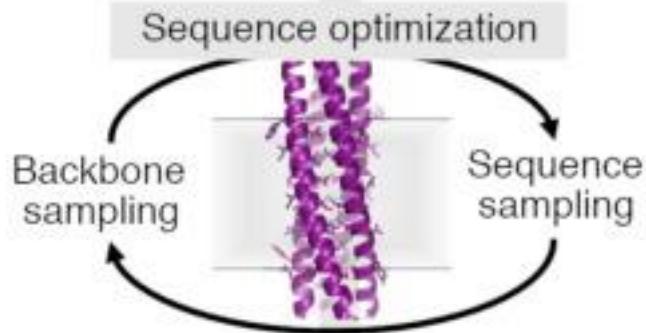
Scoring

Backbone definition

Generate “designable”
sequence-free protein
backbones



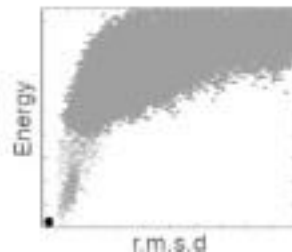
Sequence design



Design new sequences
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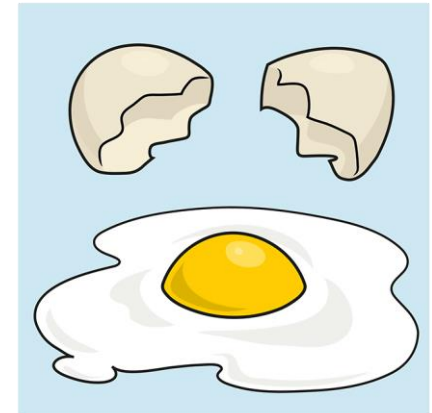
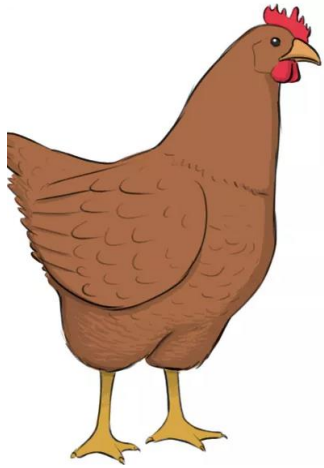
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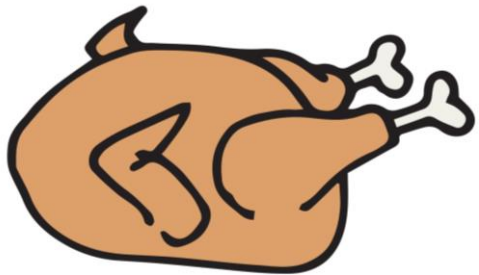


Verify sequence/structure
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De novo protein design: chicken/egg problem?

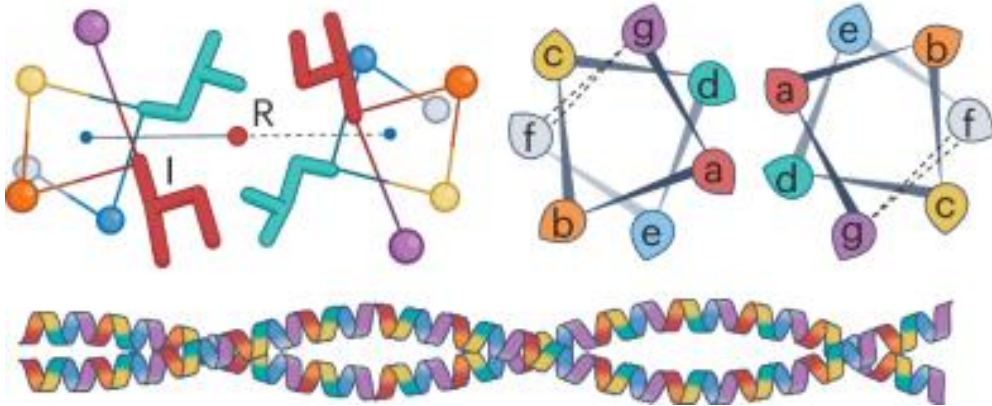


De novo protein design: chicken/egg problem?



Parametric design of alpha-helical bundles

Knob-into-hole packing of side-chains



Crick coiled coil equations

For a helix in a coiled coil with supercoil axis along z , the Cartesian coordinates of the C α of residue t are given by:

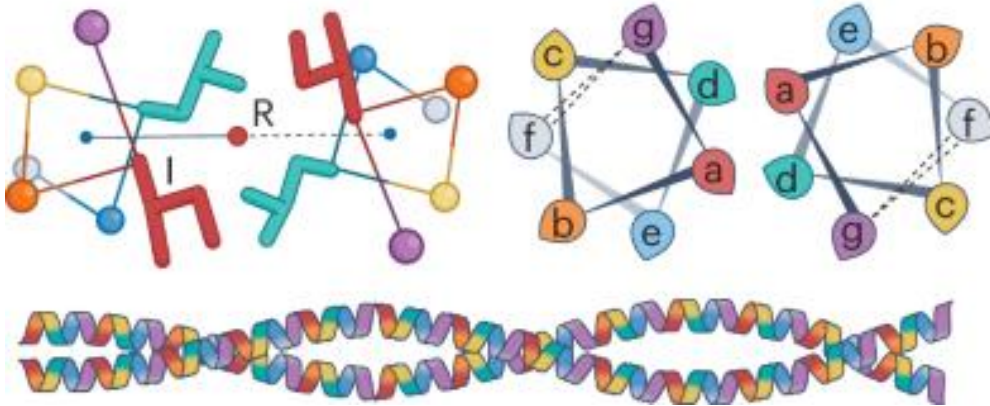
$$x = R_0 \cos(\omega_0 t + \phi_0') + R_1 \cos(\omega_0 t + \phi_0') \cos(\omega_1 t + \phi_1) - R_1 \cos(\alpha) \sin(\omega_0 t + \phi_0') \sin(\omega_1 t + \phi_1)$$

$$y = R_0 \sin(\omega_0 t + \phi_0') + R_1 \sin(\omega_0 t + \phi_0') \cos(\omega_1 t + \phi_1) + R_1 \cos(\alpha) \cos(\omega_0 t + \phi_0') \sin(\omega_1 t + \phi_1)$$

$$z = (\omega_0 R_0 / \tan(\alpha)) t - R_1 \sin(\alpha) \sin(\omega_1 t + \phi_1) + \Delta z$$

Parametric design of alpha-helical bundles

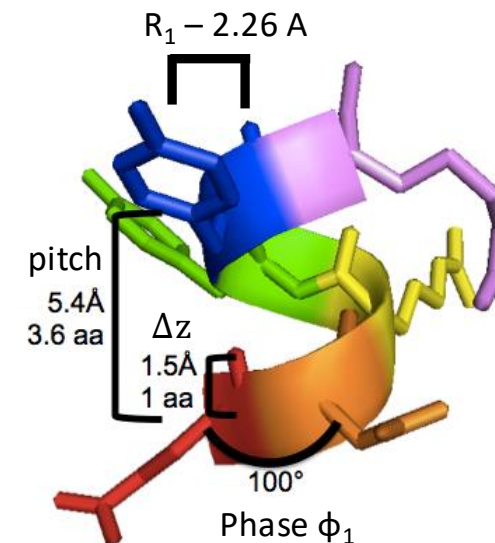
Knob-into-hole packing of side-chains



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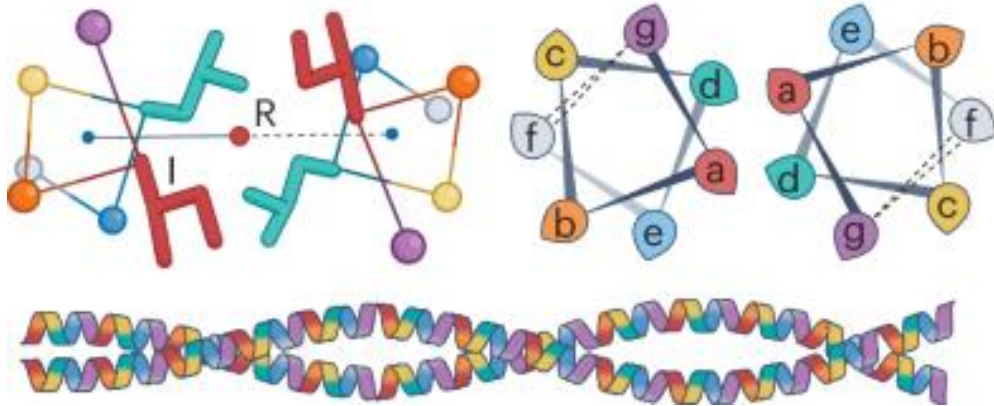


ω_1 – helical twist

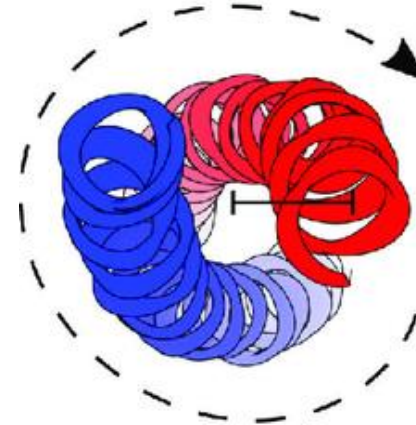
<https://doi.org/10.1016/j.jbc.2023.104579>

Parametric design of alpha-helical bundles

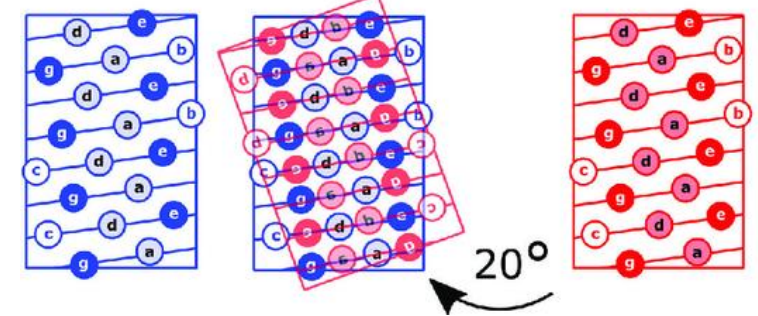
Knob-into-hole packing of side-chains



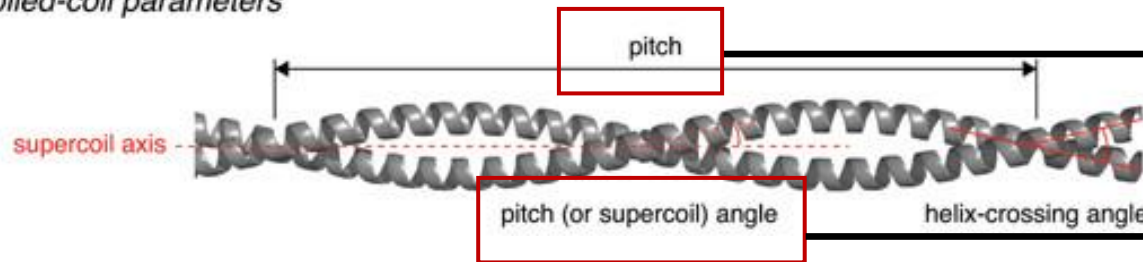
Radius



Knobs into holes packing



Coiled-coil parameters



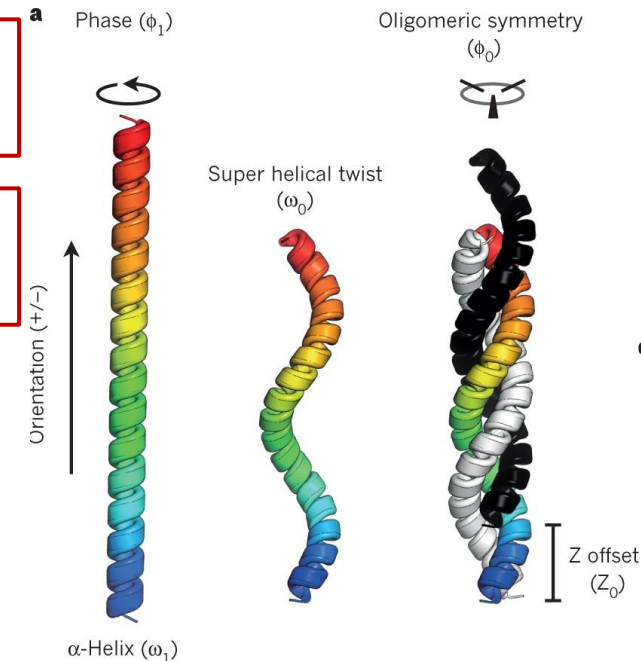
Number of residues to bring the helices back to sync.

Angle between helix and supercoil axis

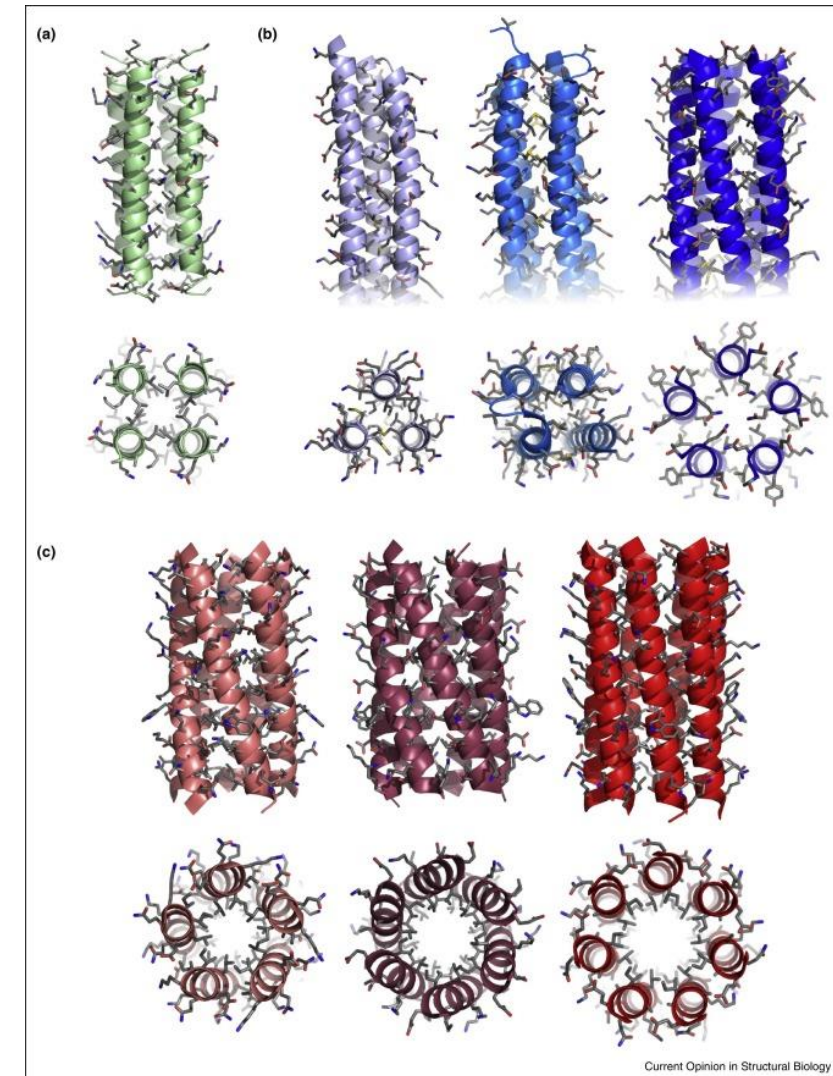
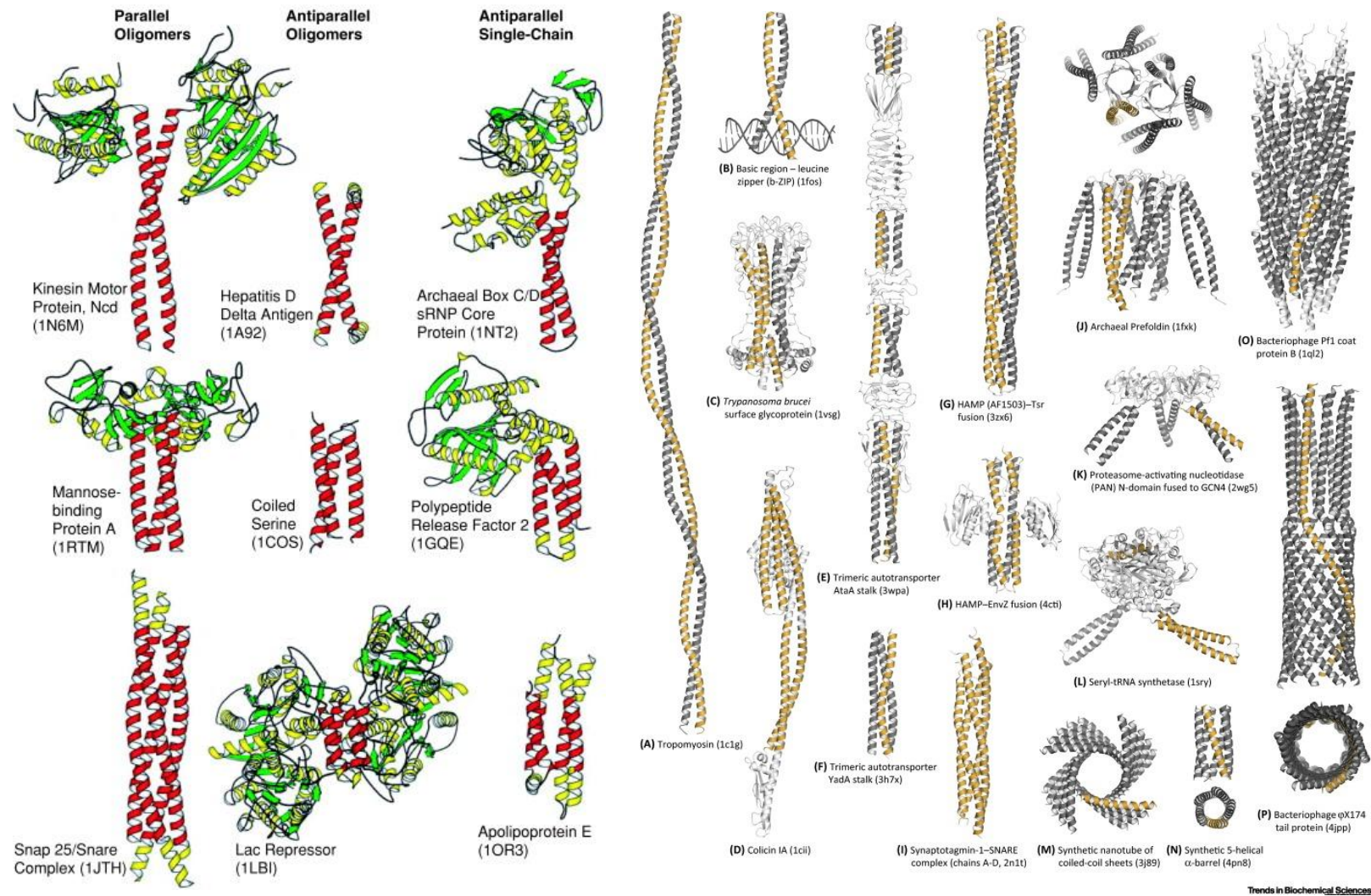
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 y &= R_0 \sin(\omega_0 t + \phi_0') + R_1 \sin(\omega_0 t + \phi_0') \cos(\omega_1 t + \phi_1) + R_1 \cos(\alpha) \cos(\omega_0 t + \phi_0') \sin(\omega_1 t + \phi_1) \\
 z &= (\omega_0 R_0 / \tan(\alpha)) t - R_1 \sin(\alpha) \sin(\omega_1 t + \phi_1) + \Delta z
 \end{aligned}$$



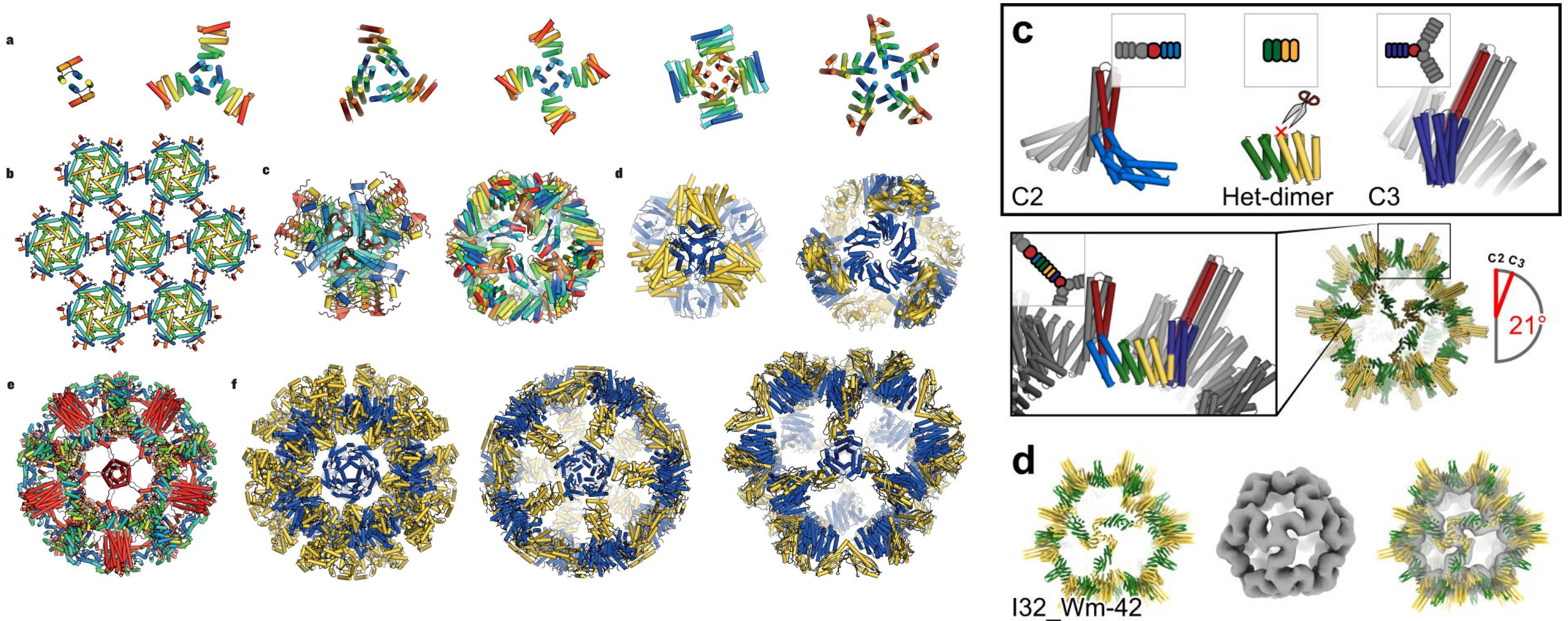
Natural vs designed alpha-helical bundles



De novo designed coiled coils and bundles

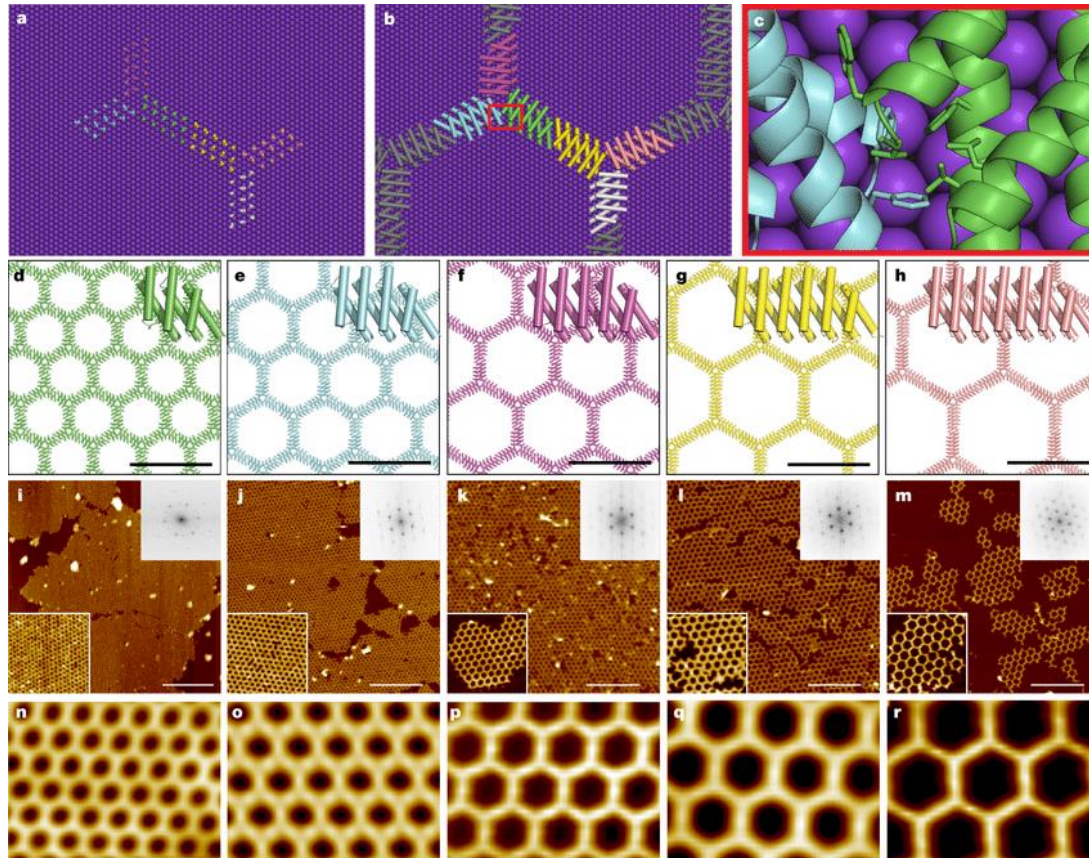
Natural helical bundles and coiled-coil domains hidden in proteins

Example 1: alpha-helical bundles as building units

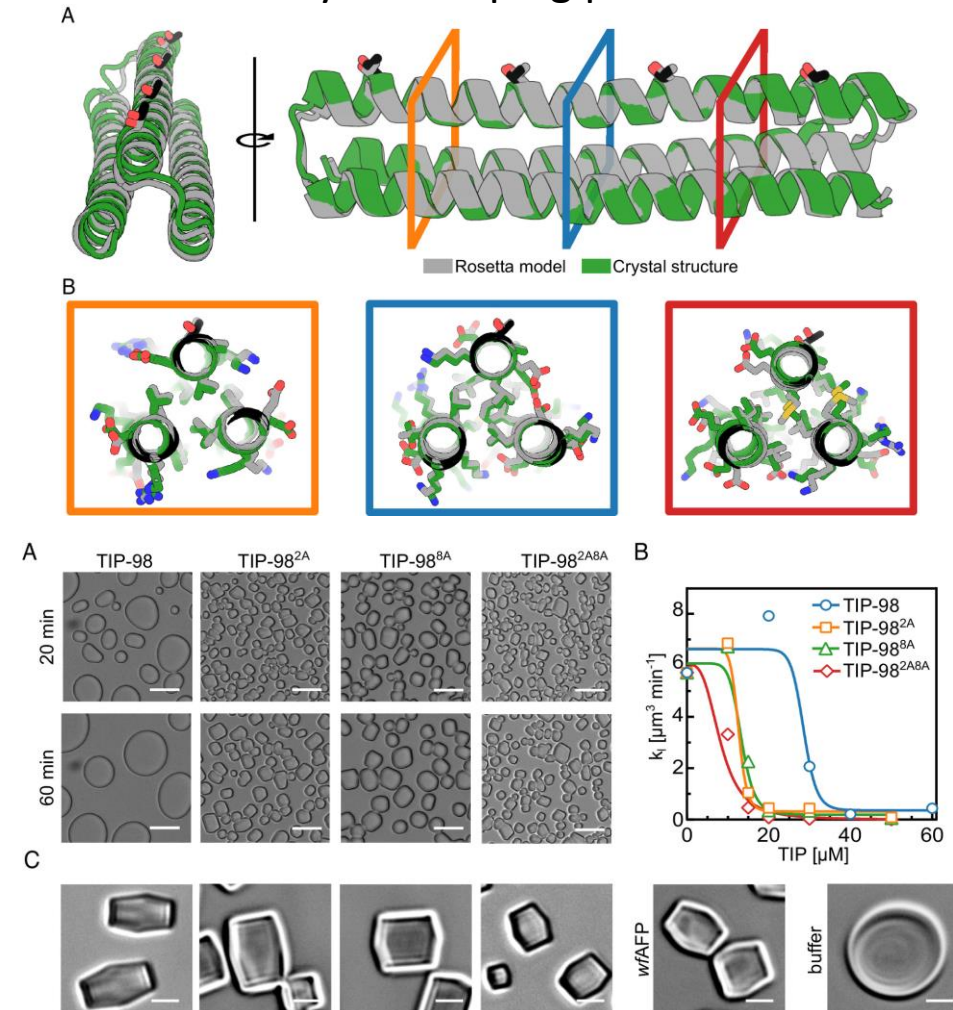


Example 2: lattice-matching proteins

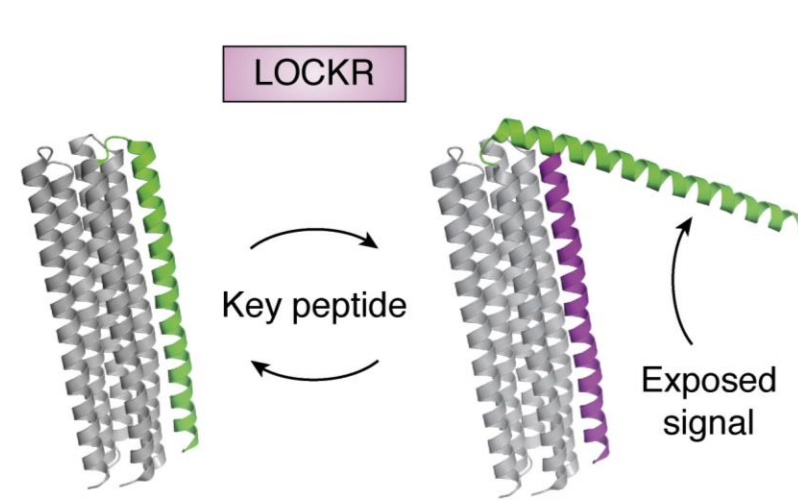
Mica-binding proteins and self-assembling material



Ice crystal-shaping proteins

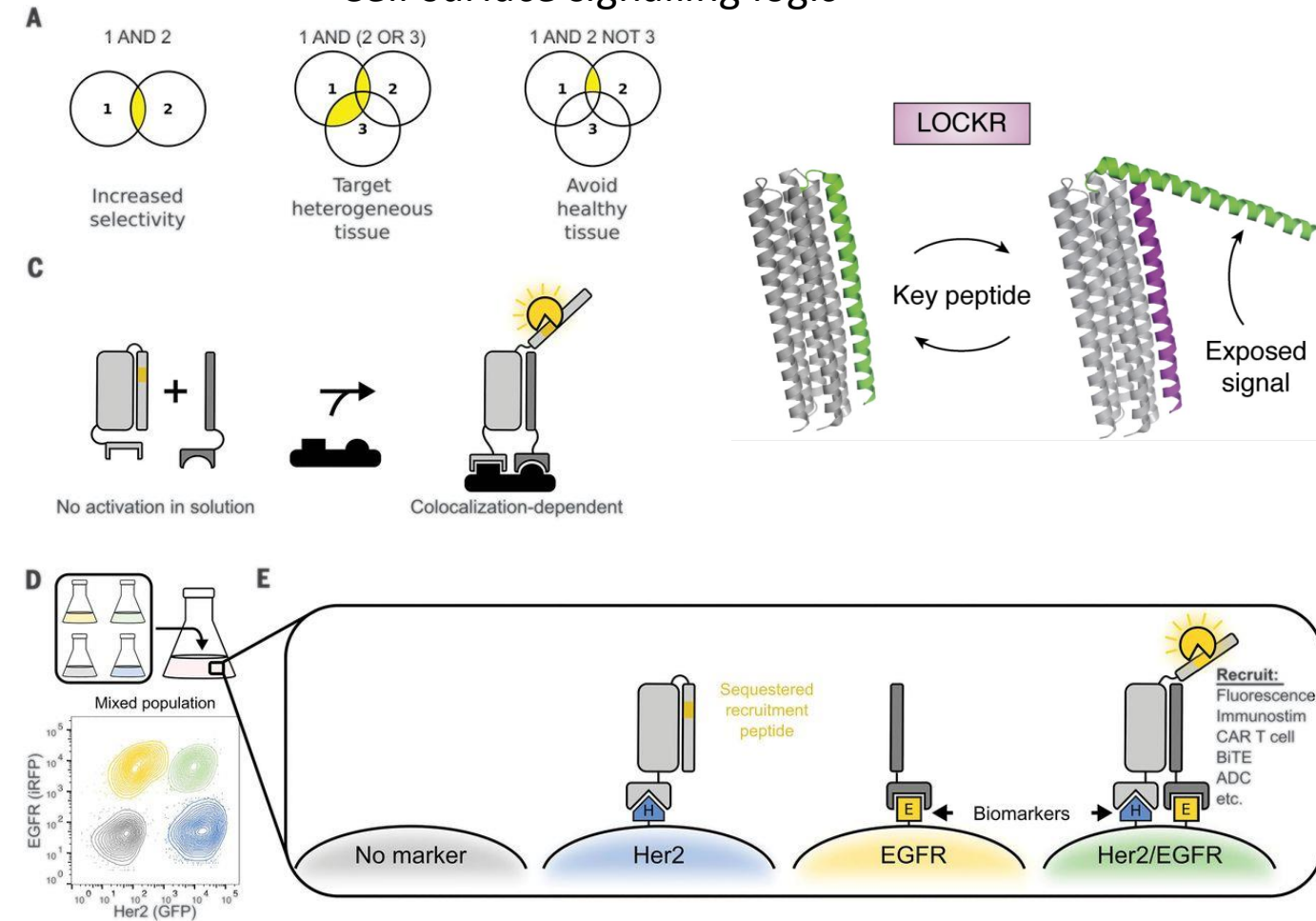


Example 3: protein logic and switches



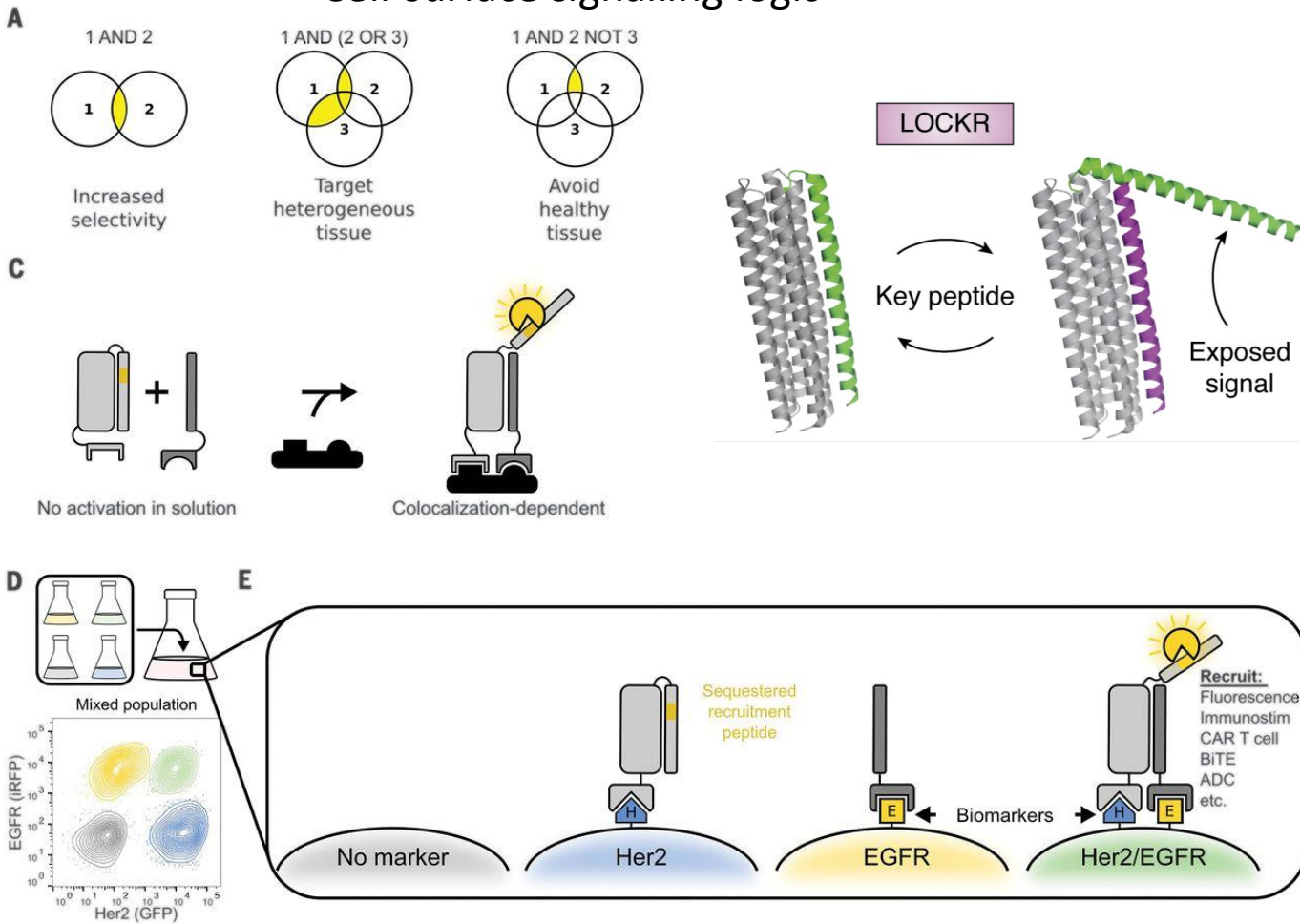
Example 3: protein logic and switches

Cell-surface signalling logic

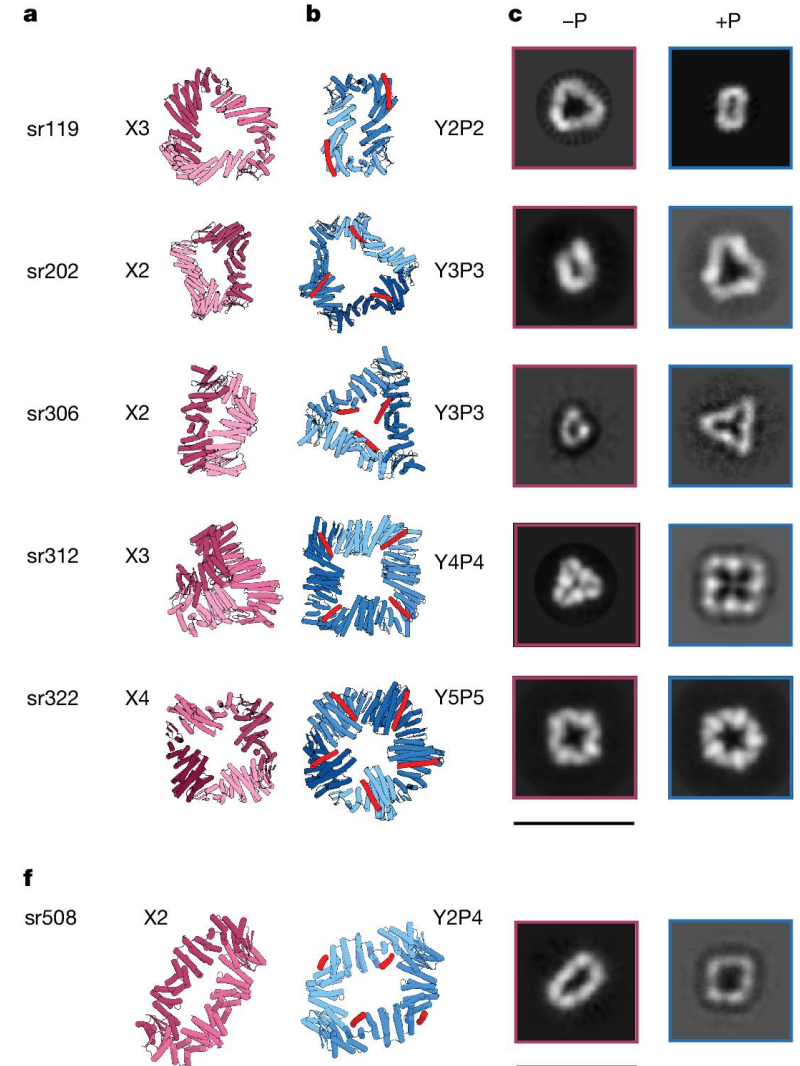


Example 3: protein logic and switches

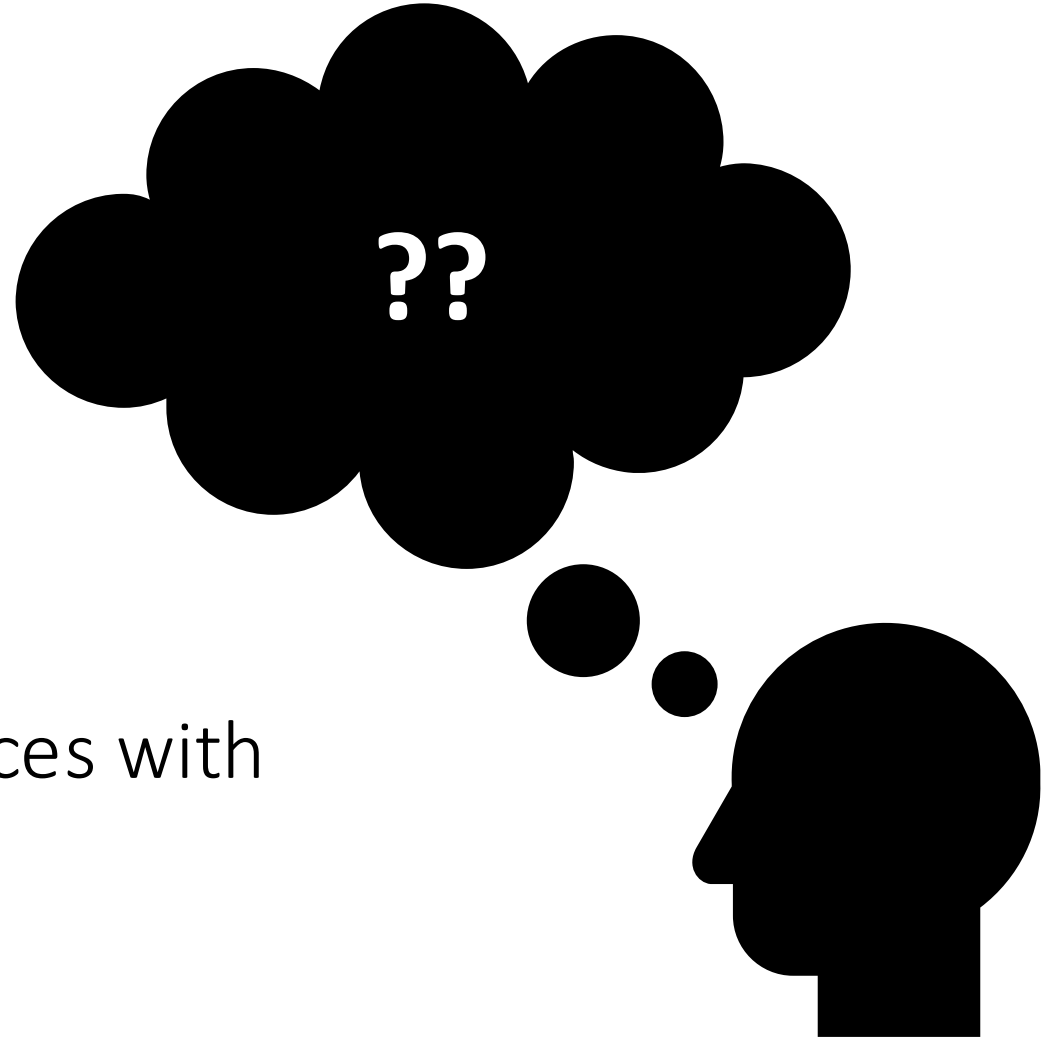
Cell-surface signalling logic

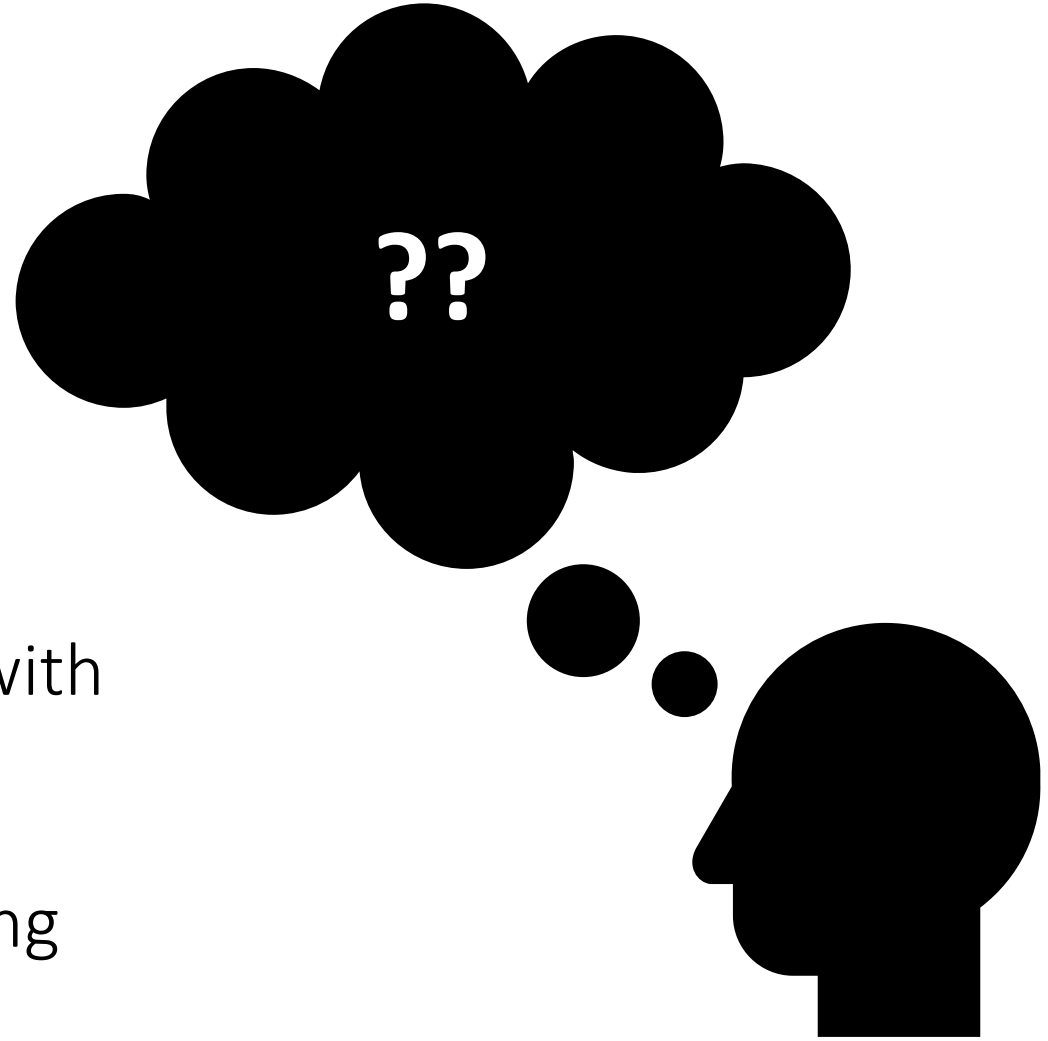


Switchable assemblies^{nsEM}



How to generate diverse sequences with
parametric design?



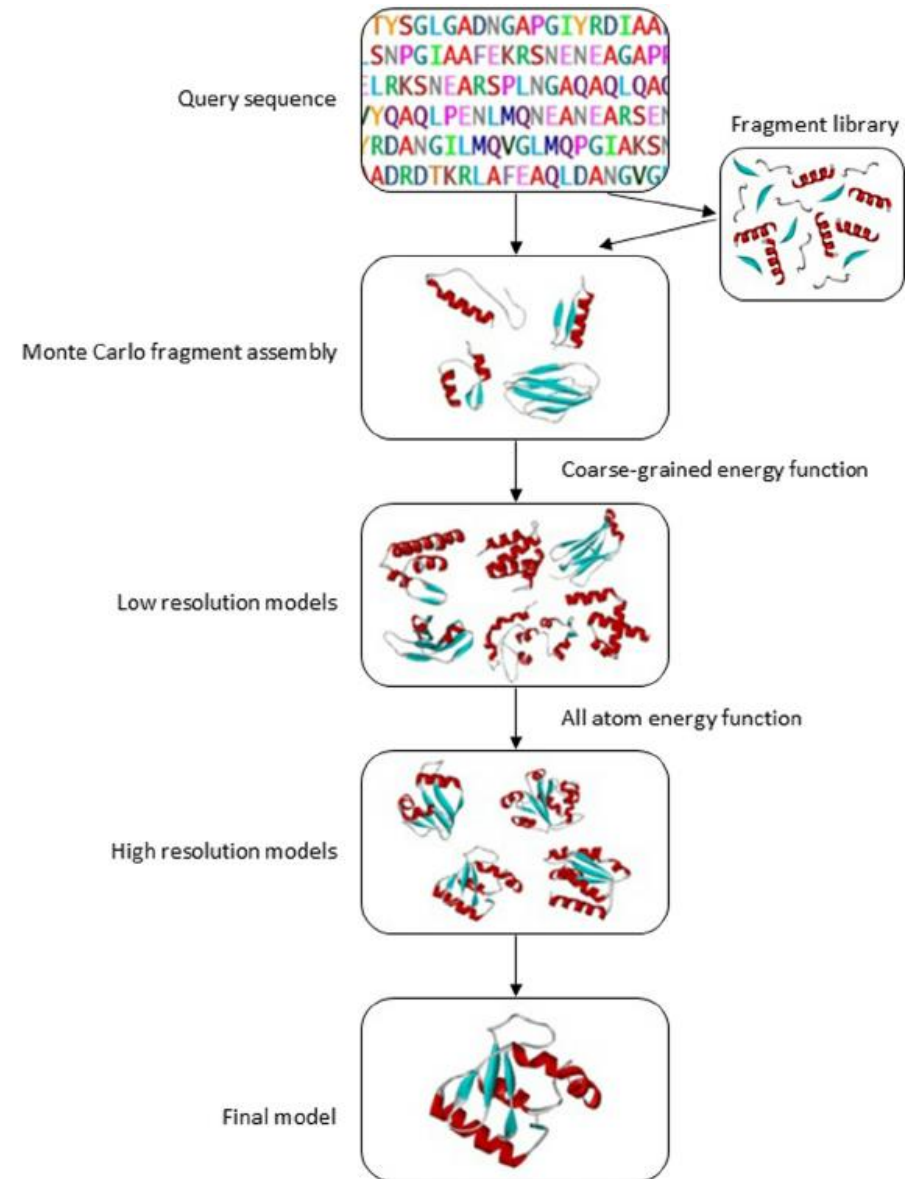
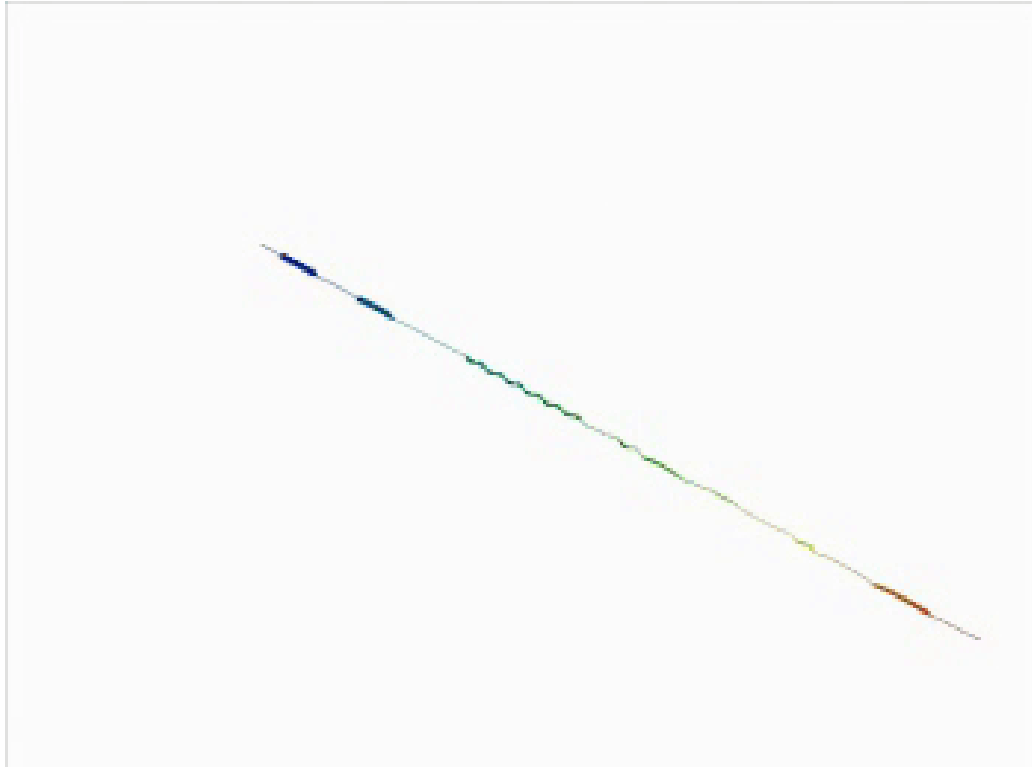


How to generate diverse sequences with parametric design?

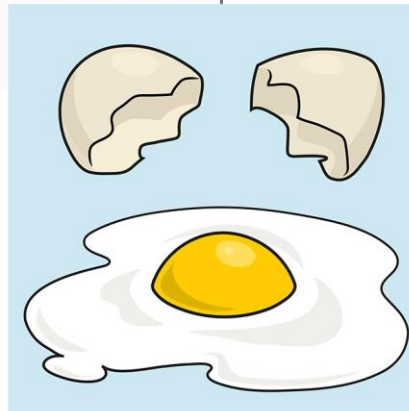
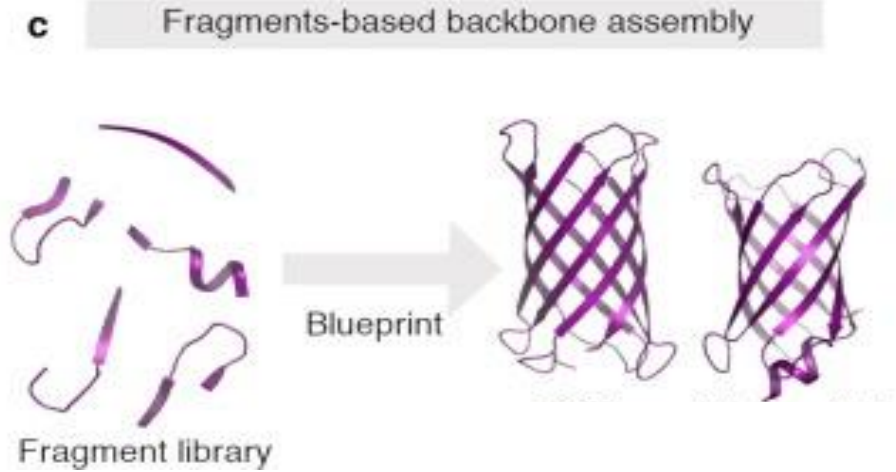
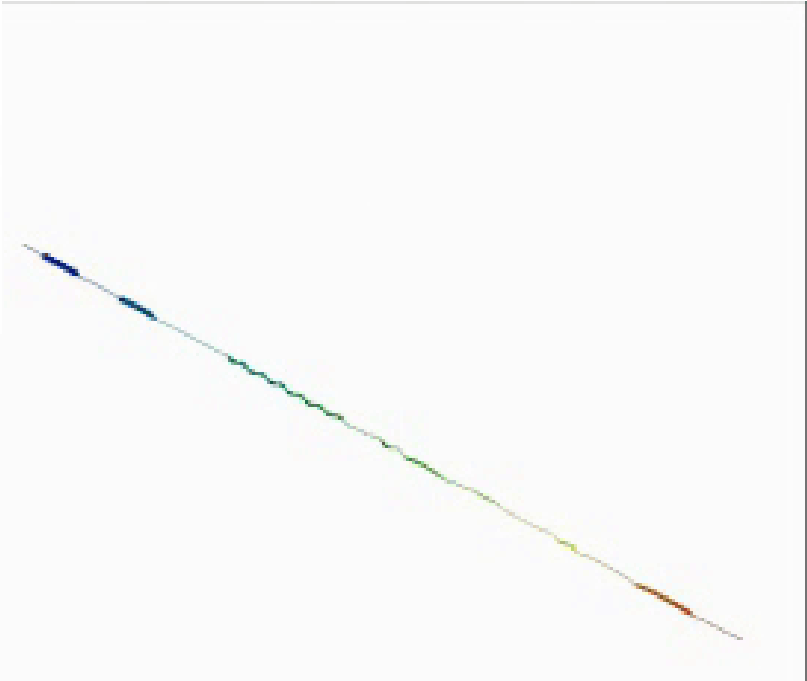
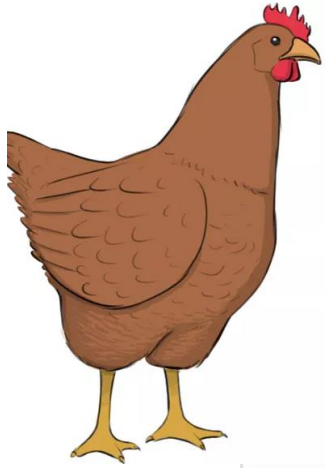
→ Superhelical parameters fine-tuning

BUT ... parameteric designs mostly unsuccessful for beta-sheet proteins

“Stealing” concepts from structure prediction



“Stealing” concepts from structure prediction



How can we pick fragments if the sequence is unknown?

Picking fragments based on local structure properties == Blueprint

residues

→ 0 V L R
→ 0 V L R
→ 0 V E B R
→ 0 V E B R
→ 0 V E B R
→ 0 V E B R
→ 0 V E B R
→ 0 V E B R
0 V E B R
0 V E B R
0 V L A R
0 V L A R
0 G L G R
0 V E B R
0 V E B R
0 V E A R
0 V E B R
0 V E B R
...

Place-holder amino acid: Valine

Blueprint + constraints

Picking fragments based on local structure properties == Blueprint

residues

→ 0 V L R
→ 0 V L R
→ 0 V E B R
→ 0 V E B R
→ 0 V E B R
→ 0 V E B R
→ 0 V E B R
→ 0 V E B R
0 V E B R
0 V E B R
0 V L A R
0 V L A R
0 G L G R
0 V E B R
0 V E B R
0 V E A R
0 V E B R
0 V E B R
....

Place-holder amino acid: Valine

Secondary structure:

E = Beta-strand

H = Alpha-helix

L = Loop

Blueprint + constraints

Picking fragments based on local structure properties == Blueprint

residues

→ 0 V L R
→ 0 V L R
→ 0 V E B R
→ 0 V E B R
→ 0 V E B R
→ 0 V E B R
→ 0 V E B R
→ 0 V E B R
0 V E B R
0 V L A R
0 V L A R
0 G L G R
0 V E B R
0 V E B R
0 V E A R
0 V E B R
0 V E B R
....

Place-holder amino acid: Valine

Secondary structure:

E = Beta-strand

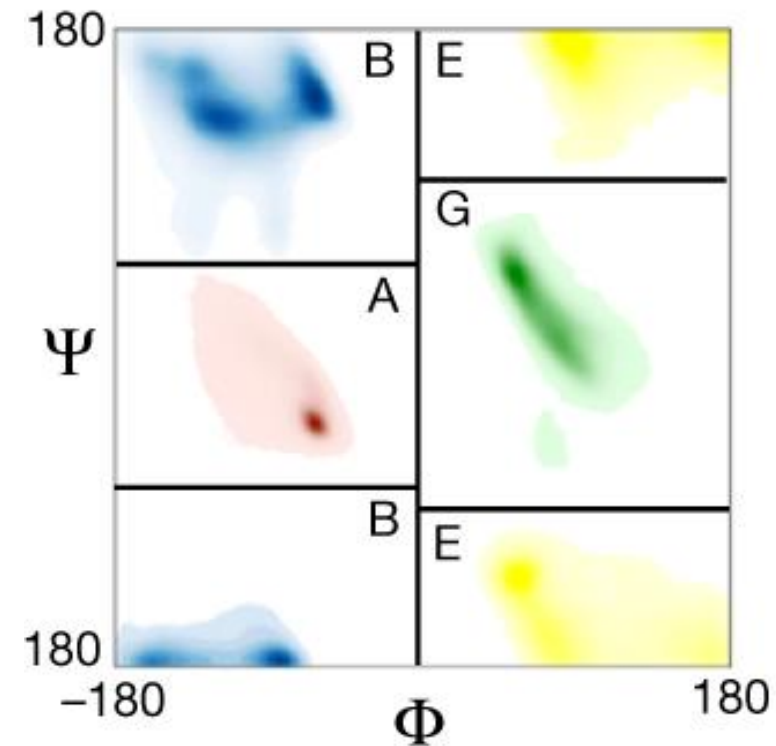
H = Alpha-helix

L = Loop

ABEGO type

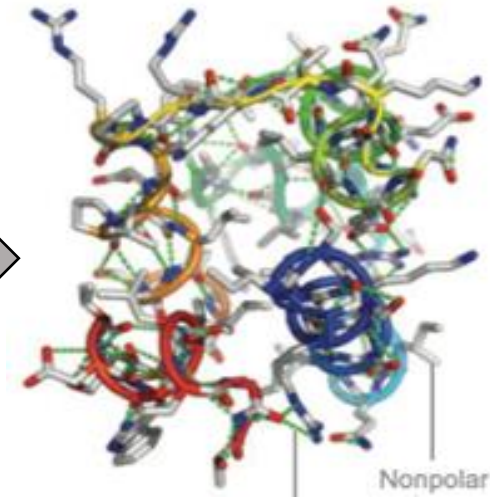
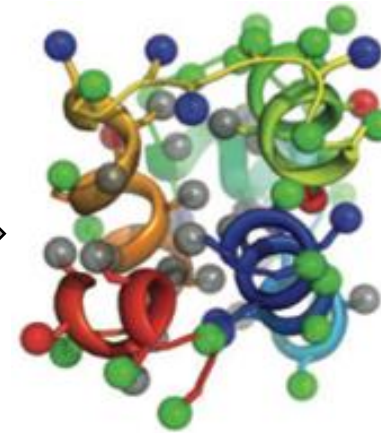
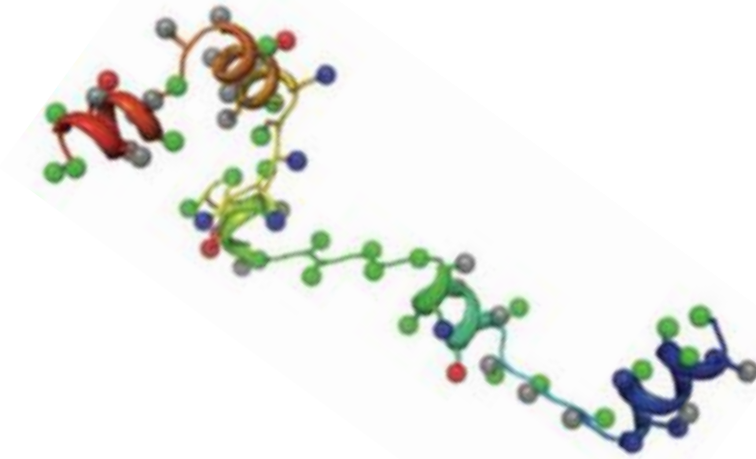
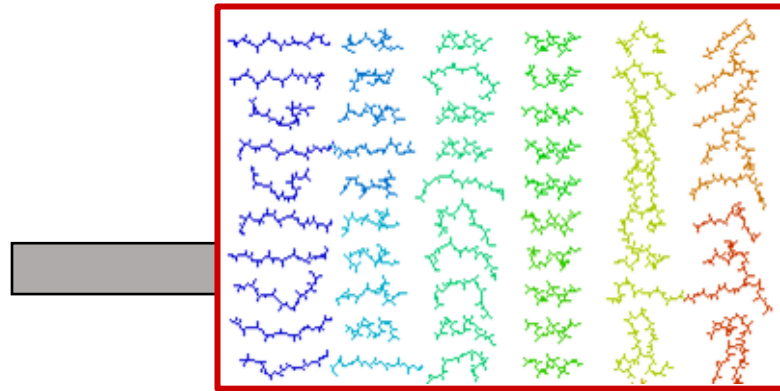
"O" = cis-peptide bond

Blueprint + constraints



Picking fragments based on local structure properties == Blueprint

0VLR
0VLR
0VEBR
0VEBR
0VEBR
0VEBR
0VEBR
0VEBR
0VEBR
0VEBR
0VLR
0VLR
0GLGR
0VEBR
0VEBR
0VEAR
0VEBR
0VEBR
....

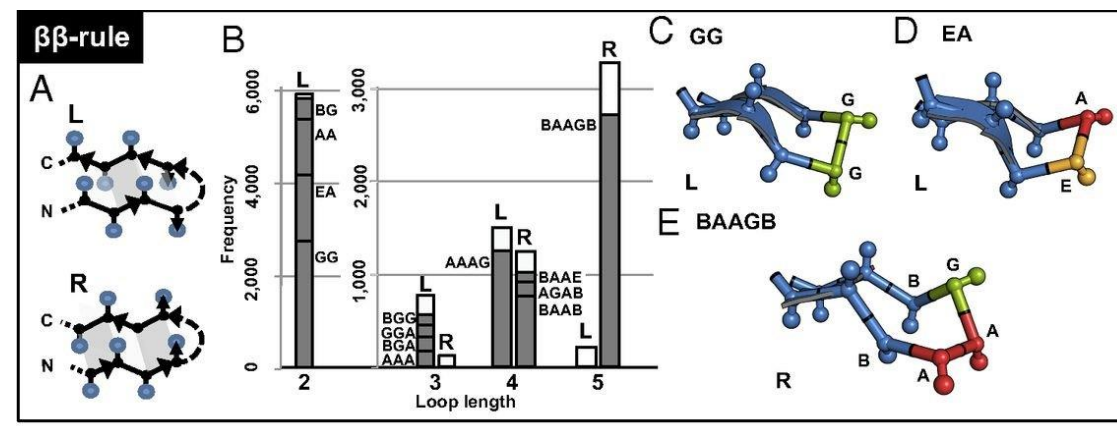
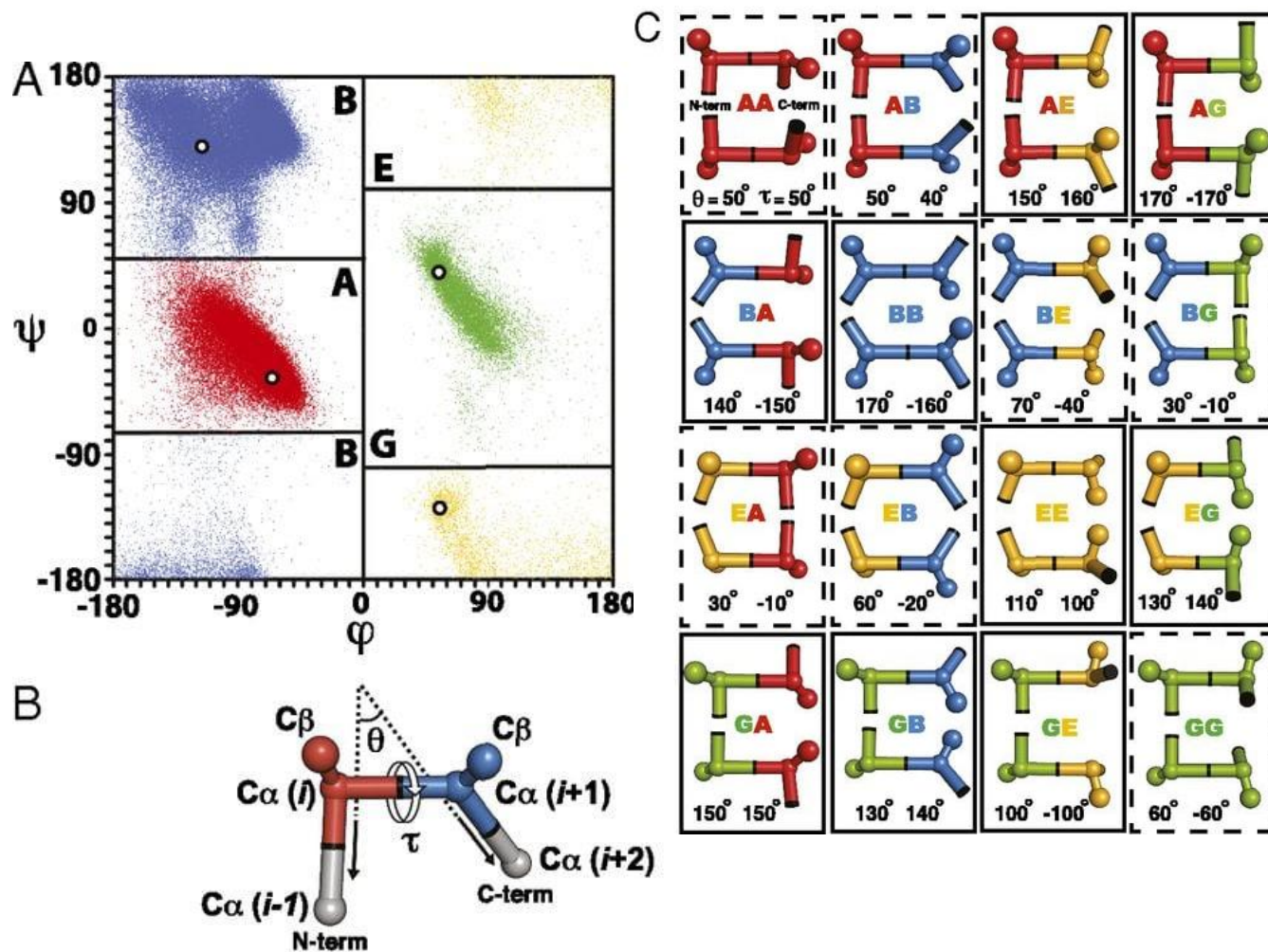


Blueprint + constraints

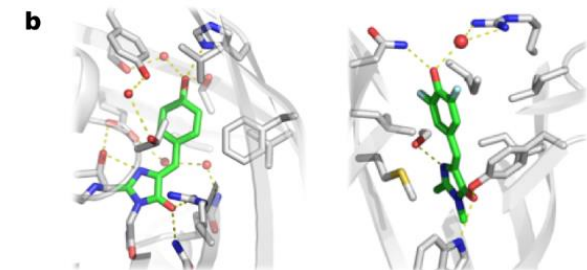
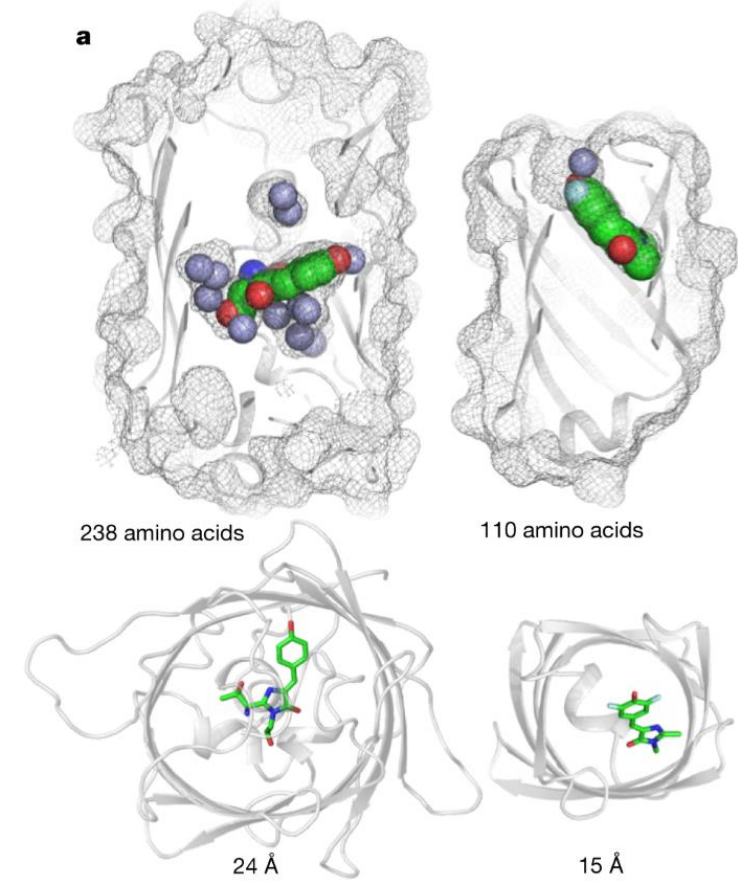
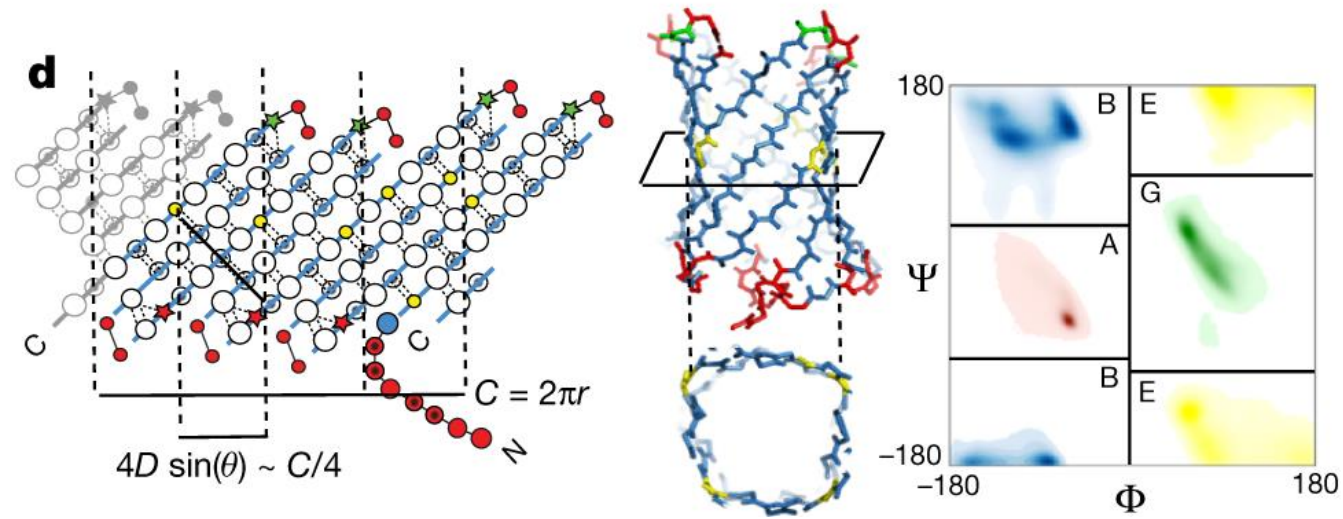
Ab initio folding

Sequence design

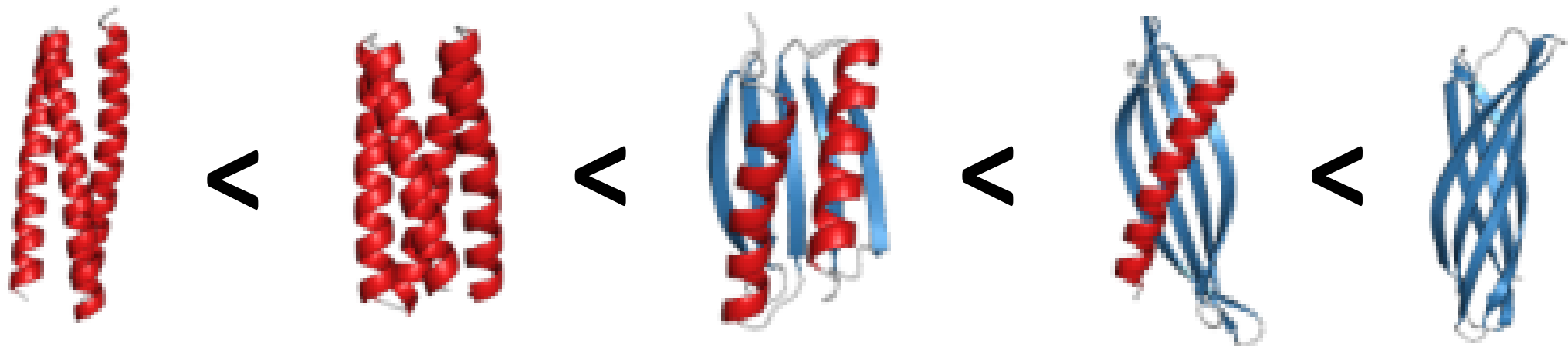
Generating designable backbones



Generating complex protein folds



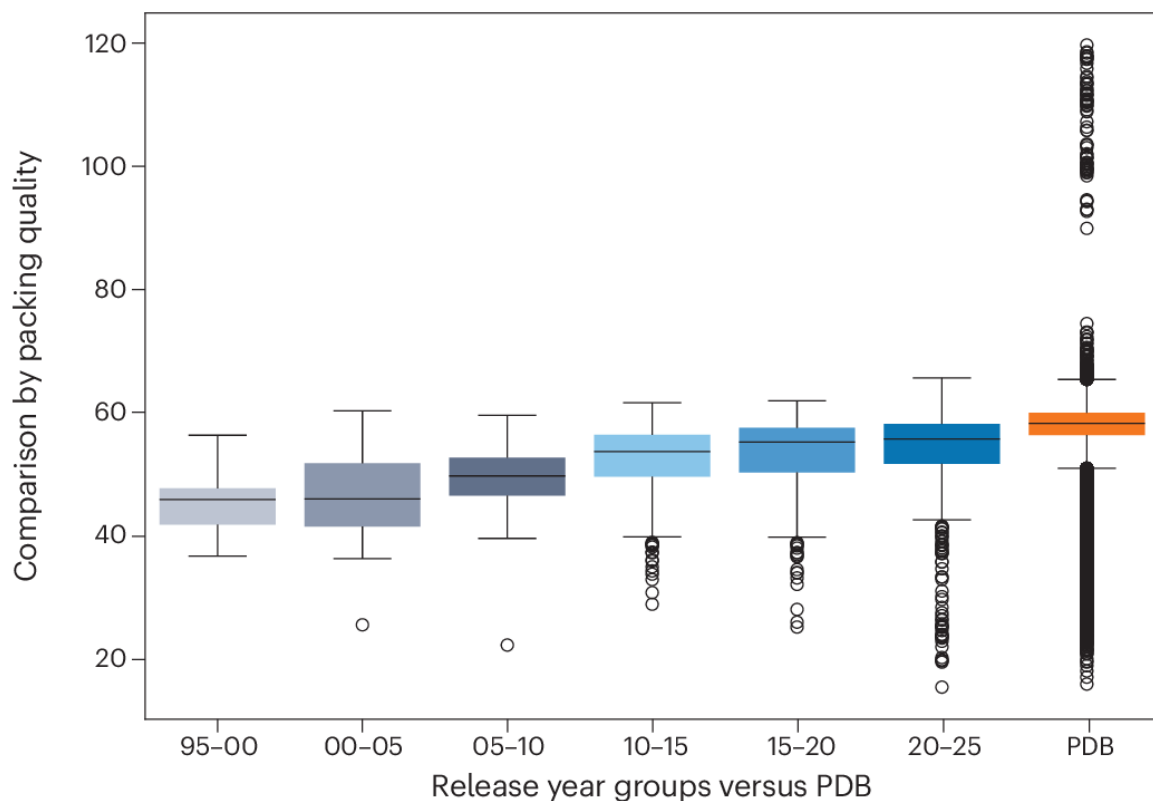
De novo design: remaining challenges



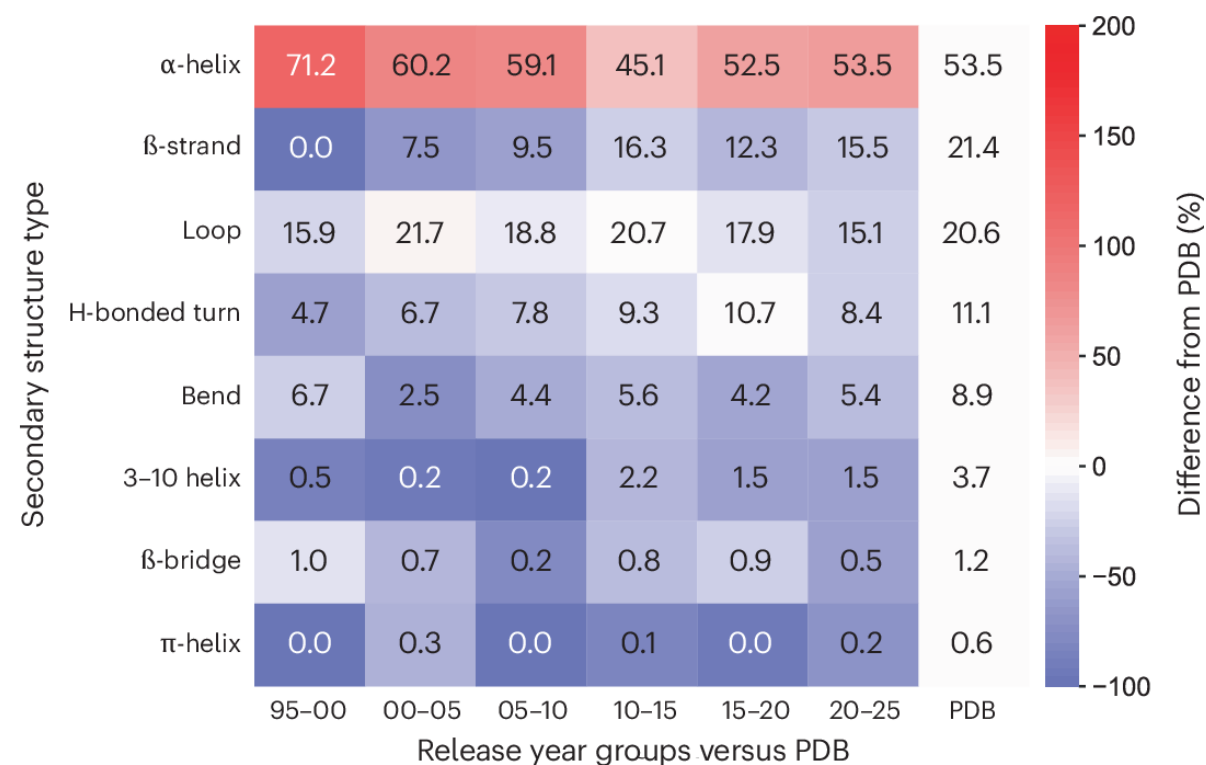
- Increases with:
- Size
 - Long-range interactions
 - Beta-sheet content
 - Membrane proteins require specific energy functions

... but remain pre-dominantly alpha-helical

d Comparison by packing quality

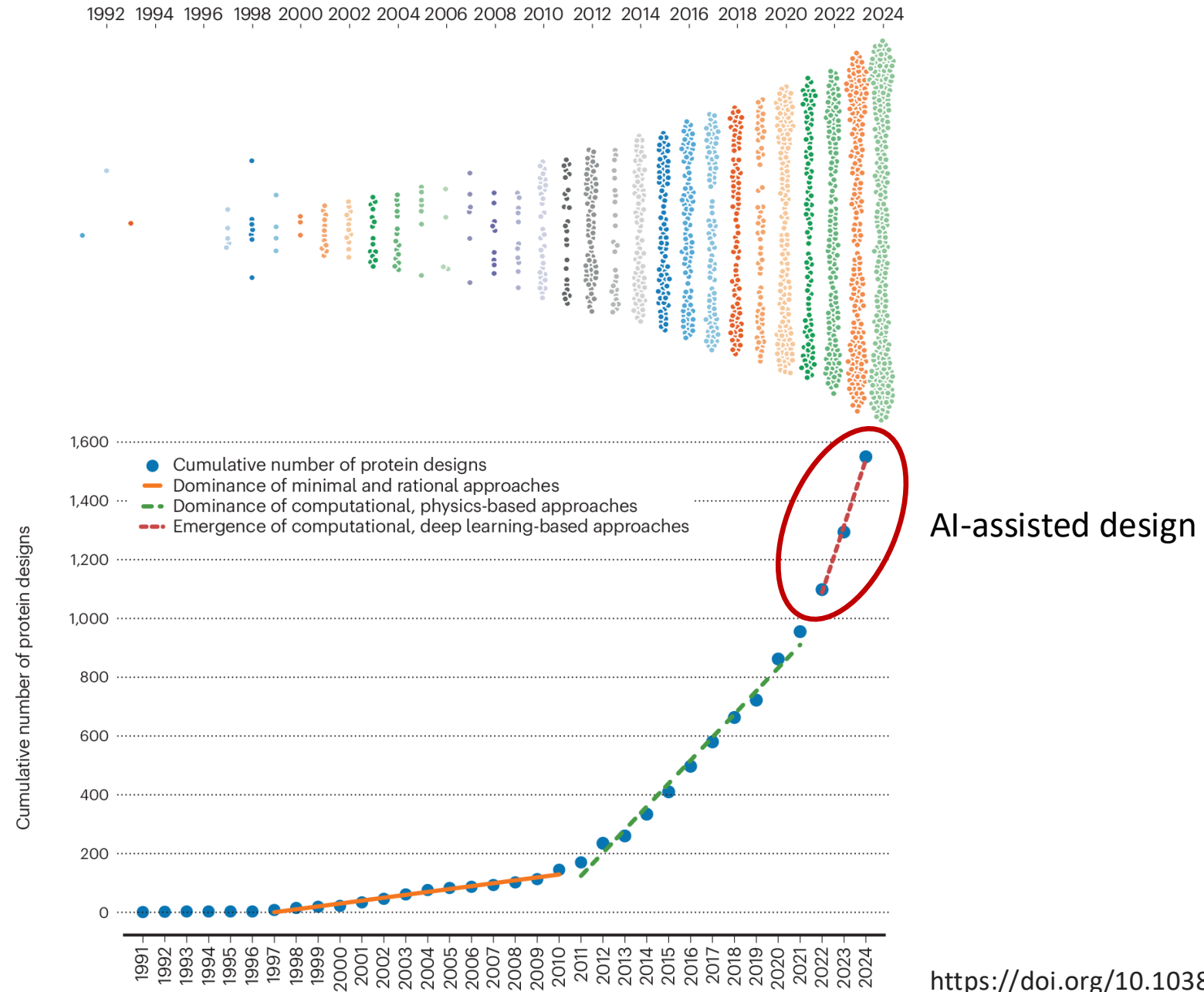


b Design secondary structure percentages by year group versus PDB



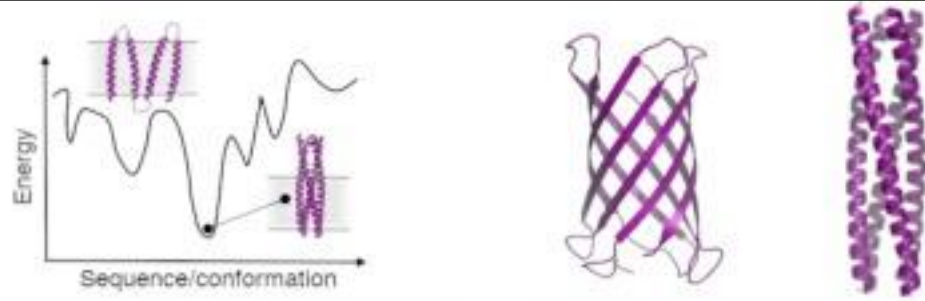
AI-assisted design

Evolution of *de novo* protein design



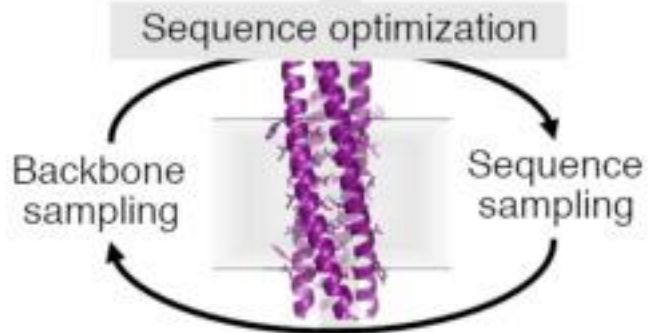
AI-era *de novo* design pipeline

Backbone generation



RFDiffusion

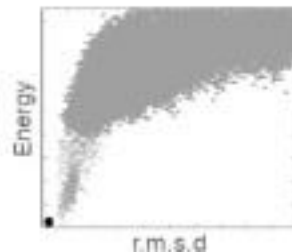
Sequence design



ProteinMPNN

Validation

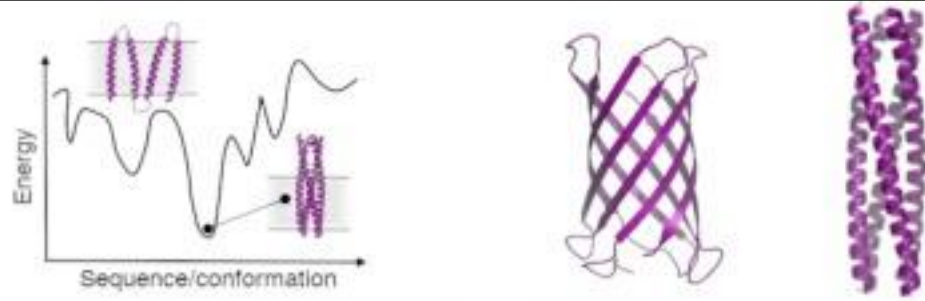
Energy landscape of the designed sequence converges on target structure



ColabFold

AI-era *de novo* design pipeline

Backbone generation

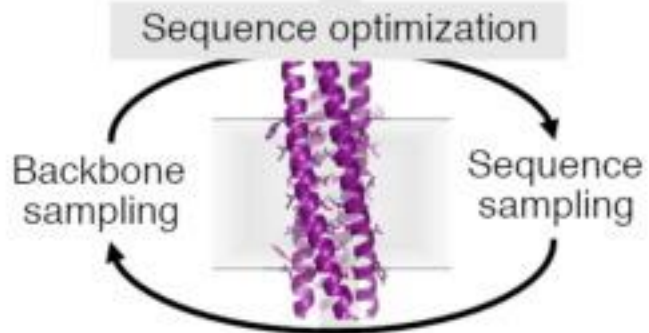


Scoring

Backbone definition

RFDiffusion

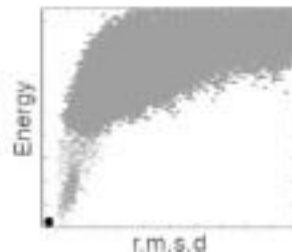
Sequence design



ProteinMPNN

Validation

Energy landscape of the designed sequence converges on target structure



ColabFold

Diffusion models for image generation

“an abstract painting of a room full of proteins learning machine learning”



Generated using DALL-E, DALL-E 3

Diffusion models for image generation

“a humanized-redbull drinking a redbull with a redbull-themed fridge full of redbull in the background”



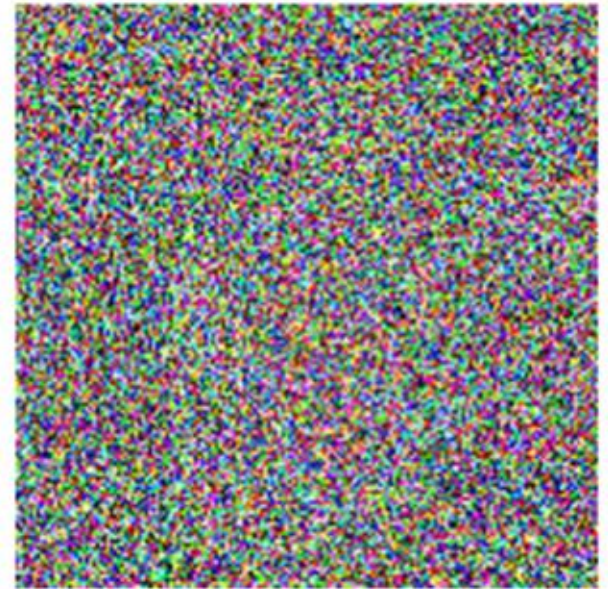
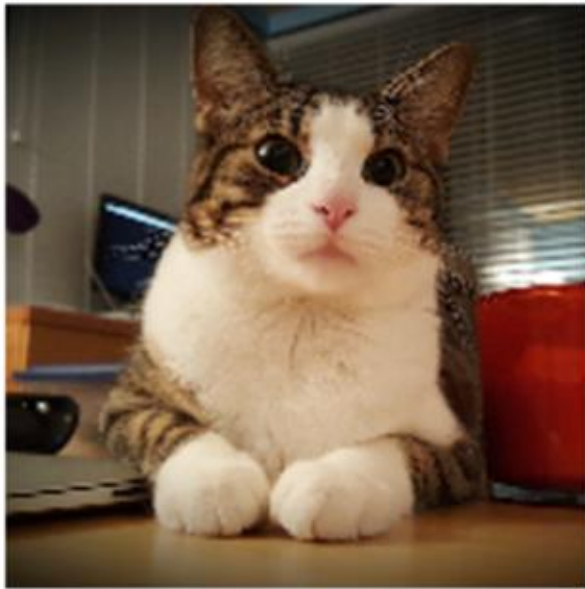
Generated using DALL-E, DALL-E 3

How do diffusion models work?

Diffusion models iteratively add noise to data and learn how to reverse (denoise) the process

Once trained, data can be endlessly generated from input (random noise)

Forward diffusion process



Generate reverse denoising process

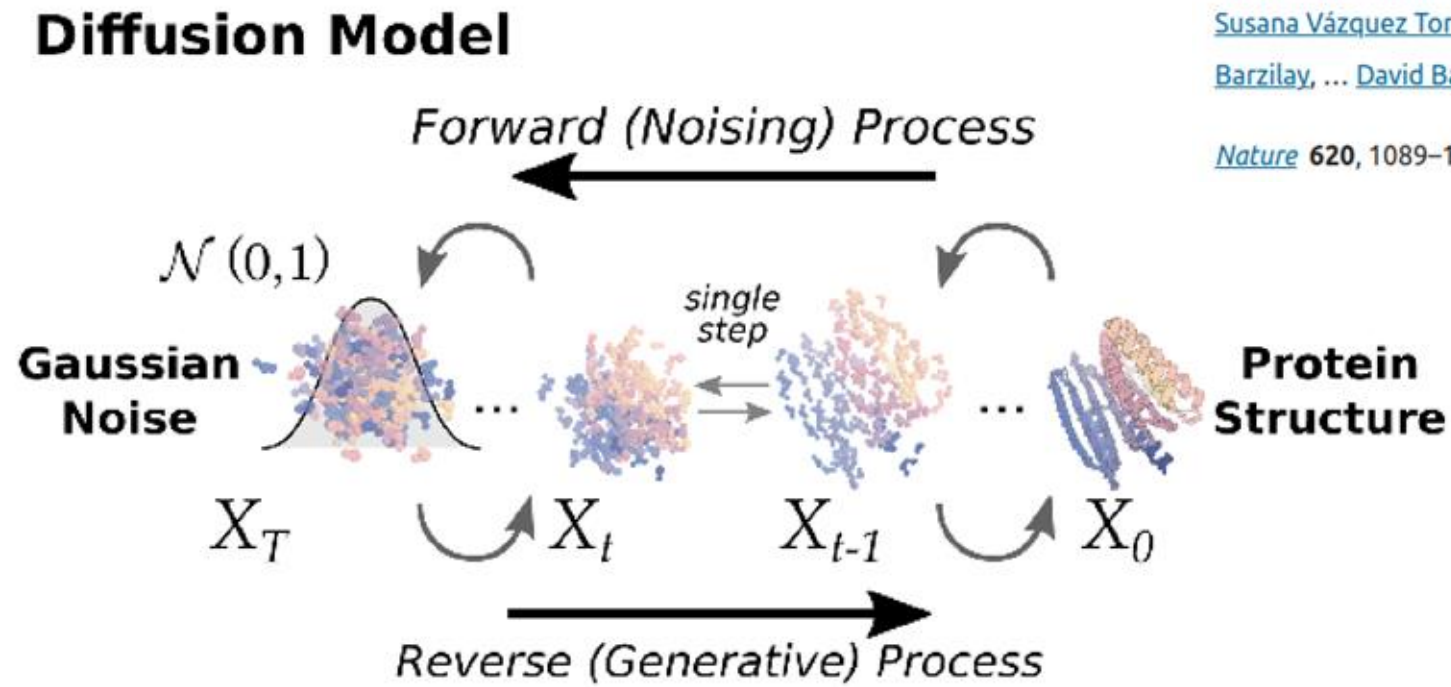
RFDiffusion model for protein generation

Article | [Open access](#) | Published: 11 July 2023

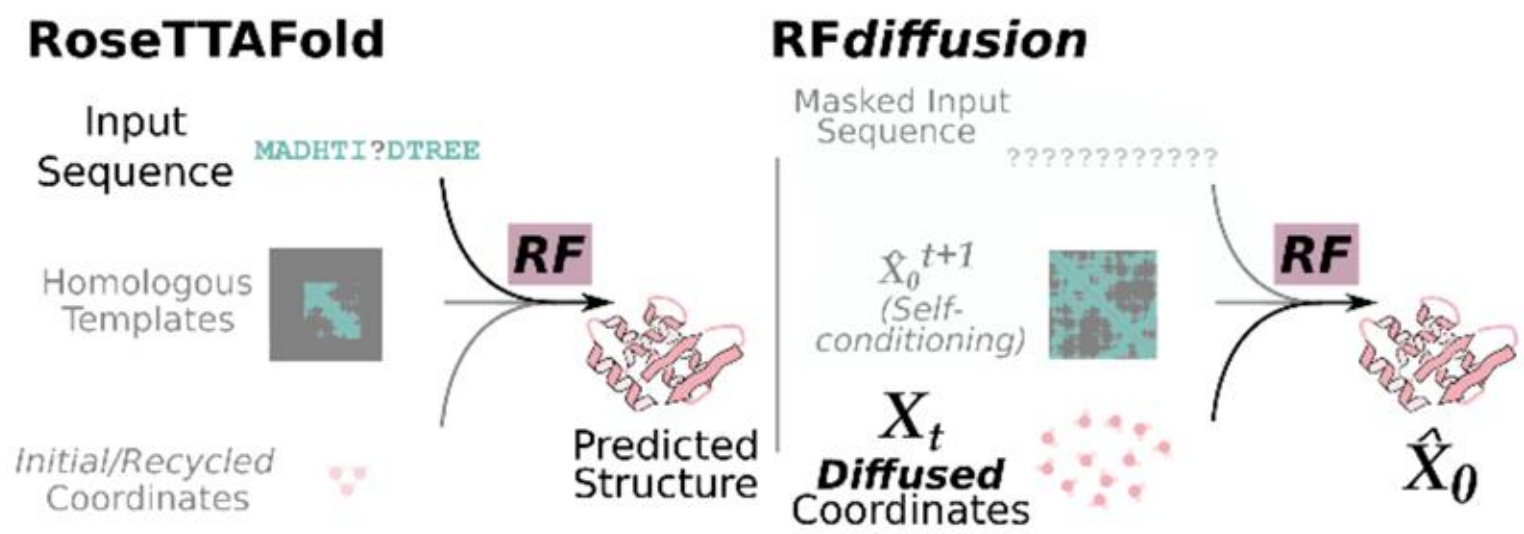
De novo design of protein structure and function with RFDiffusion

[Joseph L. Watson](#), [David Juergens](#), [Nathaniel R. Bennett](#), [Brian L. Trippe](#), [Jason Yim](#), [Helen E. Eisenach](#), [Woody Ahern](#), [Andrew J. Borst](#), [Robert J. Ragotte](#), [Lukas F. Milles](#), [Basile I. M. Wicky](#), [Nikita Hanikel](#), [Samuel J. Pellock](#), [Alexis Courbet](#), [William Sheffler](#), [Jue Wang](#), [Preetham Venkatesh](#), [Isaac Sappington](#), [Susana Vázquez Torres](#), [Anna Lauko](#), [Valentin De Bortoli](#), [Emile Mathieu](#), [Sergey Ovchinnikov](#), [Regina Barzilay](#), ... [David Baker](#)  [+ Show authors](#)

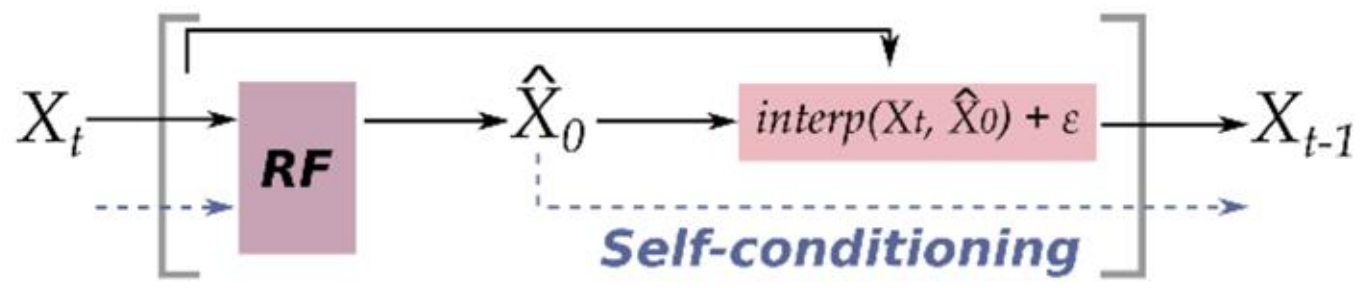
[Nature](#) 620, 1089–1100 (2023) | [Cite this article](#)



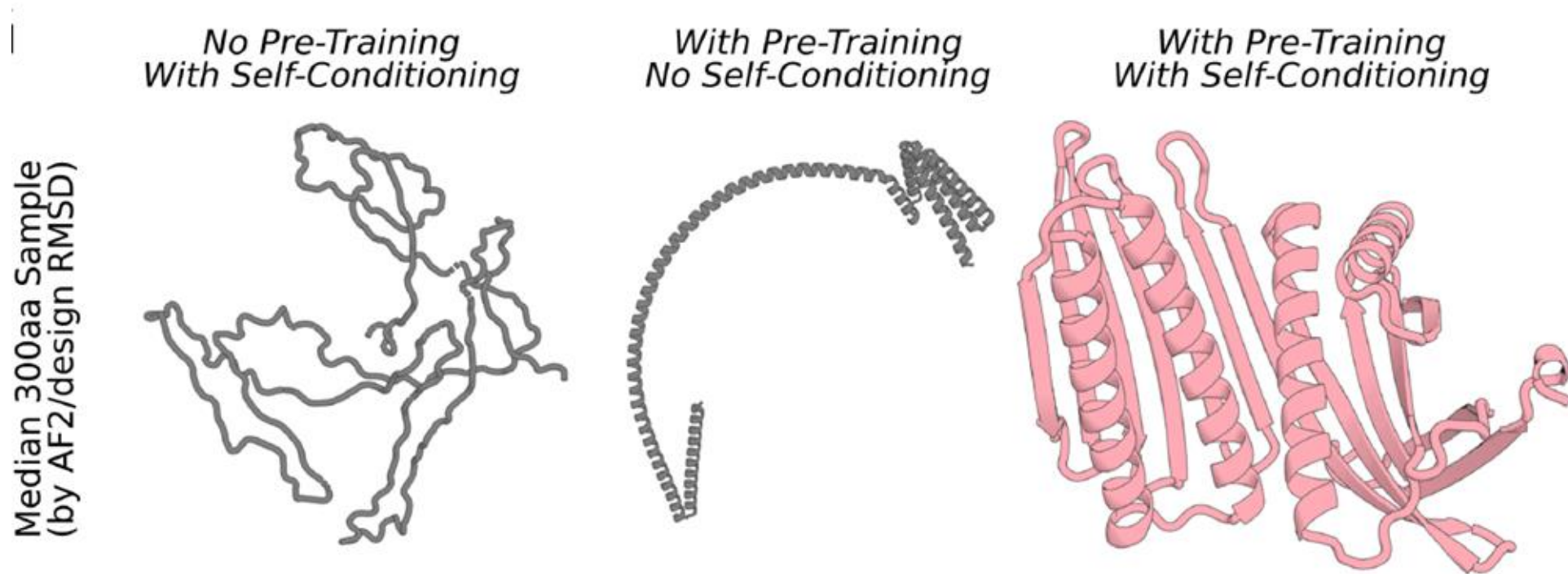
RFDiffusion model for protein generation



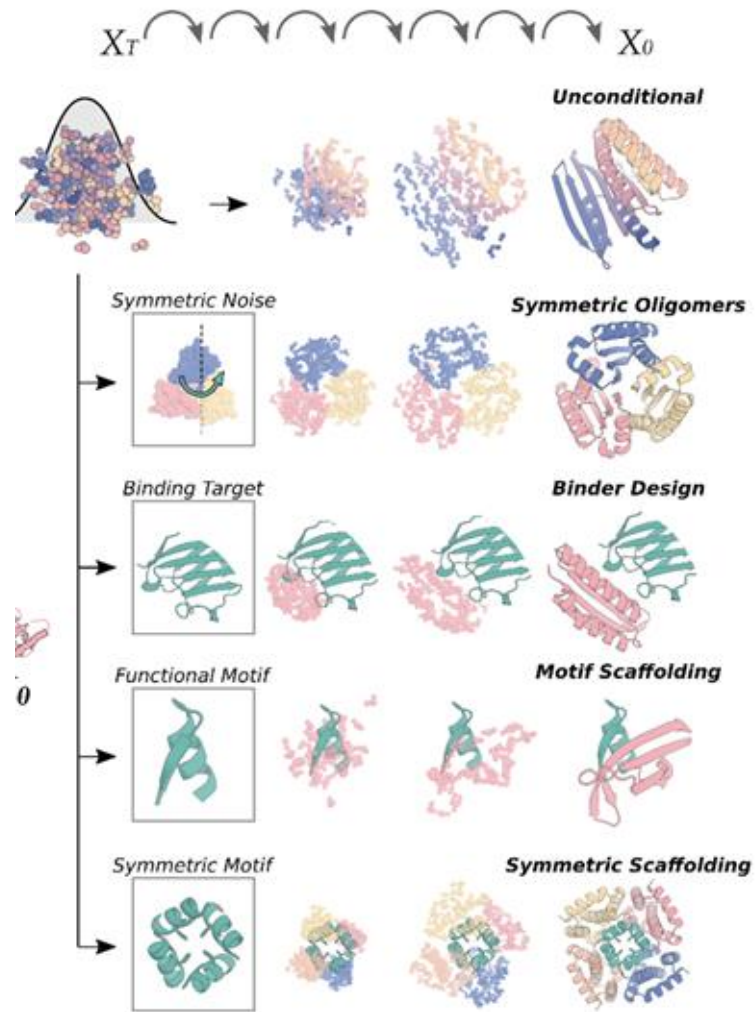
Single RFDiffusion step



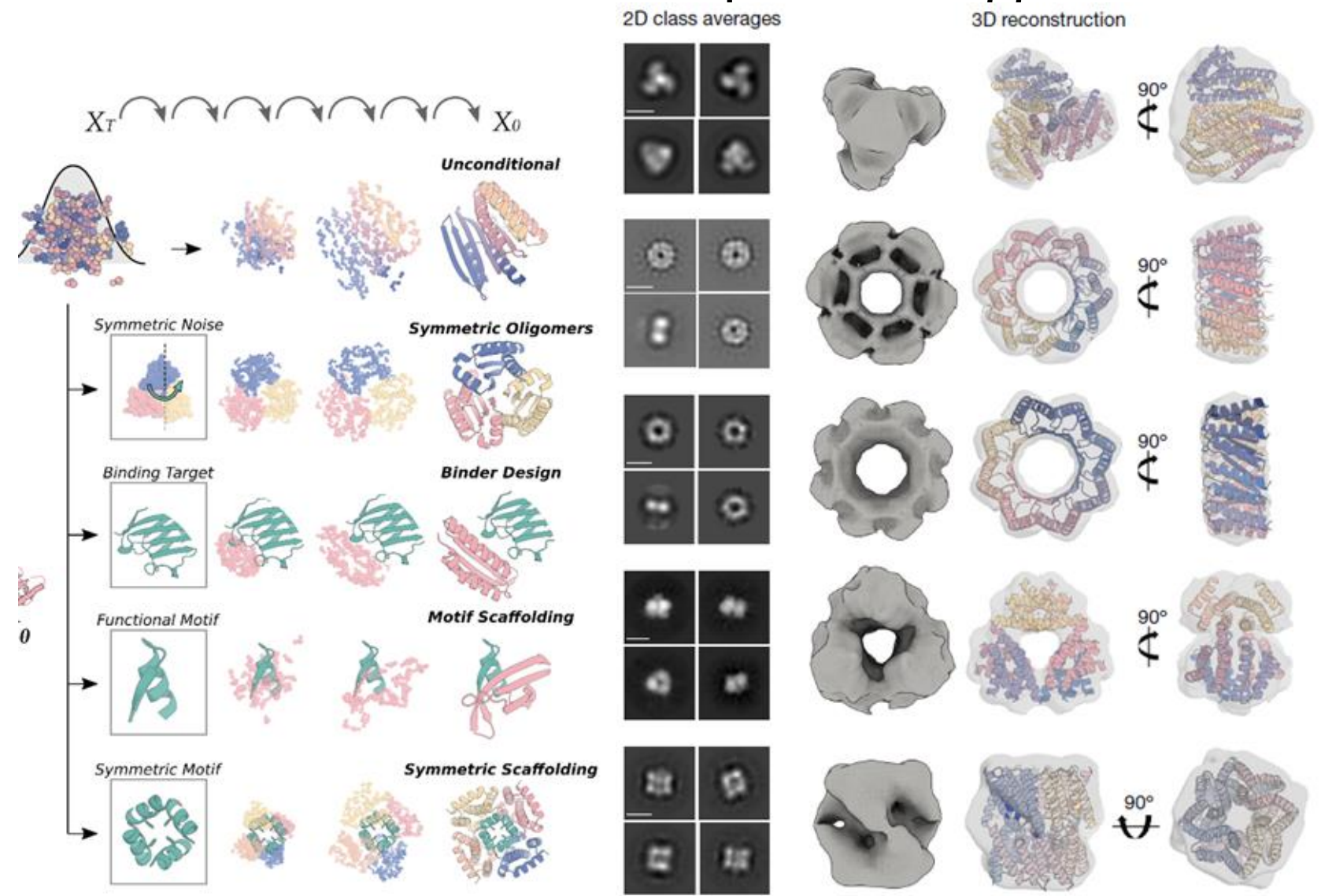
Performance is worse without pre-training & self-conditioning



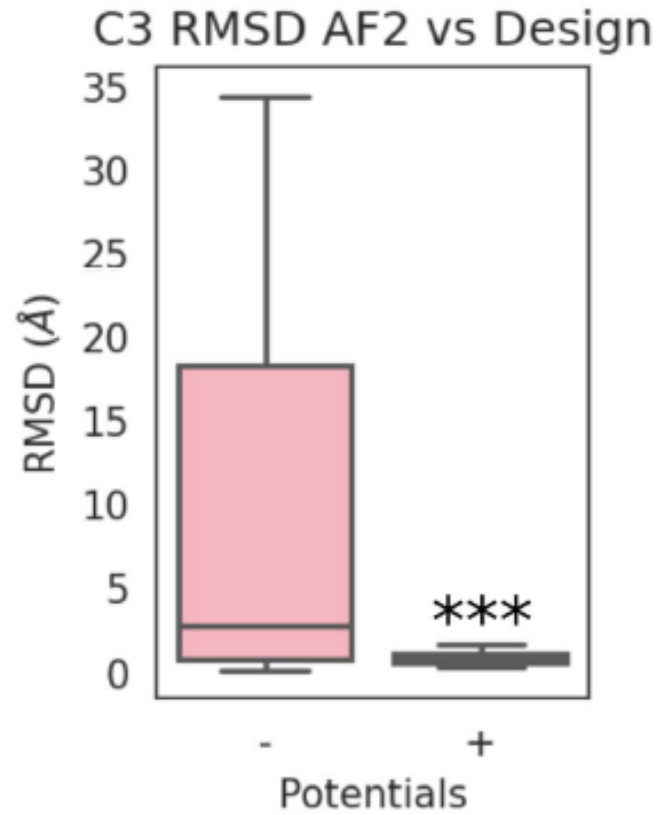
RFDiffusion model for protein generation



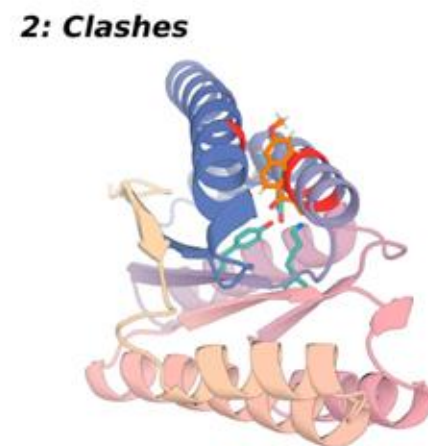
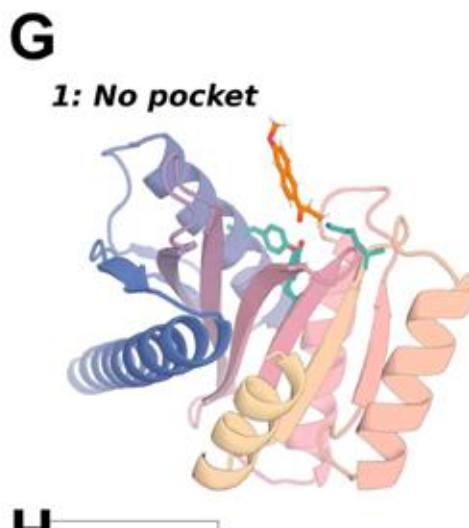
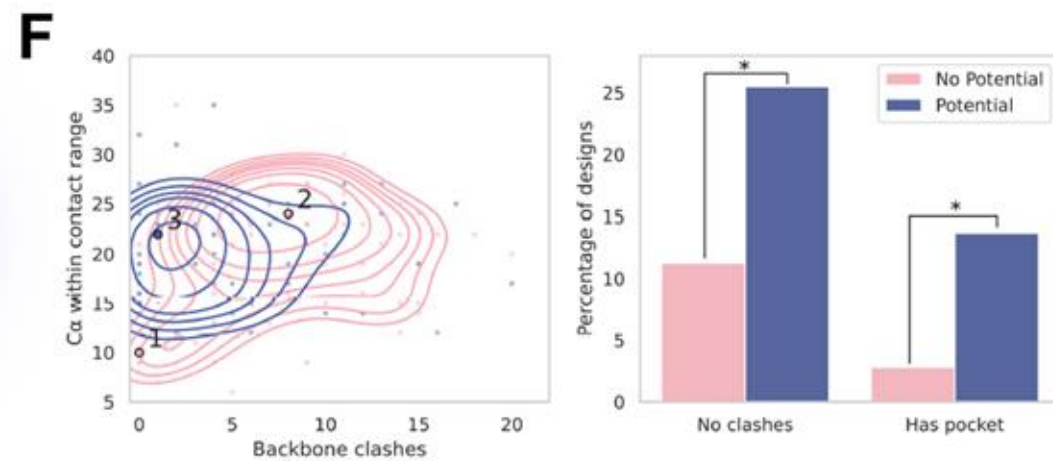
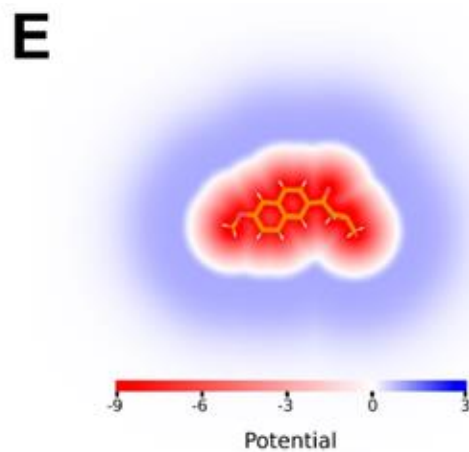
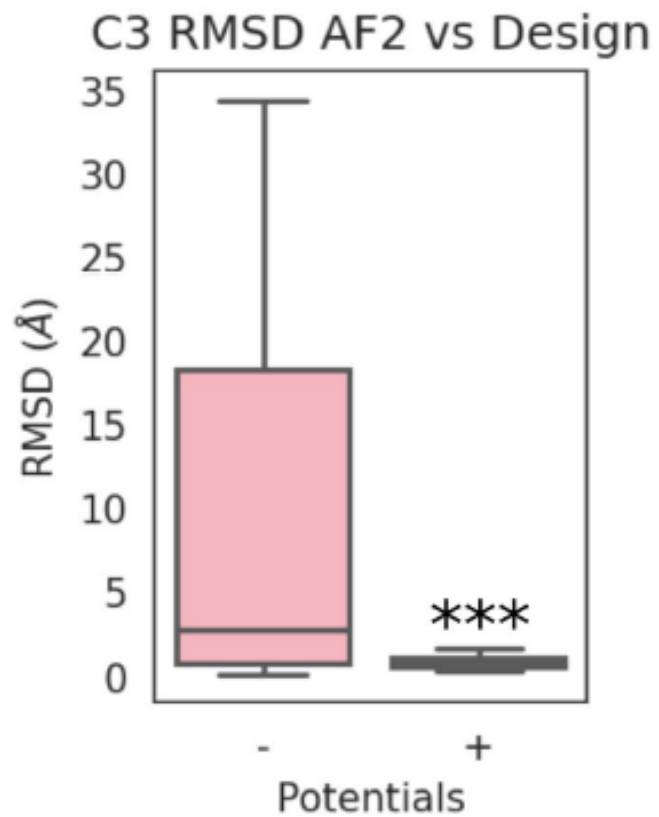
RFDiffusion model for protein generation



Guiding potentials

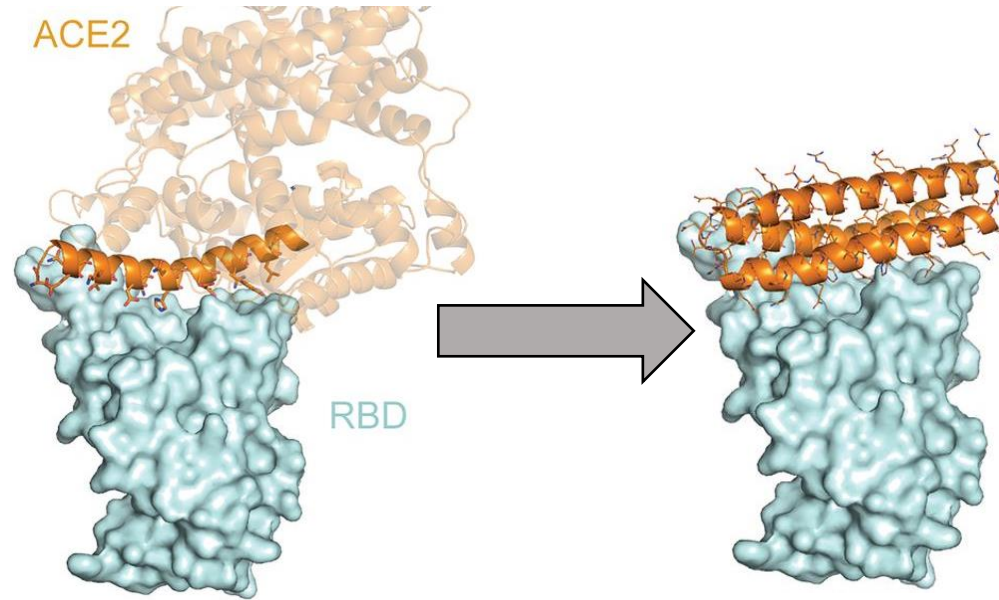


Guiding potentials

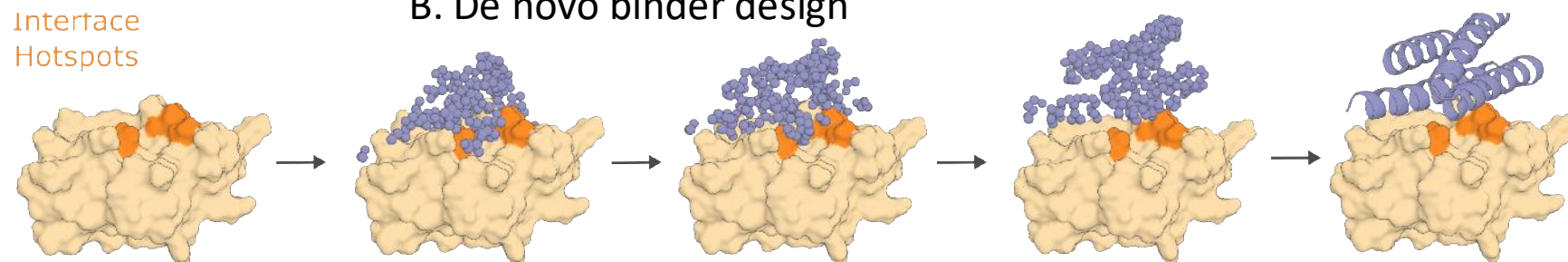


De novo design of protein binders

A. Motif scaffolding



B. De novo binder design

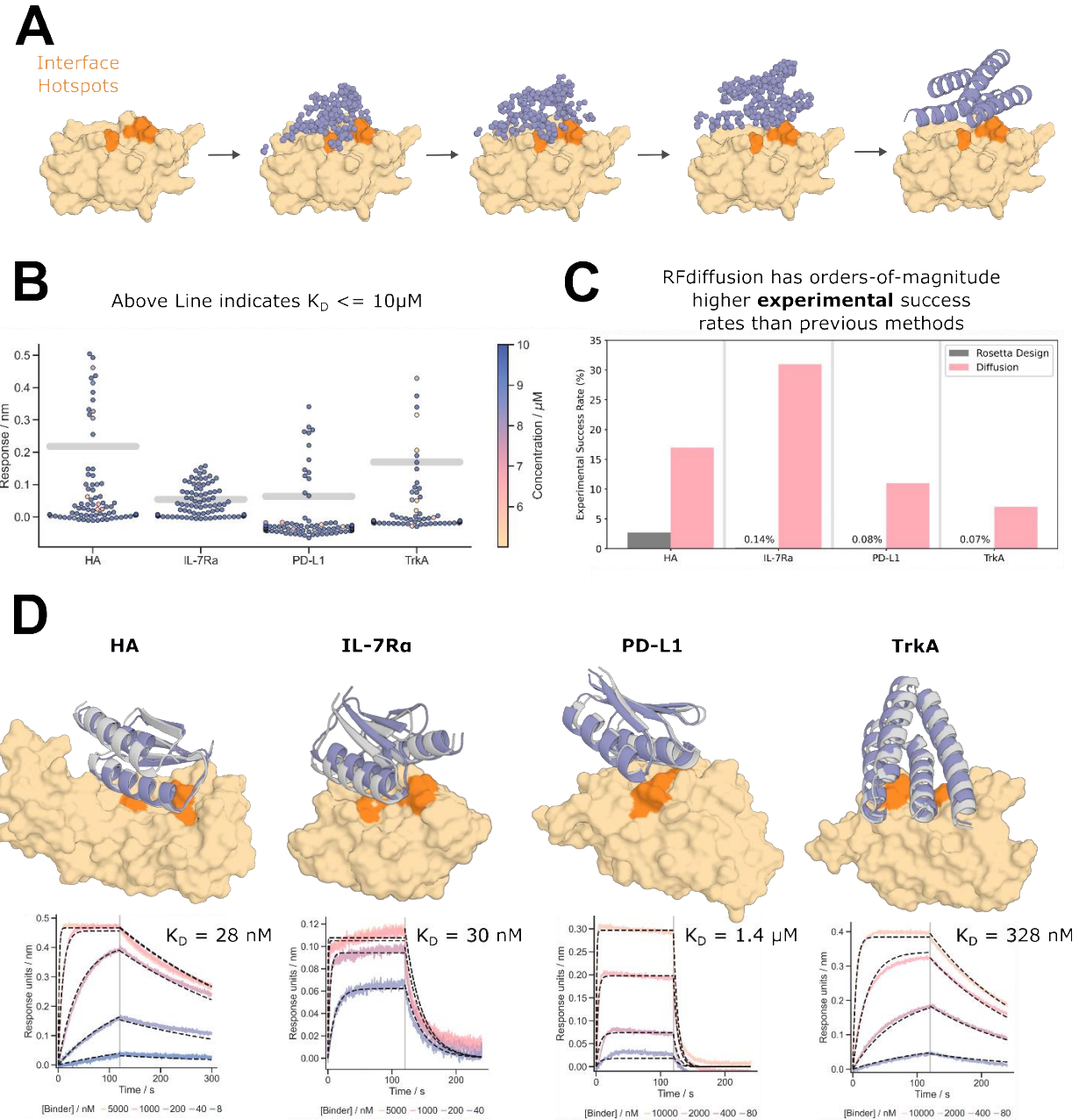


De novo designed protein-protein interfaces

4 targets

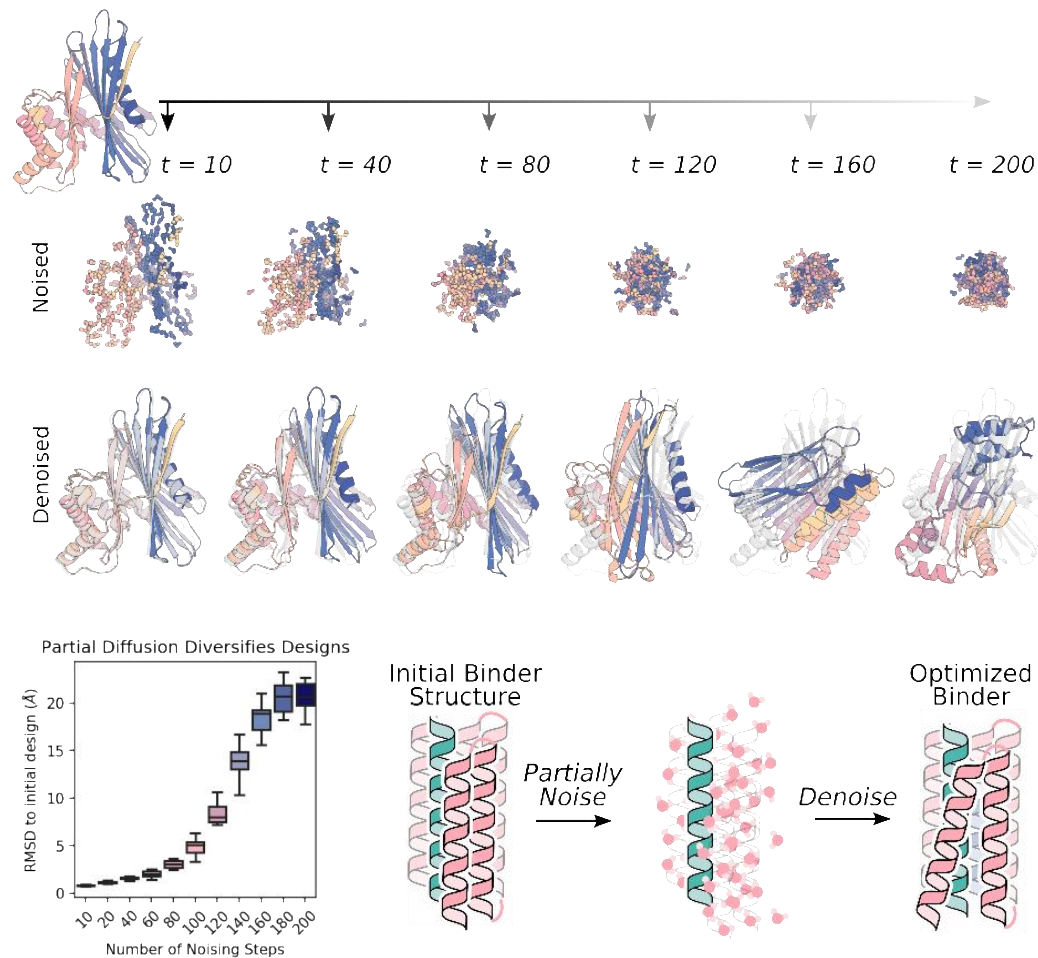
Binders obtained for all targets (95 designs tested per target)

~18% success rate (< 1% success rate before, required yeast display libraries).



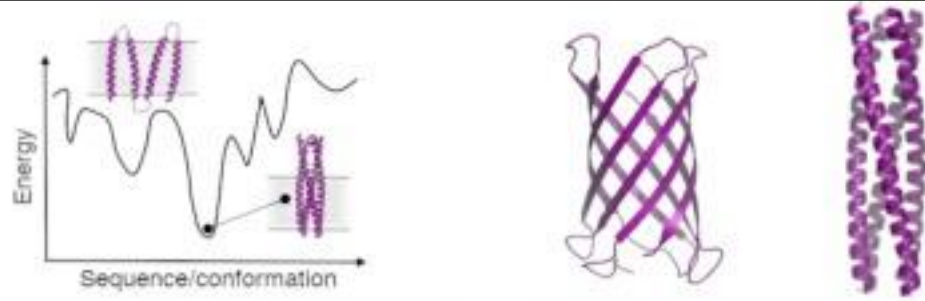
Partial diffusion (refinement)

A Partial noising and denoising using RFdiffusion



2. Sequence design with **ProteinMPNN**

Backbone generation

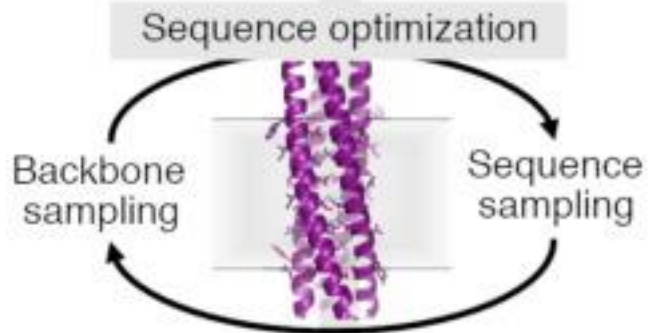


Scoring

Backbone definition

RFDiffusion

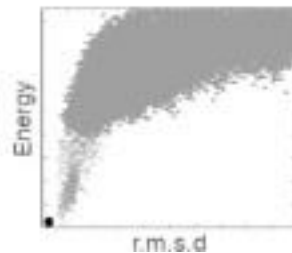
Sequence design



ProteinMPNN

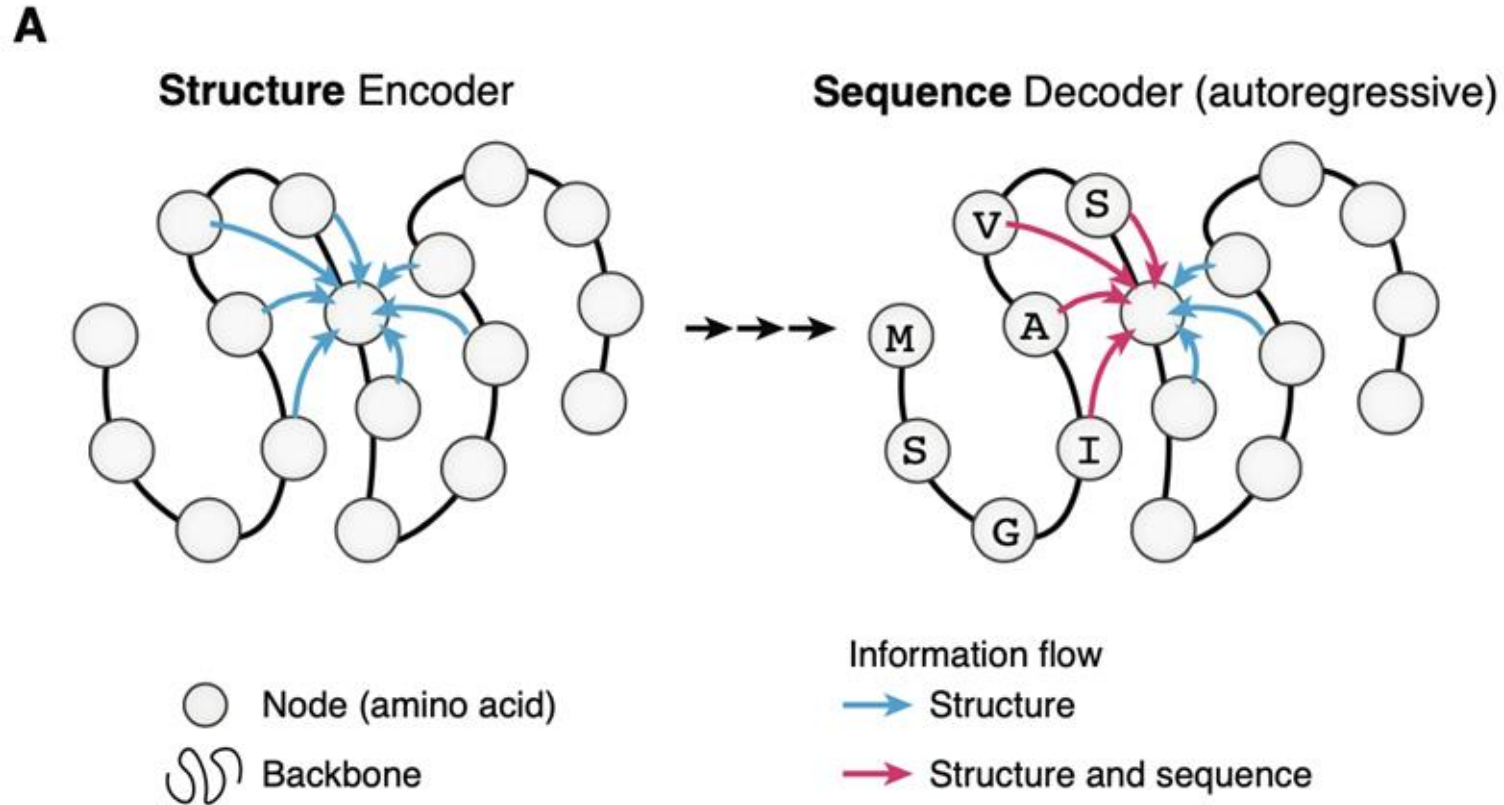
Validation

Energy landscape of the designed sequence converges on target structure



ColabFold

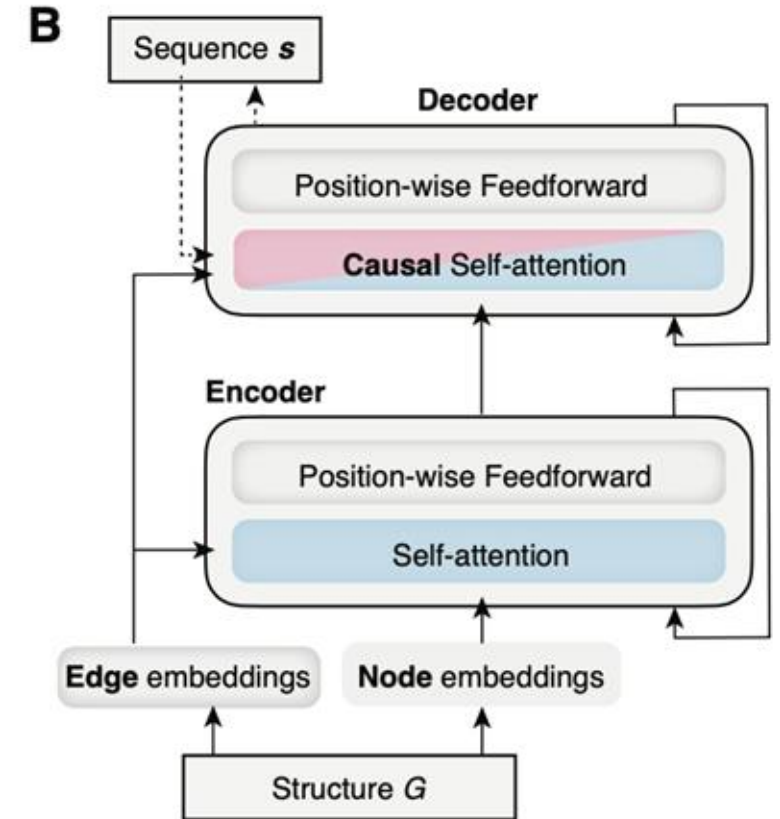
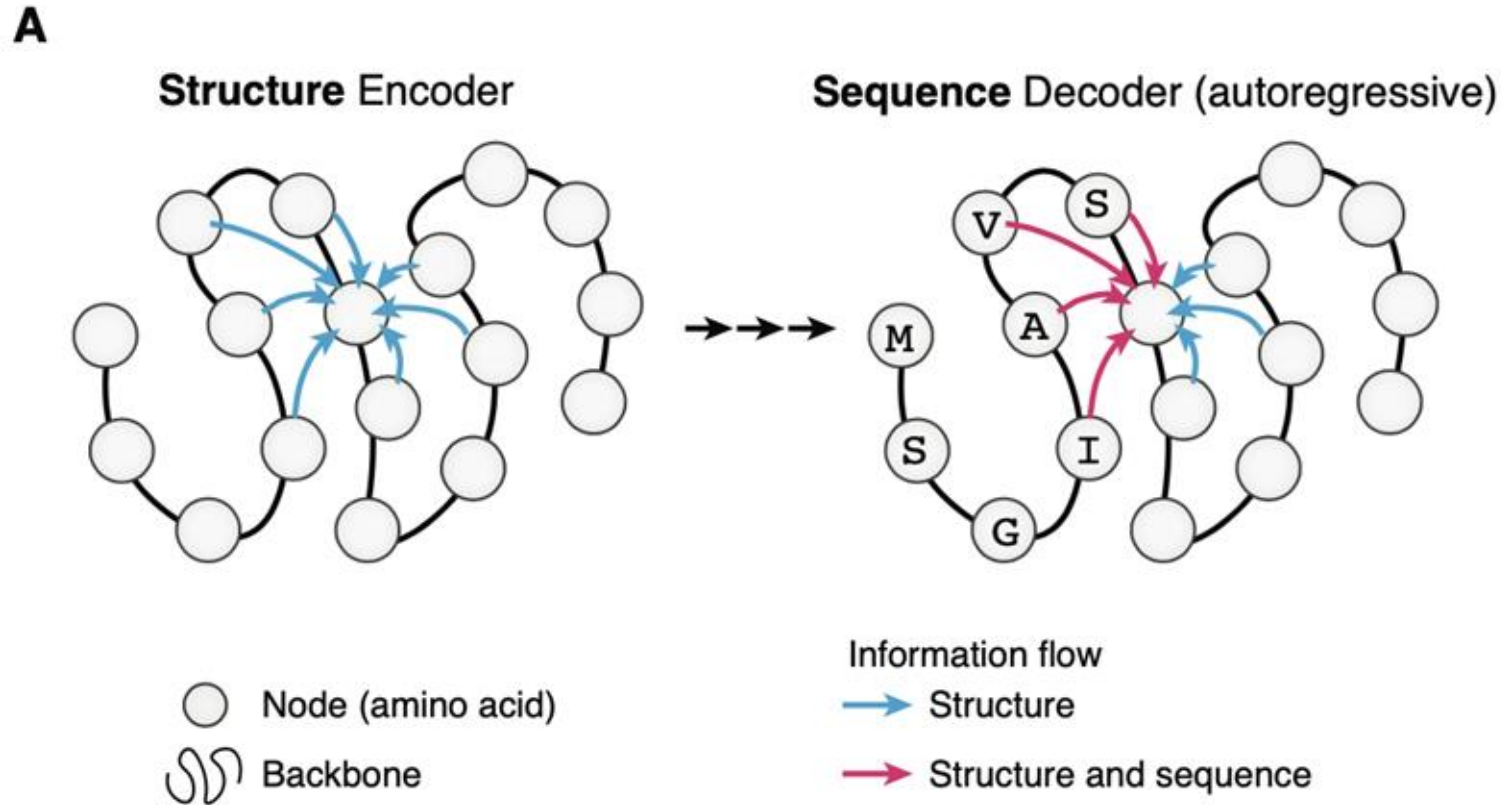
The MPNN architecture for protein design built by John Ingraham et al.



Ingraham, et al., *NeurIPS* 2019

https://proceedings.neurips.cc/paper_files/paper/2019/file/f3a4ff4839c56a5f460c88cce3666a2b-Paper.pdf

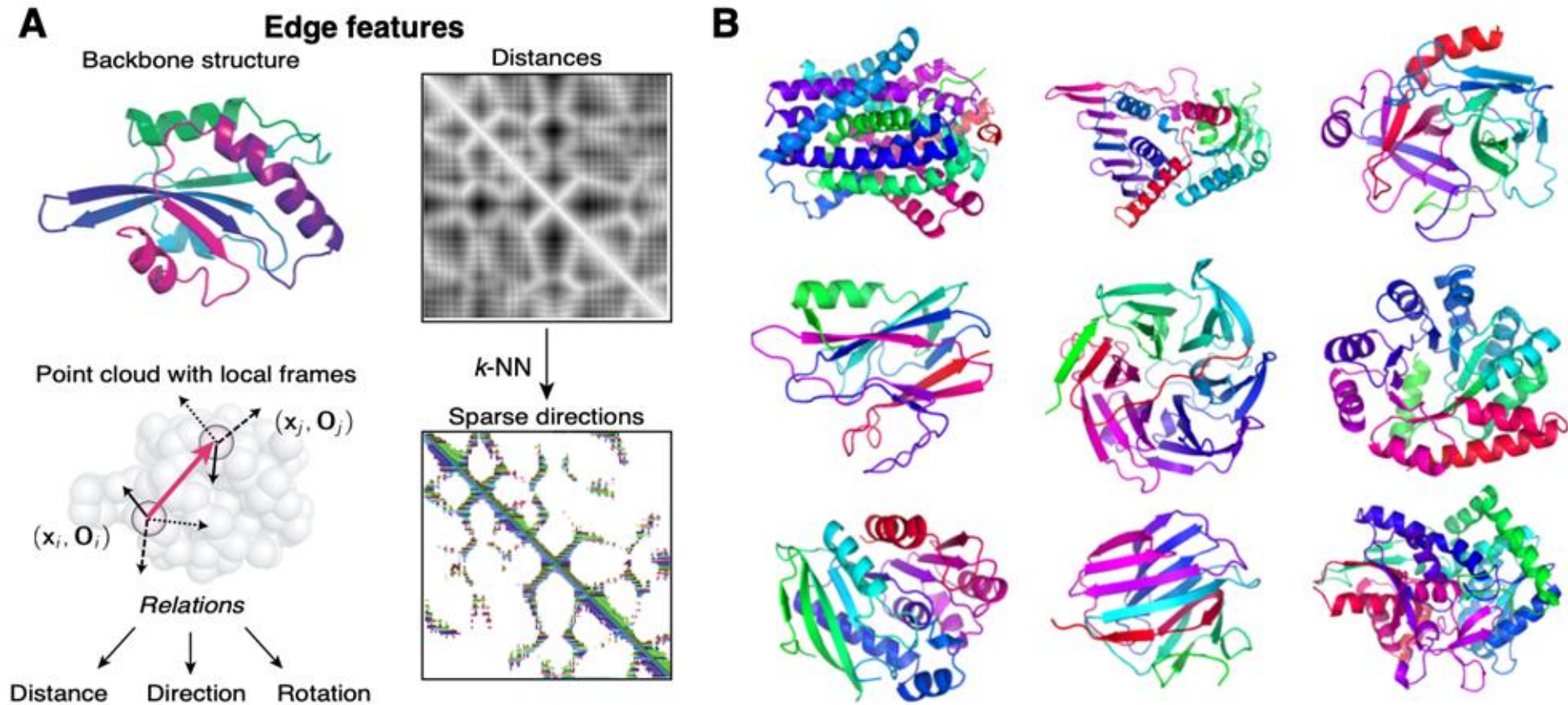
The MPNN architecture for protein design built by John Ingraham et al.



Ingraham, et al., *NeurIPS* 2019

https://proceedings.neurips.cc/paper_files/paper/2019/file/f3a4ff4839c56a5f460c88cce3666a2b-Paper.pdf

The MPNN architecture for protein design built by John Ingraham et al.



Ingraham, et al., *NeurIPS* 2019

https://proceedings.neurips.cc/paper_files/paper/2019/file/f3a4ff4839c56a5f460c88cce3666a2b-Paper.pdf

The MPNN architecture for protein design built by John Ingraham et al.

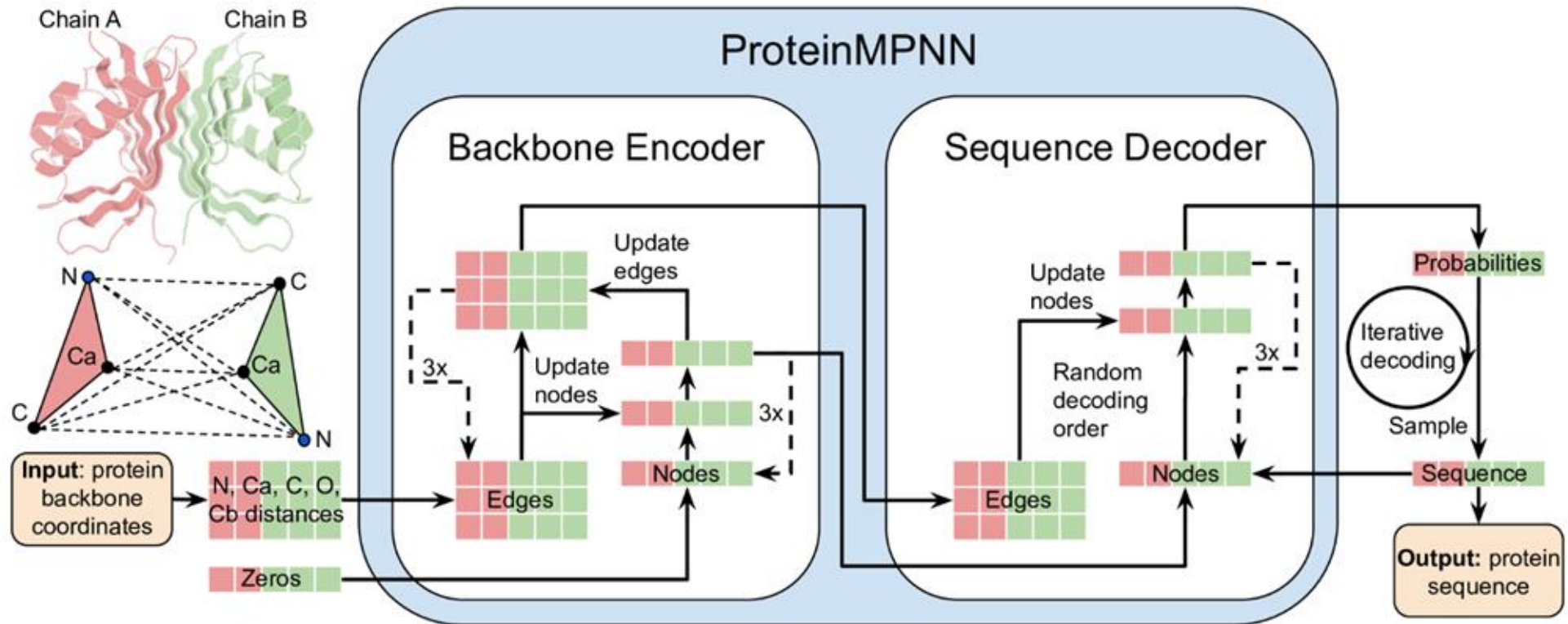
Method	Recovery (%)	Speed (AA/s) CPU	Speed (AA/s) GPU
Rosetta 3.10 fixbb	17.9	4.88×10^{-1}	N/A
Ours ($T = 0.1$)	27.6	2.22×10^2	1.04×10^4

(a) Single chain test set (103 proteins)

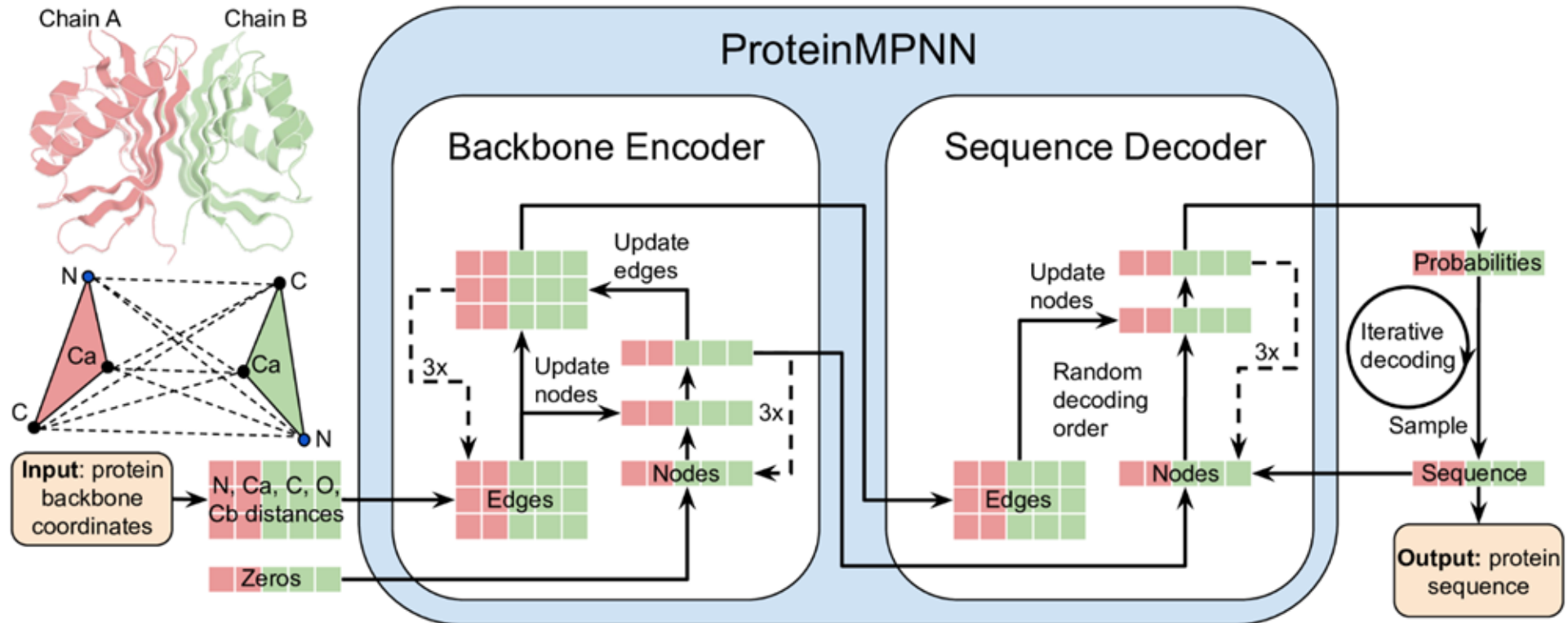
Method	Recovery (%)
Rosetta, fixbb 1	33.1
Rosetta, fixbb 2	38.4
Ours ($T = 0.1$)	39.2

(b) Ollikainen benchmark (40 proteins)

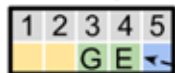
ProteinMPNN predicts protein sequences given a backbone structure



ProteinMPNN predicts protein sequences given a backbone structure



Fixed left to right decoding



Chain A

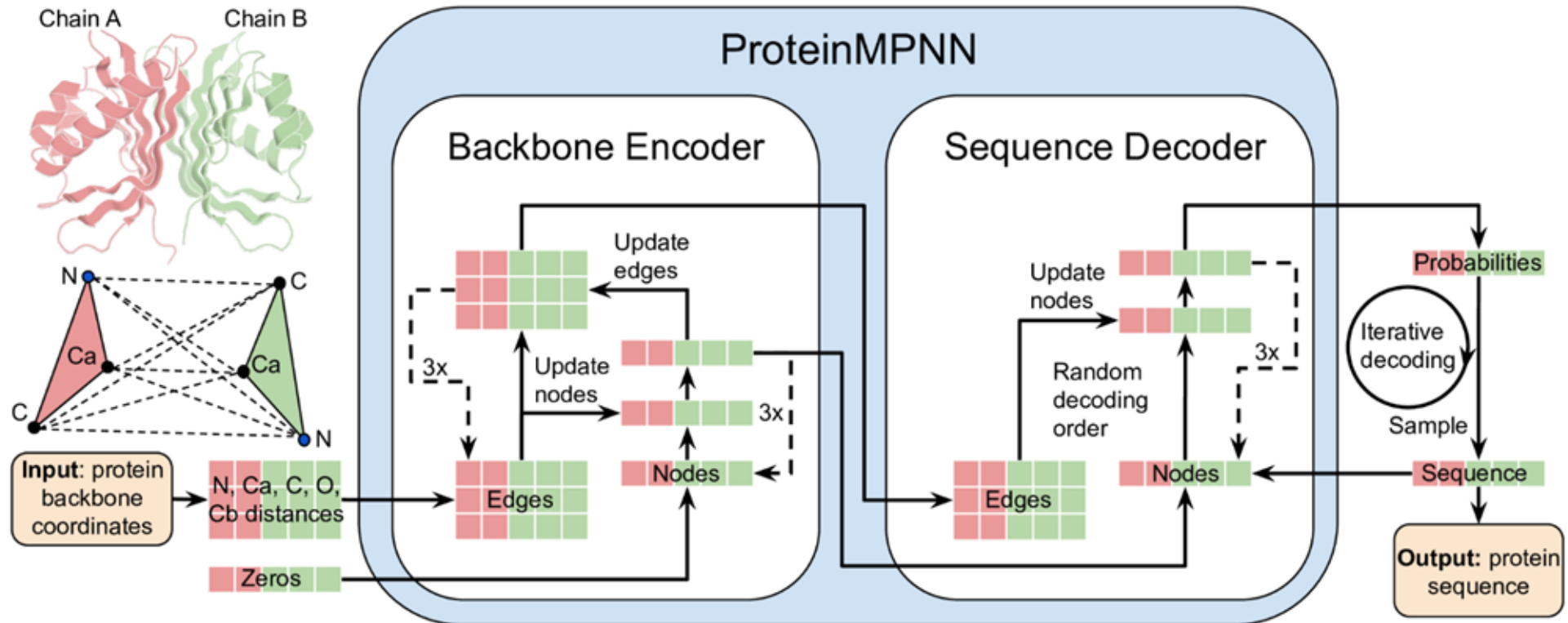
ProteinMPNN decoding



Chain A

- autoregressive decoding order
- fixed amino acids (context)
- sequence context not used
- sequence context used

ProteinMPNN predicts protein sequences given a backbone structure



Fixed left to right decoding

1 2 3 4 5

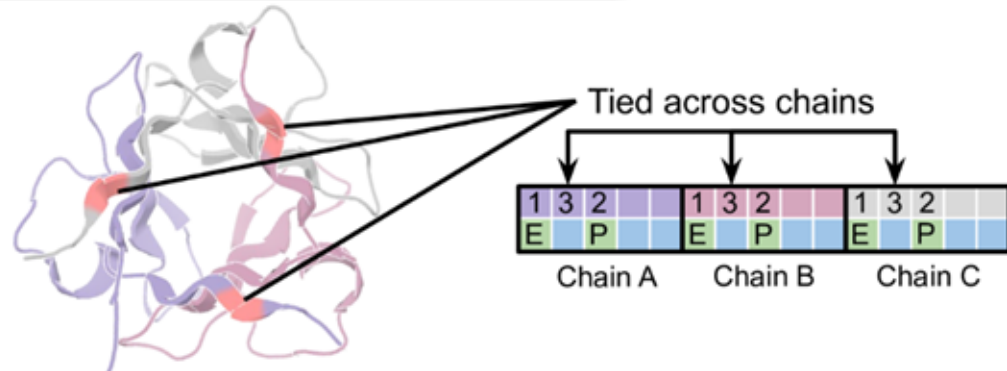
Chain A

ProteinMPNN decoding

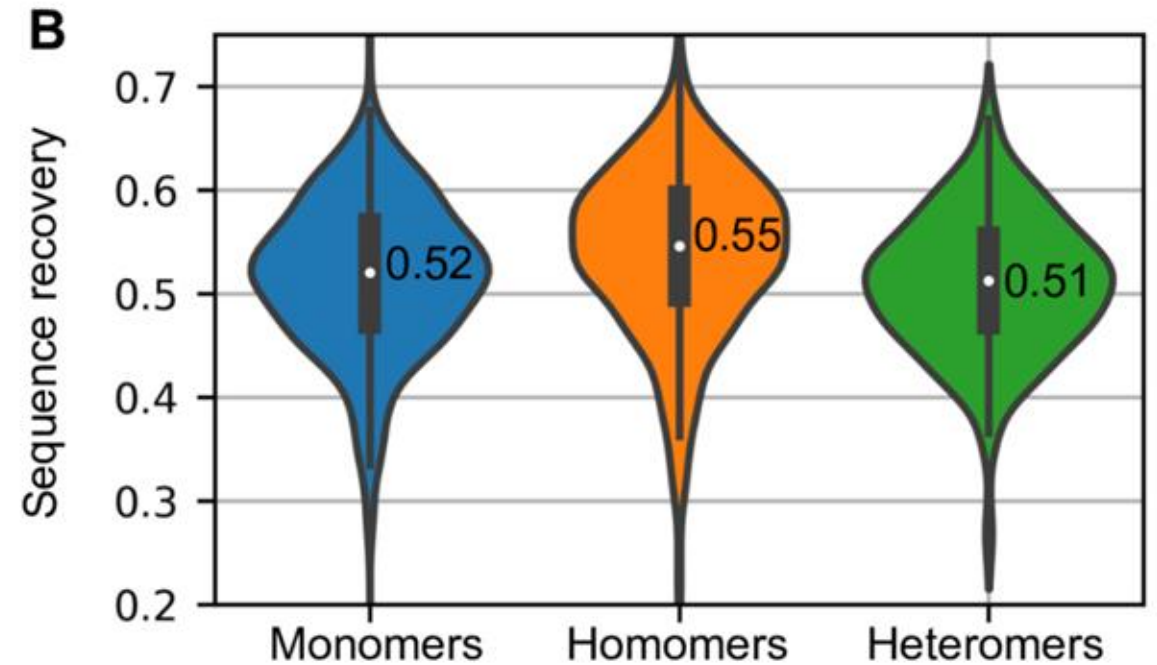
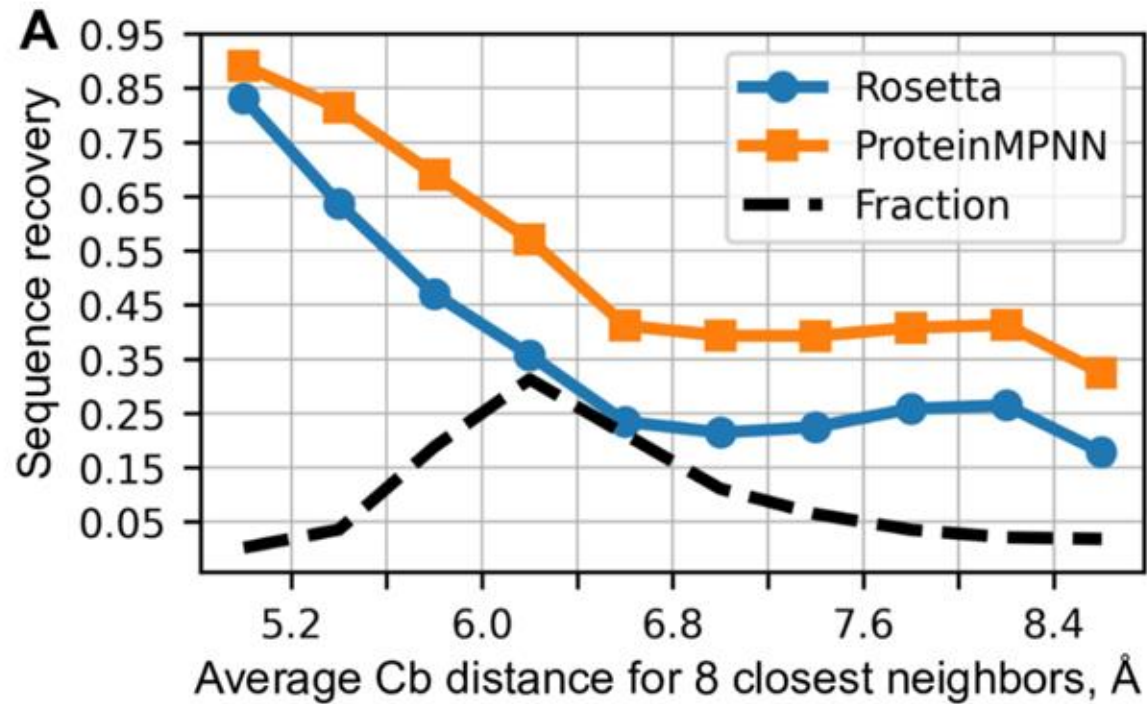
3 5 1 2 4

Chain A

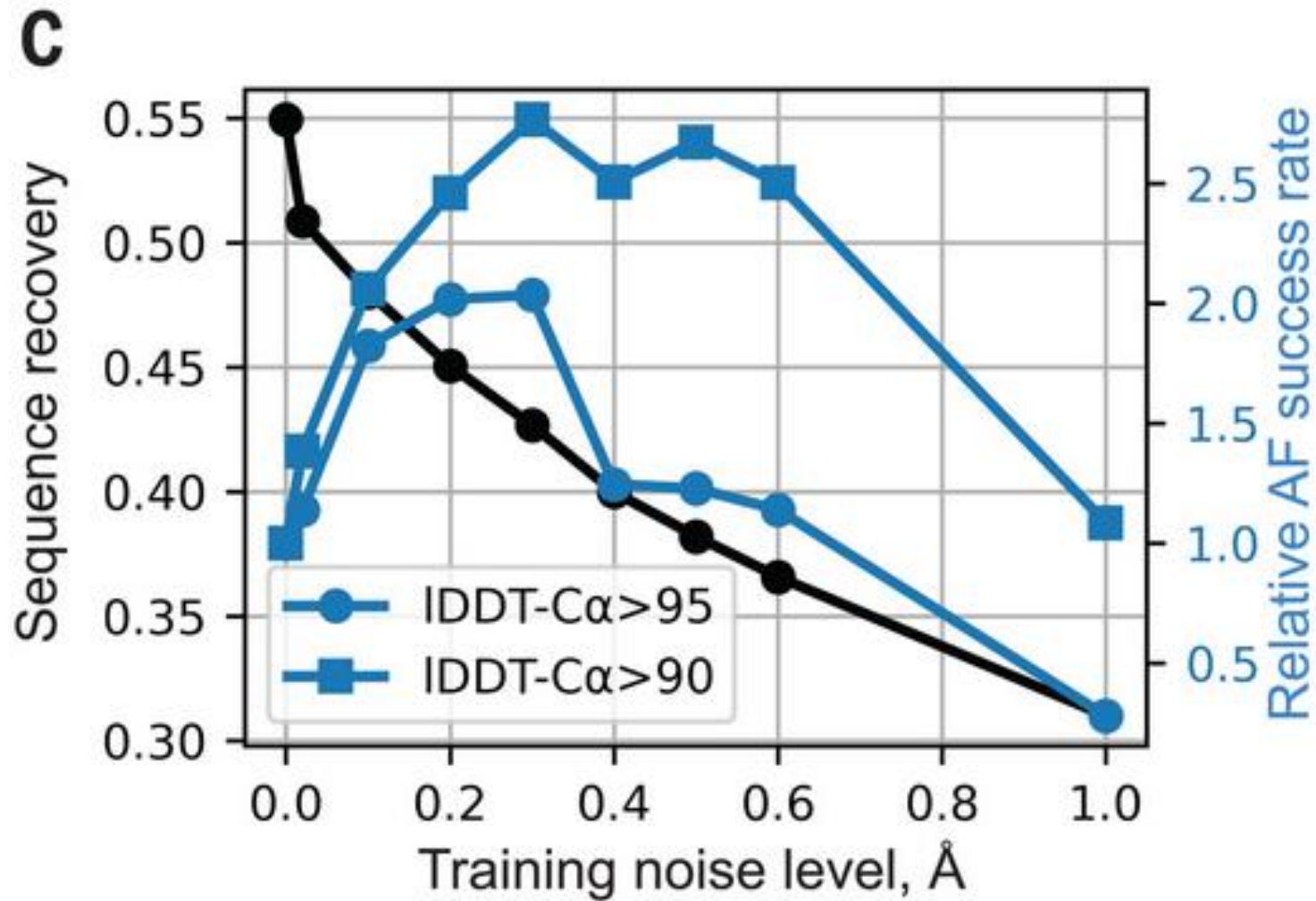
- autoregressive decoding order
- fixed amino acids (context)
- sequence context not used
- sequence context used



ProteinMPNN performance



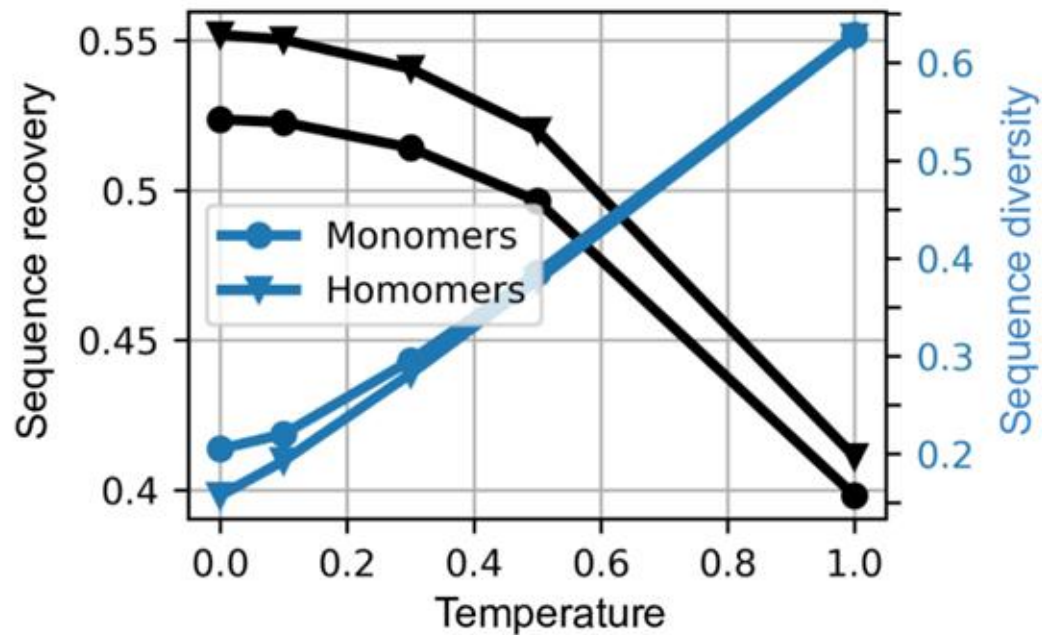
Influence of backbone noise added during training



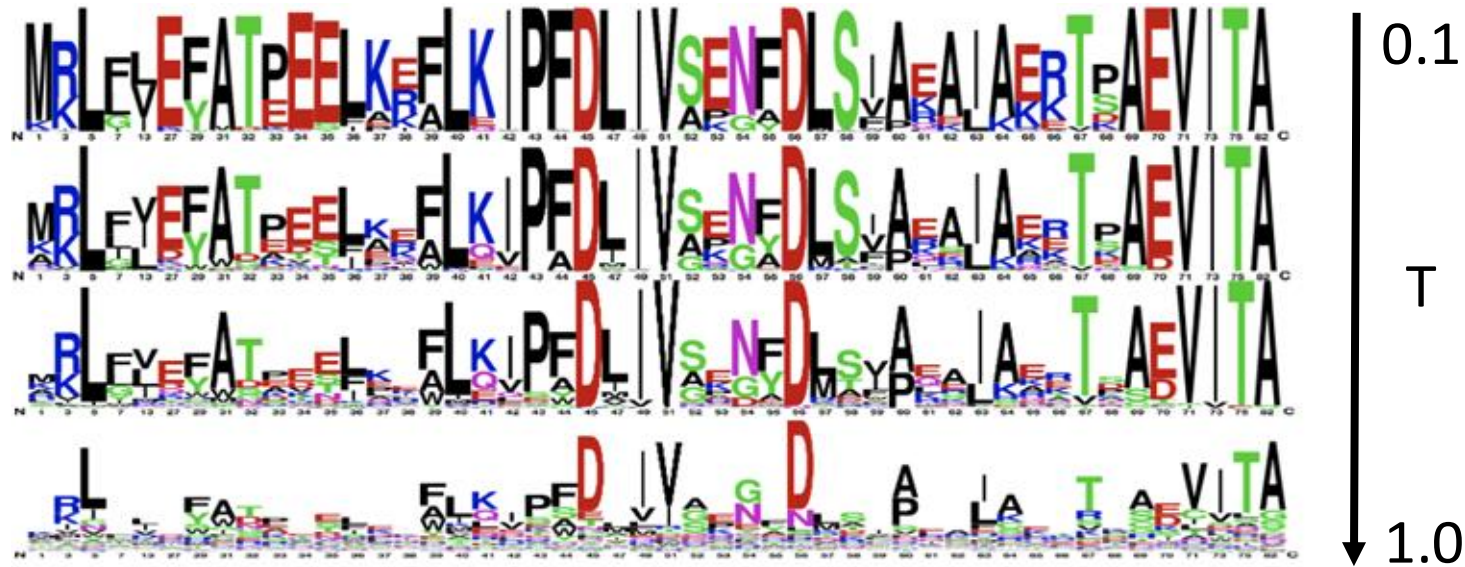
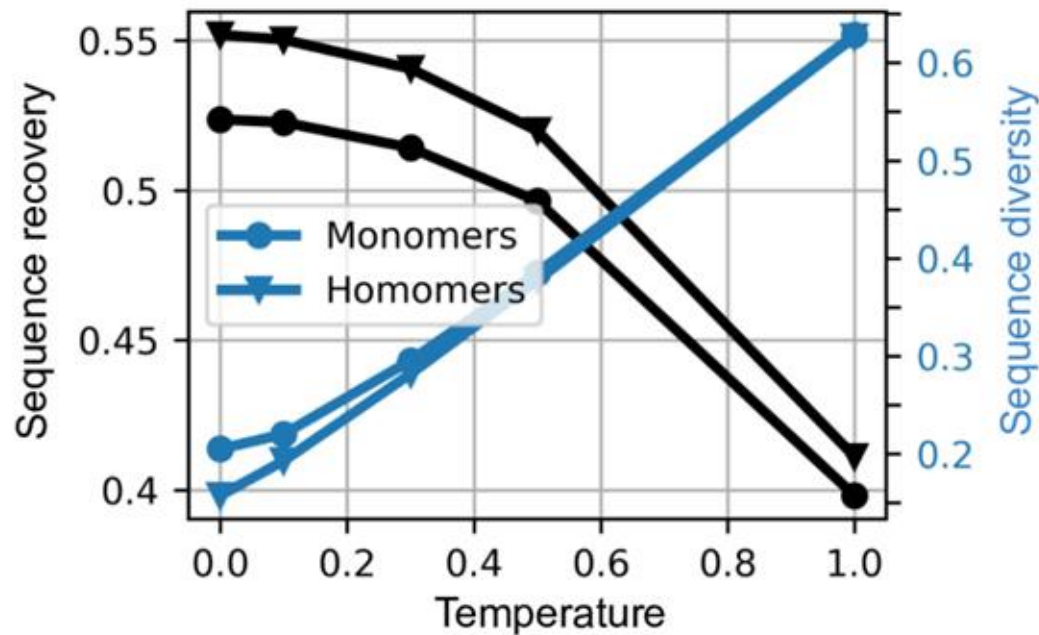
ProteinMPNN performance breakdown – single chain

Noise level when training: 0.00A/ 0.02A	Modification	Number of Parameters	PDB Test Accuracy	PDB Test Perplexit y	AlphaFold Model Accuracy
Baseline model	None	1.381 mln	41.2/ 40.1	6.51/ 6.77	41.4/ 41.4
Experiment 1	Add N, Ca, C, Cb, O distances	1.430 mln	49.0/ 46.1	5.03/ 5.54	45.7/ 47.4
Experiment 2	Update encoder edges	1.629 mln	43.1/ 42.0	6.12/ 6.37	43.3/ 43.0
Experiment 3	Combine 1 and 2	1.678 mln	50.5/ 47.3	4.82/ 5.36	46.3/ 47.9
Experiment 4	Experiment 3 with random instead of forward decoding	1.678 mln	50.8/ 47.9	4.74/ 5.25	46.9/ 48.5

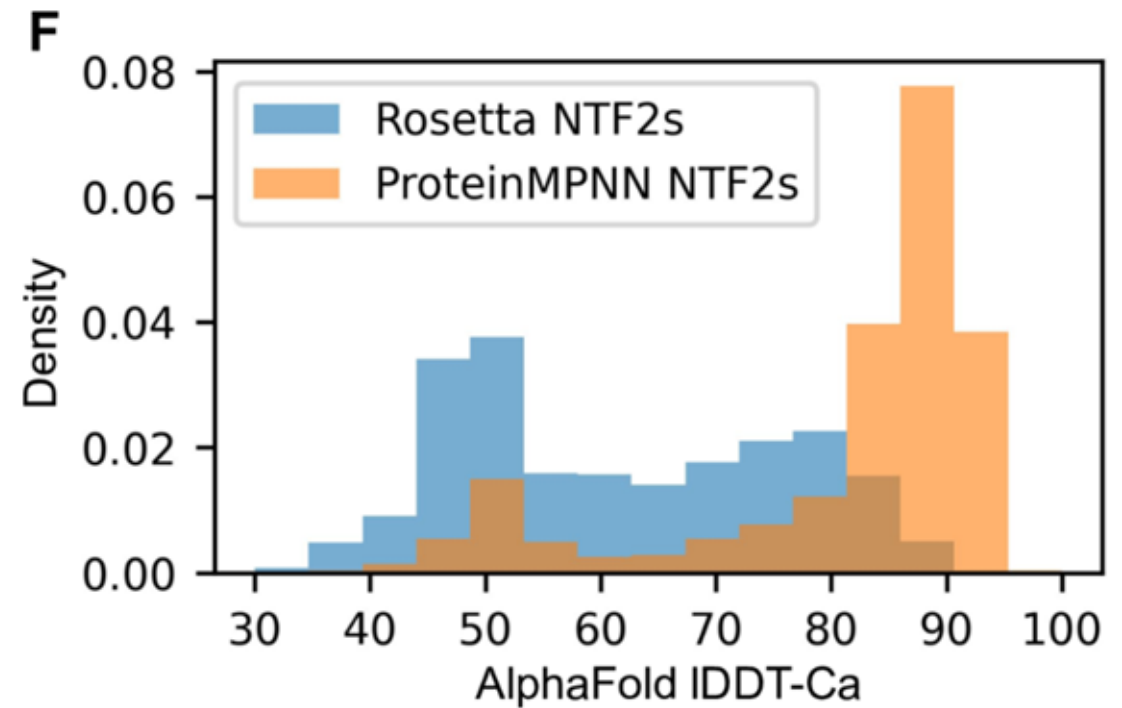
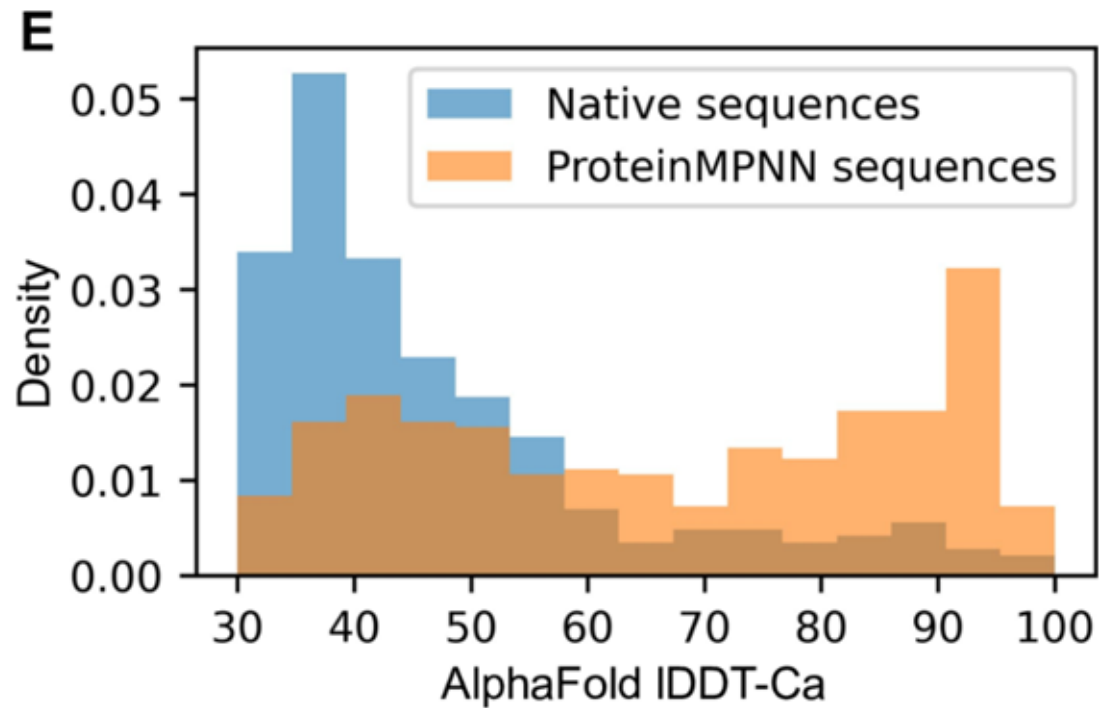
Increasing ProteinMPNN sampling temperature increases sequence diversity (and entropy)



Increasing ProteinMPNN sampling temperature increases sequence diversity (and entropy)



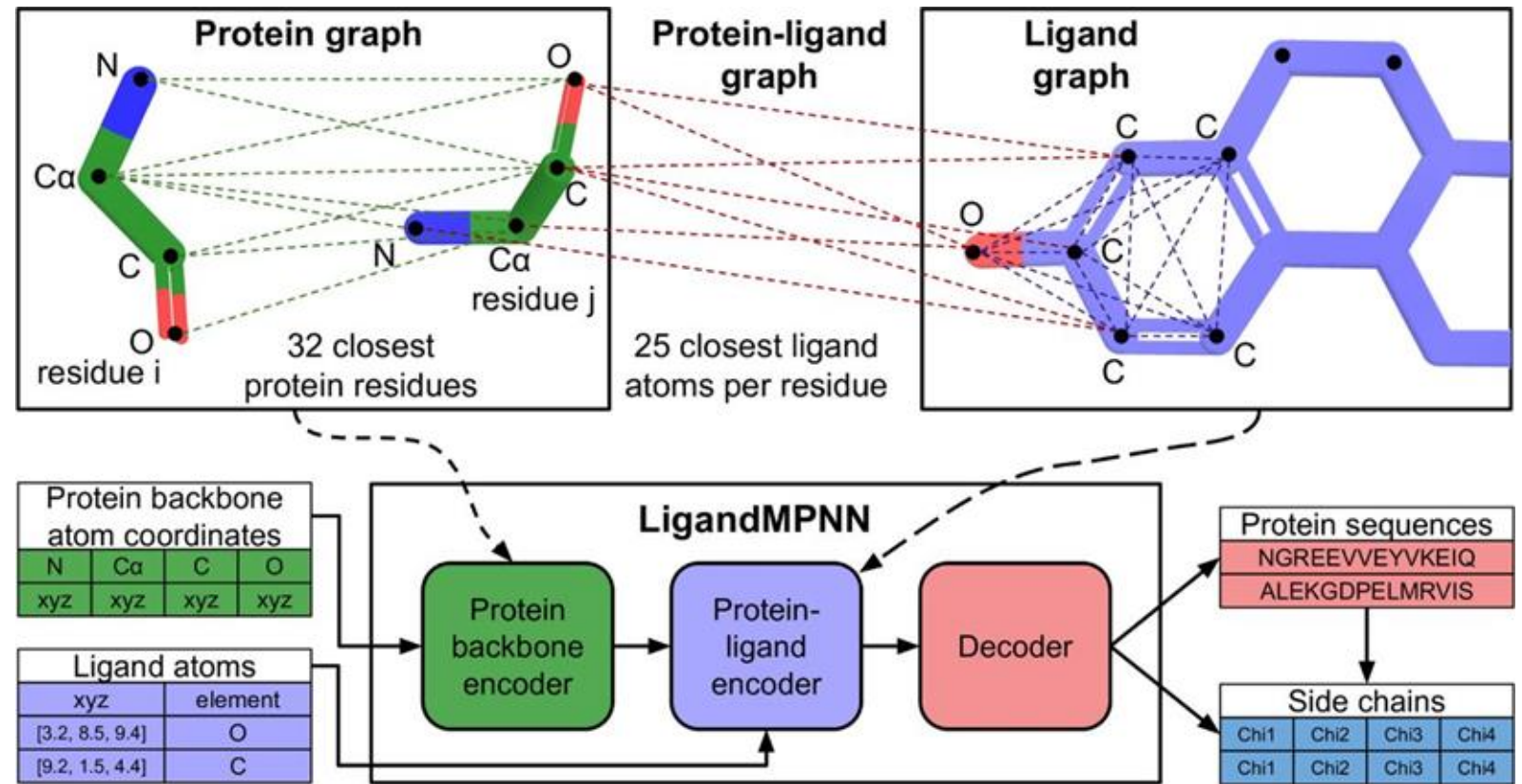
AlphaFold performance on ProteinMPNN sequences



LigandMPNN as extension of ProteinMPNN

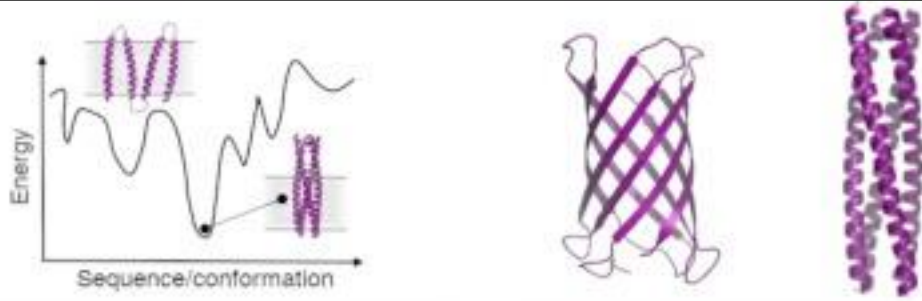
LigandMPNN is built on top of the ProteinMPNN architecture and allows us to input protein backbones with ligands (small molecules) bound

Potentially, this allows us to design sequences that will bind tightly to small molecules



3. Prioritizing designs in silico with Colabfold

Backbone generation

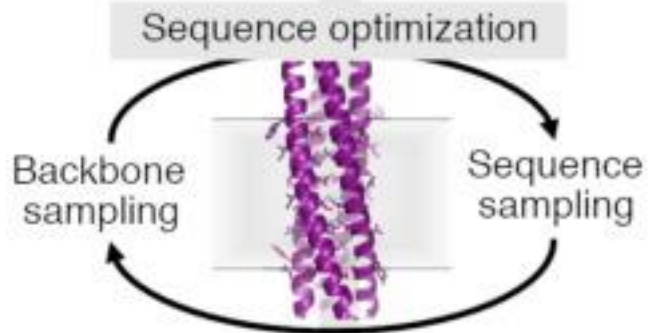


Scoring

Backbone definition

RFDiffusion

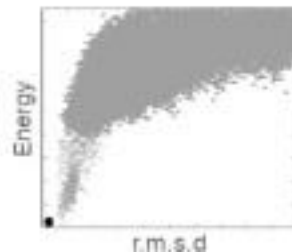
Sequence design



ProteinMPNN

Validation

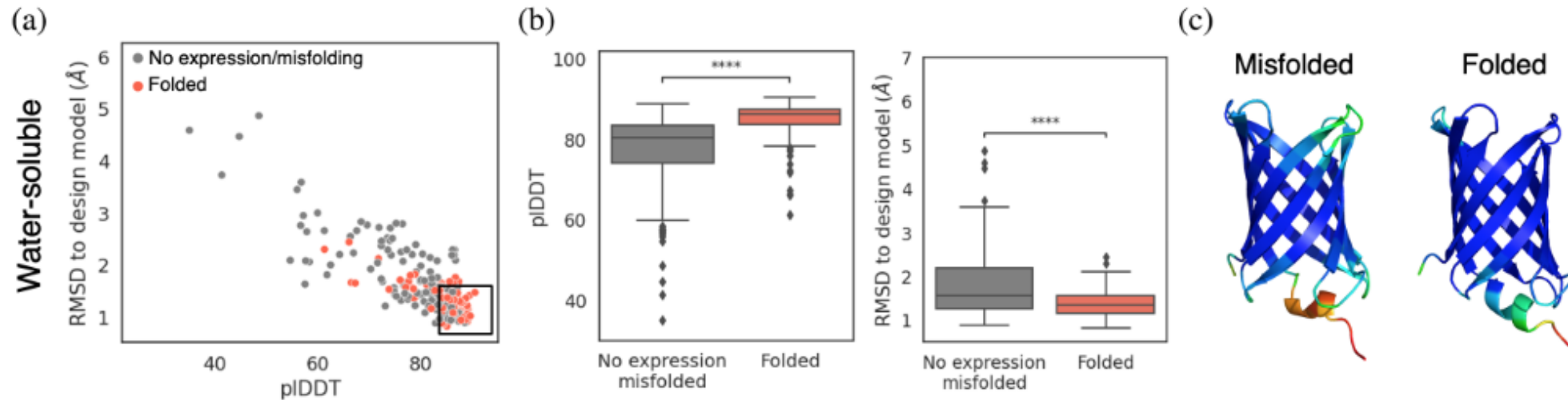
Energy landscape of the designed sequence converges on target structure



ColabFold

Rationale: AlphaFold can predict which designs are more likely to fold

IF structure predicted from a single sequence (no MSA, only possible for de novo designed proteins)



ColabFold: making protein folding accessible to all

<https://github.com/sokrypton/ColabFold>

Utilisation gratuite (limitée) des GPU de Google

✓ ColabFold v1.5.5: AlphaFold2 using MMseqs2

Easy to use protein structure and complex prediction using [AlphaFold2](#) and [Alphafold2-multimer](#). Sequence alignments/templates are generated through [MMseqs2](#) and [HHsearch](#). For more details, see [bottom](#) of the notebook, checkout the [ColabFold GitHub](#) and [Nature Protocols](#).

Old versions: [v1.4](#), [v1.5.1](#), [v1.5.2](#), [v1.5.3-patch](#)

[Mirdita M, Schütze K, Moriwaki Y, Heo L, Ovchinnikov S, Steinegger M. ColabFold: Making protein folding accessible to all. Nature Methods, 2022](#)



> Input protein sequence(s), then hit Runtime -> Run all



query_sequence: "PIAQIHILEGRSDEQKETLIREVSEAISRSLDAPLTSVRVIITEMAKGHFGIGGELASK"

- Use : to specify inter-protein chainbreaks for **modeling complexes** (supports homo- and hetro-oligomers). For example **PI...SK:PI...SK** for a homodimer

jobname: "test"

num_relax: 0

- specify how many of the top ranked structures to relax using amber

template_mode: none

- none = no template information is used. pdb100 = detect templates in pdb100 (see [notes](#)). custom - upload and search own templates (PDB or mmCIF format, see [notes](#))

[Afficher le code](#)

ColabFold: making protein folding accessible to all

<https://github.com/sokrypton/ColabFold>

0 s

MSA options (custom MSA upload, single sequence, pairing mode)

msa_mode: mmseqs2_uniref_env

pair_mode: mmseqs2_uniref_env → Include metagenomic databases

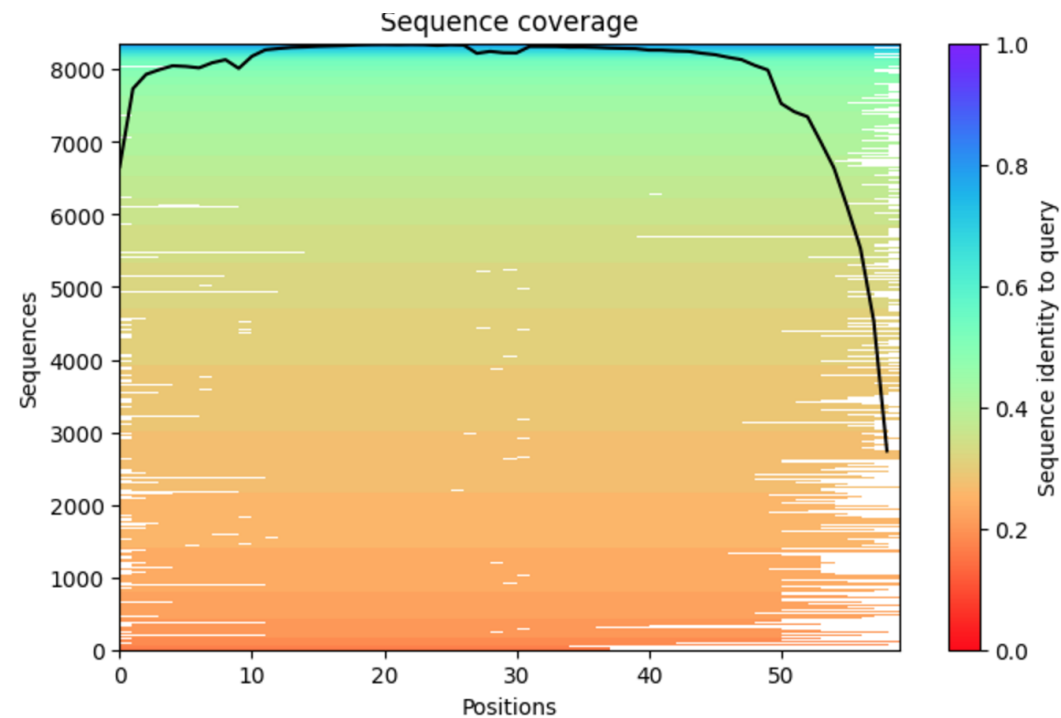
- "unpaired"

[Afficher le code](#)

mmseqs2_uniref → Uniref only

single_sequence → No MSA (de novo design ONLY!)

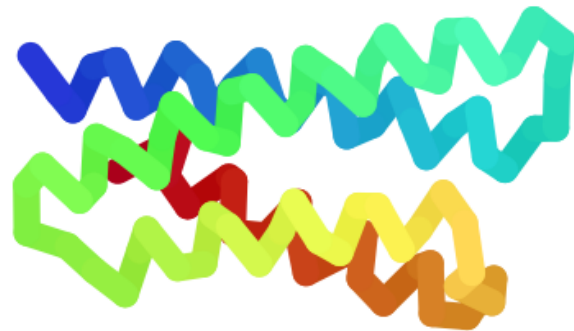
custom → Upload own MSA



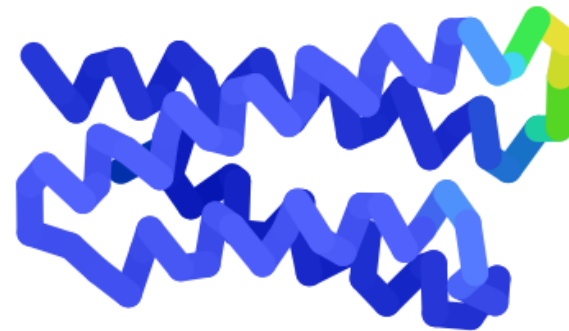
2024-11-24 19:59:24,743 Setting max_seq=512, max_extra_seq=5120

Example from day1 – using AlphaFold in Colab

colored by N→C



colored by pLDDT



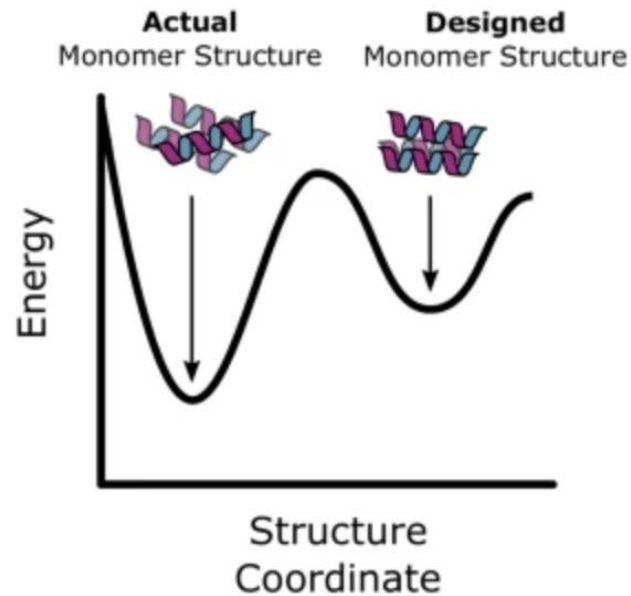
Reminder: Neoleukin-4 from Day 1

`Msa_mode is single_sequence`

BUT binders can have several failure modes

Interface OK, but protein
misfolds/aggregates

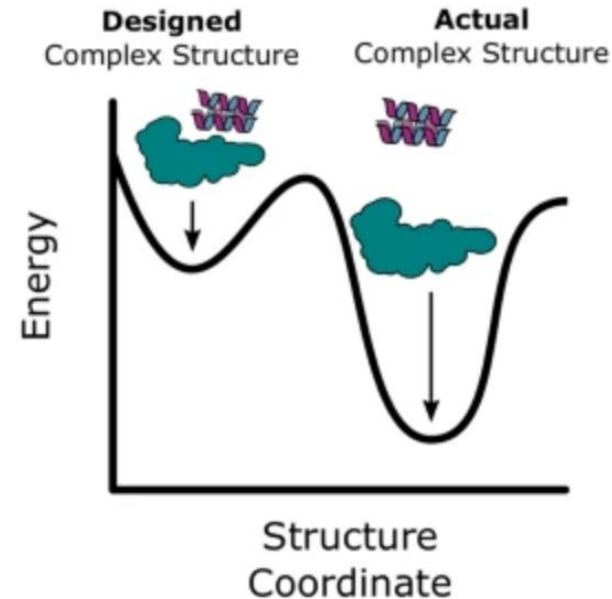
Type 1 Failure



AF2 prediction from
single sequence

Protein folds but does not bind

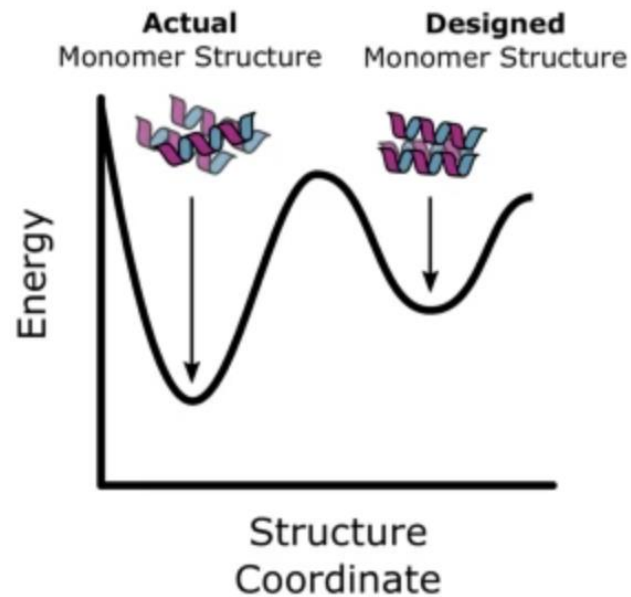
Type 2 Failure



BUT binders can have several failure modes

Interface OK, but protein
misfolds/aggregates

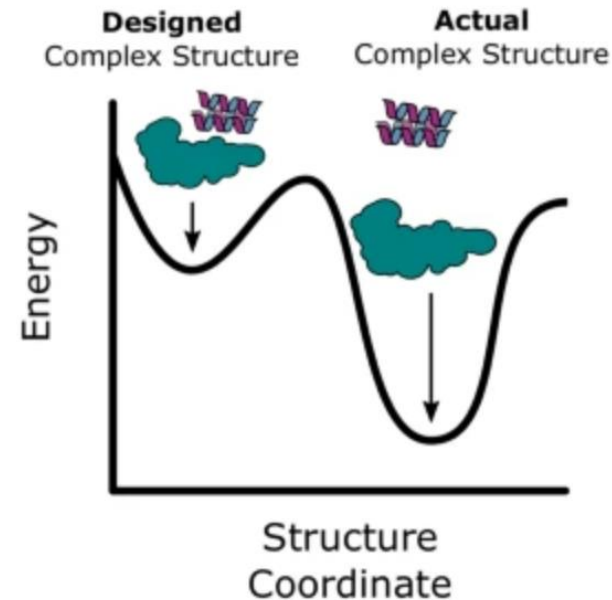
Type 1 Failure



AF2 prediction from
single sequence

Protein folds but does not bind

Type 2 Failure



AF2 prediction from
single sequence with
initial guess

AF2 with initial guess

Designed sequence

MGTF...TKLE

+

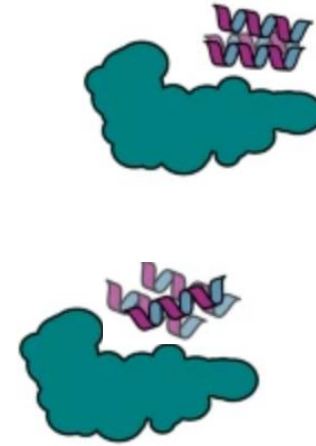
Target sequence

MKPRQ...SRHV

AF structure prediction
Single sequence!



Target structure as
template



Questions?

Thank you!

Learning how to design alpha-helical bundles

